

# Infinitesimal study of group schemes

SGA 3, Exposé VIIA

English draft

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## About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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# Exposé VIIA. Infinitesimal study of group schemes

by P. Gabriel

In Exposé II we had confined ourselves to the study of differential invariants of the first order, and we had not approached certain phenomena peculiar to characteristic  $p > 0$  or to characteristic 0. Our purpose in part A of this exposé is to fill this gap.

Moreover, the infinitesimal study of arbitrary order of a group scheme is related to that of the associated formal group; the purpose of the second part of this exposé is to present the first definitions and properties concerning formal groups.

## A) Differential operators and Lie $p$ -algebras<sup>\*</sup>

### 1. Differential operators

In this section, as well as in sections 2 and 3,  $S$  denotes a fixed scheme and the products considered are cartesian products in the category of  $S$ -schemes.<sup>(1)</sup> If  $X$  is an  $S$ -scheme, we denote by  $p_{X/S}$ ,  $p_X$ , or simply  $p$  the structural morphism from  $X$  to  $S$ .

**1.1.** — Let  $u : Y \rightarrow X$  be a morphism of  $S$ -schemes, and equip the direct image  $u_*(\mathcal{O}_Y)$  of the structure sheaf of  $Y$  with the  $\mathcal{O}_X$ -module structure induced by  $u$ . The sheaf  $\mathcal{H} = \mathcal{H}om_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y))$  of homomorphisms of  $p_X^{-1}(\mathcal{O}_S)$ -modules from  $\mathcal{O}_X$  to  $u_*(\mathcal{O}_Y)$  is thus naturally equipped with an  $\mathcal{O}_X$ -bimodule structure: if  $U$  is an open subset of  $X$ , and  $f$  and  $d$  are sections of  $\mathcal{O}_X$  and  $\mathcal{H}$  over  $U$ , then  $fd$  and  $df$  are respectively the morphisms  $g \mapsto fd(g)$  and  $g \mapsto d(fg)$  from  $\mathcal{O}_X$  to  $u_*(\mathcal{O}_Y)$ . From now on we shall write  $(\text{ad } f)(d)$  instead of  $fd - df$ .

**Definition 1.1.1.** An  $S$ -deviation of order  $\leq n$  is by definition a pair  $D = (u, d)$  consisting of a morphism of  $S$ -schemes  $u : Y \rightarrow X$  and a morphism of  $p_X^{-1}(\mathcal{O}_S)$ -modules  $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  such that, for every open subset  $U$  of  $X$  and every sequence of  $n + 1$  sections  $f_0, \dots, f_n \in \mathcal{O}_X(U)$ , the following equality holds in  $\text{Hom}_{p_U^{-1}(\mathcal{O}_S)}(\mathcal{O}_U, u_*(\mathcal{O}_Y)|_U)$ :

$$(*_n) \quad (\text{ad } f_0)(\text{ad } f_1) \cdots (\text{ad } f_n)(d) = 0.$$

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<sup>\*</sup>Part A of the present exposé had not been treated seriously in the oral lectures.

<sup>(1)</sup>**N.D.E.** In particular, if  $X$  and  $Y$  are two  $S$ -schemes,  $X \times_S Y$  is denoted simply by  $X \times Y$ . On the other hand, let us point out that for the contents of sections 1 and 2 one may refer to [DG70], § II.4, nos. 5–6; see also [Ja03], § I.7.

(2) In this case we shall also say that  $d$  is an  $S$ -deviation of  $u$  of order  $\leq n$ . In particular, an  $S$ -deviation of  $u$  of order  $\leq 0$  is a morphism of  $\mathcal{O}_X$ -modules from  $\mathcal{O}_X$  to  $u_*(\mathcal{O}_Y)$ , i.e. an element of  $\Gamma(Y, \mathcal{O}_Y)$ .

**Definition 1.1.2.** A morphism of  $p^{-1}(\mathcal{O}_S)$ -modules  $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  is an  $S$ -deviation of  $u$  if, for every point  $y$  of  $Y$ , there exist an open neighborhood  $U$  of  $u(y)$  in  $X$  and an open neighborhood  $V$  of  $y$  in  $Y$  satisfying the following conditions:

- a)  $u(V) \subset U$ ;
- b) if  $v : V \rightarrow U$  is the morphism induced by  $u$ , there is an integer  $n$  such that the morphism  $\mathcal{O}_U \rightarrow v_*(\mathcal{O}_V)$  induced by  $d$  is an  $S$ -deviation of  $v$  of order  $\leq n$ .<sup>(3)</sup>

If  $d$  is an  $S$ -deviation of  $u$ , we also say that the pair  $D = (u, d)$  is an  $S$ -deviation, and we shall sometimes write  $Y \xrightarrow{D} X$  or  $Y \xrightarrow[u]{d} X$ .

When  $d$  is the algebra homomorphism  $u^\sharp : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  corresponding to the morphism  $u : Y \rightarrow X$ , we shall also write  $u$  instead of  $D$ .

**Remarks 1.1.3.** <sup>(4)</sup> Let  $\text{D\'ev}(u)$  (resp.  $\text{D\'ev}^{\leq n}(u)$ ) be the set of  $S$ -deviations of  $u$  (resp.  $S$ -deviations of  $u$  of order  $\leq n$ ). It is equipped with a natural  $\mathcal{O}_Y(Y)$ -module structure: if  $\lambda \in \mathcal{O}_Y(Y)$ , then  $\lambda d$  is the deviation sending  $f$  to  $\lambda d(f)$ , for every section  $f$  of  $\mathcal{O}_X$  over an open subset  $U$ .

For every open subset  $V$  of  $Y$ , put  $\mathcal{D}\text{\'ev}(u)(V) = \text{D\'ev}(u|_V)$ , i.e.  $\mathcal{D}\text{\'ev}(u)(V)$  is the set of

$$\begin{aligned} d_V \in \text{Hom}_{p^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, (u|_V)_*(\mathcal{O}_V)) &\simeq \text{Hom}_{p^{-1}(\mathcal{O}_S)}((u|_V)^{-1}\mathcal{O}_X, \mathcal{O}_V) \\ &\simeq \mathcal{H}om_{p^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)(V) \end{aligned}$$

such that, for every open subset  $U$  of  $X$ , the map  $d_V(U) : \mathcal{O}_X(U) \rightarrow \mathcal{O}_Y(u^{-1}(U) \cap V)$  satisfies  $(*)_n$ . This defines a presheaf of  $\mathcal{O}_Y$ -modules on  $Y$ , and one sees easily that it is a sheaf (more precisely, a subsheaf of  $\mathcal{H}om_{p^{-1}(\mathcal{O}_S)}(u^{-1}\mathcal{O}_X, \mathcal{O}_Y)$ ).

**1.2.** — Now consider two  $S$ -deviations  $D = (u, d)$  and  $E = (v, e)$ :

$$Z \xrightarrow[e]{v} Y \xrightarrow[u]{d} X.$$

As  $U$  ranges over the open subsets of  $X$ , the composite maps

$$\Gamma(U, \mathcal{O}_X) \xrightarrow{d(U)} \Gamma(u^{-1}U, \mathcal{O}_Y) \xrightarrow{e(u^{-1}U)} \Gamma(v^{-1}u^{-1}U, \mathcal{O}_Z)$$

define an  $S$ -deviation of  $uv$  which we shall denote by  $de$ ; when  $d$  is of order  $\leq m$  and  $e$  of order  $\leq n$ ,  $de$  is of order  $\leq m + n$ . We shall also write

$$(\dagger) \quad D \circ E = (uv, de)$$

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(2)**N.D.E.** One sees easily that this is equivalent to saying that, for every  $x \in X$  and  $f_0, \dots, f_n, g \in \mathcal{O}_{X,x}$ , one has  $(\text{ad } f_0)(\text{ad } f_1) \cdots (\text{ad } f_n)(d_x)(g) = 0$ . On the other hand, recall that the adjunction isomorphism

$$\theta : \text{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \text{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}(\mathcal{O}_X), \mathcal{O}_Y)$$

associates to every morphism of  $p_X^{-1}(\mathcal{O}_S)$ -modules  $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  the morphism  $d' = \varepsilon \circ u^{-1}(d)$ , where  $\varepsilon$  is the canonical morphism  $u^{-1}u_*(\mathcal{O}_Y) \rightarrow \mathcal{O}_Y$ . Conversely, for every  $p_Y^{-1}(\mathcal{O}_S)$ -morphism  $d' : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$ , one has  $\theta^{-1}(d') = u_*(d') \circ \eta$ , where  $\eta$  is the canonical morphism  $\mathcal{O}_X \rightarrow u_*u^{-1}(\mathcal{O}_X)$ . It follows that  $d$  satisfies  $(*)_n$  if and only if  $d'$  satisfies:

$$(*'_n) \quad (\text{ad } f_0) \cdots (\text{ad } f_n)(d')(g) = 0$$

for every open subset  $V$  of  $Y$  and  $f_0, \dots, f_n, g \in u^{-1}(\mathcal{O}_X)(V)$ .

(3)**N.D.E.** If  $X$  and  $u$  are quasi-compact, every  $S$ -deviation of  $u$  is therefore of order  $\leq n$ , for some integer  $n$ .

(4)**N.D.E.** These remarks have been added; they will be useful in 1.3, 1.4 and 2.1.

<sup>(5)</sup> and we shall say that  $D \circ E$  or  $DE$  is the *composite  $S$ -deviation*. When  $d = u^\natural$  (i.e.  $D = u$  with the convention of 1.1),  $DE$  is also called the *image of  $E$  under  $u$* .

The map  $(D, E) \mapsto D \circ E$  which we have just defined will henceforth allow us to speak of the *category of  $S$ -deviations*, whose objects are the  $S$ -schemes and whose morphisms are the  $S$ -deviations.<sup>(6)</sup>

**Definition 1.2.0.** <sup>(7)</sup> Let  $w : Z \rightarrow X$  be an  $S$ -morphism. An  $S$ -derivation of  $w$ , or  $S$ -derivation from  $\mathcal{O}_X$  to  $w_*(\mathcal{O}_Z)$ , is a morphism of  $p^{-1}(\mathcal{O}_S)$ -modules  $d : \mathcal{O}_X \rightarrow w_*(\mathcal{O}_Z)$  such that, for every open subset  $U$  of  $X$  and  $f, g \in \mathcal{O}_X(U)$ ,

$$d(fg) = w^\natural(f)d(g) + w^\natural(g)d(f).$$

Then  $d$  is a deviation of  $w$  of order  $\leq 1$ , which vanishes on the unit section of  $\mathcal{O}_X$ . The set of  $S$ -derivations of  $w$  will be denoted by  $\text{Dér}_S(w)$ ; it is an  $\mathcal{O}(Z)$ -module.

With the notation of 1.2, take  $Y$  equal to  $I_Z = \text{Spec } \mathcal{O}_Z[t]$ , where  $t^2 = 0$ , and  $v$  equal to the zero section  $\tau : Z \rightarrow I_Z$ , defined by the morphism of  $\mathcal{O}_Z$ -algebras  $\mathcal{O}_Z[t] \rightarrow \mathcal{O}_Z$  sending  $t$  to 0, and take  $e$  equal to the morphism of  $\mathcal{O}_Z$ -modules  $\sigma : \mathcal{O}_Z[t] \rightarrow \mathcal{O}_Z$  defined by  $\sigma(1) = 0$  and  $\sigma(t) = 1$ ,<sup>(8)</sup> which it is convenient to denote by  $\partial_t$ .

If  $u : I_Z \rightarrow X$  is a morphism satisfying  $w = u \circ s$ , then  $\sigma \circ u^\natural$  is an  $S$ -derivation from  $\mathcal{O}_X$  to  $w_*(\mathcal{O}_Z)$ . Conversely, to every  $S$ -derivation  $d$  one associates the morphism  $u : I_Z \rightarrow X$  such that  $u = w$  on the underlying spaces, and

$$u^\natural(f) = w^\natural(f) + d(f)t,$$

for every section  $f$  of  $\mathcal{O}_X$  over an open subset  $U$ . One thus obtains:

**Lemma 1.2.1.** *Let  $E = (\tau, \partial_t)$  be the deviation of  $\tau : Z \rightarrow I_Z$  defined above. For every  $S$ -morphism  $w : Z \rightarrow X$ , the map  $u \mapsto u \circ E$  is a bijection between the  $S$ -morphisms  $u : I_Z \rightarrow X$  such that  $u \circ s = w$ , and the  $S$ -derivations of  $w$ .*

**1.2.2.** — Let  $d$  be an  $S$ -deviation of  $u : Y \rightarrow X$ . On the one hand,  $d$  is clearly an  $S'$ -deviation of  $u$  for every morphism  $s : S \rightarrow S'$ .

On the other hand, let  $t : T \rightarrow S$  be a morphism with target  $S$ , and let  $u_T : Y_T \rightarrow X_T$  be the morphism deduced from  $u$  by base change, and  $t_Y : Y_T \rightarrow Y$  and  $t_X : X_T \rightarrow X$  the canonical projections. There then exists one and only one  $T$ -deviation of  $u_T$ , which we shall denote by  $d_T$  or  $d \times T$ , satisfying the equality  $t_X d_T = dt_Y$ , in the sense of  $(\dagger)$  above, i.e. for

<sup>(5)</sup>**N.D.E.** Care must be taken that, with this notation,  $de$  denotes the composite “ $d$  followed by  $e$ ”.

<sup>(6)</sup>**N.D.E.** Often one considers only the  $S$ -deviations of the morphism  $\text{id}_X$ , which form the algebra of  $S$ -differential operators of  $X$ , cf. 1.4 below. However, the more general setting of  $S$ -deviations provides a convenient “functorial” language for proving statements such as: “if  $G$  is an  $S$ -group, the algebra of left-invariant  $S$ -differential operators on  $G$  is isomorphic to the algebra of  $S$ -deviations of the unit section  $\varepsilon : S \rightarrow G$ , cf. 2.1 and 2.4 below.

<sup>(7)</sup>**N.D.E.** This paragraph has been expanded, assigning to this definition (resp. to the following lemma) the number 1.2.0 (resp. 1.2.1).

<sup>(8)</sup>**N.D.E.** What follows has been added, i.e. the notation  $\partial_t$  has been introduced.

every open subset  $U$  of  $X$  there is a commutative diagram:<sup>(9)</sup>

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{t_X^\sharp} & \mathcal{O}(U \times T) \\ d(U) \downarrow & & \downarrow d_T(U \times T) \\ \mathcal{O}(u^{-1}U) & \xrightarrow{t_Y^\sharp} & \mathcal{O}(u^{-1}U \times T). \end{array}$$

If one puts  $D = (u, d)$ , one also writes  $D_T = (u_T, d_T)$ , and we shall say that  $d_T$  and  $D_T$  are deduced from  $d$  and  $D$  by base change.

**1.2.3.** — Let, for example,  $u : Y \rightarrow X$  and  $v : Z \rightarrow T$  be two  $S$ -morphisms, and  $d$  and  $e$   $S$ -deviations of  $u$  and  $v$ . There is a commutative diagram

$$\begin{array}{ccc} X \times T & \xleftarrow{u_T} & Y \times T \\ v_X \uparrow & \swarrow u \times v & \uparrow v_Y \\ X \times Z & \xleftarrow{u_Z} & Y \times Z \end{array}$$

and we shall denote by  $d \times e$  (product of  $d$  and  $e$ ) the  $S$ -deviation of  $u \times v$  equal to  $d_T e_Y = e_X d_Z$  (with the convention  $(\dagger)$  above), i.e. for every open subset  $U$  of  $X \times T$ , denoting by  $W$  the open subset  $v_Y^{-1}u_T^{-1}U = u_Z^{-1}v_X^{-1}U$ , there is a commutative diagram:

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{d_T(U)} & \mathcal{O}(u_T^{-1}U) \\ e_X(U) \downarrow & \searrow (d \times e)(U) & \downarrow e_Y(u_T^{-1}U) \\ \mathcal{O}(v_X^{-1}U) & \xrightarrow{d_Z(v_X^{-1}U)} & \mathcal{O}(W). \end{array}$$

If one puts  $D = (u, d)$  and  $E = (v, d)$ , one also writes  $D \times E = (u \times v, d \times e)$ .

**1.3.** — <sup>(10)</sup> Let  $u : Y \rightarrow X$  be a morphism of  $S$ -schemes. Recall that the adjunction isomorphism:

$$\mathrm{Hom}_{p_X^{-1}(\mathcal{O}_S)}(\mathcal{O}_X, u_*(\mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{Hom}_{p_Y^{-1}(\mathcal{O}_S)}(u^{-1}(\mathcal{O}_X), \mathcal{O}_Y)$$

associates to every morphism of  $p^{-1}(\mathcal{O}_S)$ -modules  $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  the morphism  $d' = \varepsilon \circ u^{-1}(d)$ , where  $\varepsilon$  is the canonical morphism  $u^{-1}u_*(\mathcal{O}_Y) \rightarrow \mathcal{O}_Y$ .

Denote by  $\mathcal{J}_u$  (resp.  $\mathcal{J}_u$ ) the kernel of the algebra homomorphism  $u^\sharp : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  (resp.  $u^\flat : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$ ), and let  $d : \mathcal{O}_X \rightarrow u_*(\mathcal{O}_Y)$  be a morphism of  $p^{-1}(\mathcal{O}_S)$ -modules. If  $U$  is an open subset of  $X$  and  $f_0, \dots, f_n, g \in \mathcal{O}_X(U)$ , one sees easily by induction on  $n$  that condition  $(*_n)$  is equivalent to the following equality (cf. EGA IV<sub>4</sub>, 16.8.8.2):

$$(**_n) \quad 0 = \sum_{I \subset [0, n]} (-1)^{|I|} u^\sharp(f_{[0, n] - I}) d(f_I g),$$

where  $f_I$  denotes the product of the  $f_i$ , for  $i \in I$ . It follows that if  $d$  satisfies  $(*_n)$ , then  $d$  vanishes on the ideal  $\mathcal{J}_u^{n+1}$ .

<sup>(9)</sup>**N.D.E.** Explicitly, if  $V$  is an affine open subset of  $S$  and  $U$  (resp.  $U'$ ) an affine open subset of  $X$  (resp.  $T$ ) over  $V$ , so that  $\mathcal{O}_{X \times T}(U \times U') = \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U')$ , then  $d_T(U \times U')$  is the composite:

$$\begin{aligned} \mathcal{O}_X(U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') & \xrightarrow{d(U) \otimes \mathrm{id}} \mathcal{O}_Y(u^{-1}U) \otimes_{\mathcal{O}_S(V)} \mathcal{O}_T(U') \\ & \longrightarrow \mathcal{O}_{Y \times T}(u^{-1}U \times U'). \end{aligned}$$

The author has left to the reader the verification that  $d_T$  is well defined, and the editors do likewise.

<sup>(10)</sup>**N.D.E.** The original has been expanded in this paragraph; see also N.D.E. (2) in 1.1.1.

Now suppose that  $Y$  is equal to  $S$ ; then  $u : S \rightarrow X$  is a section of  $p : X \rightarrow S$ , hence is an immersion (cf. EGA I, 5.3.13). Then, on the one hand,  $\varepsilon : u^{-1}u_*\mathcal{O}_S \rightarrow \mathcal{O}_S$  is an isomorphism, so that  $u^{-1}(\mathcal{J}_u) = \mathcal{J}_u$ . On the other hand, there is an isomorphism:

$$(\star) \quad u^{-1}(\mathcal{O}_X) \simeq \mathcal{O}_S \oplus \mathcal{J}_u.$$

Suppose that  $d$  vanishes on  $\mathcal{J}_u^{n+1}$ . Then  $d' = \varepsilon \circ u^{-1}(d)$  vanishes on  $\mathcal{J}_u^{n+1}$ , and therefore  $d'$  satisfies the analogues  $(**'_n)$  and  $(*'_n)$  of  $(**_n)$  and  $(*_n)$ , when  $f_0, \dots, f_n \in \mathcal{J}_u(u^{-1}(U))$ . Moreover, since  $(\text{ad } a)(\phi) = 0$  for every  $a \in \mathcal{O}_S(u^{-1}(U))$  and every morphism of  $\mathcal{O}_{u^{-1}(U)}$ -modules  $\phi : u^{-1}(\mathcal{O}_U) \rightarrow \mathcal{O}_{u^{-1}(U)}$ , one deduces from  $(\star)$  that  $d'$  satisfies the analogue  $(*'_n)$  of  $(*_n)$ . It follows that  $d$  satisfies  $(*_n)$ . Consequently, one has obtained:

**Lemma.** *If  $u : S \rightarrow X$  is a section of  $p : X \rightarrow S$ , then  $d$  is an  $S$ -deviation of  $u$  of order  $\leq n$  if and only if  $d'$  vanishes on  $\mathcal{J}_u^{n+1}$ .*

This interpretation may be generalized as follows. Let  $u : Y \rightarrow X$  be an arbitrary  $S$ -morphism, and  $\Gamma u$  the graph of  $u$ , that is, the morphism  $Y \rightarrow Y \times X$  with components  $\text{id}_Y$  and  $u$ . For every  $S$ -deviation  $d$  of  $u$  of order  $\leq n$ , one obtains by composition:

$$Y \xrightarrow{\text{diag.}} Y \times Y \xrightarrow[u_Y]{d_Y} Y \times X$$

a  $Y$ -deviation of  $\Gamma u$  of order  $\leq n$ , which we shall denote by  $\Gamma d$  (the *graph* of  $d$ ).

Conversely, to every  $Y$ -deviation  $e$  of  $\Gamma u$  one associates the composite  $S$ -deviation  $e_X = \text{pr}_2 \circ e$ :

$$Y \xrightarrow[\Gamma u]{e} Y \times X \xrightarrow{\text{pr}_2} X.$$

One sees immediately that  $(\Gamma d)_X = d$ , and the equality  $\Gamma e_X = e$  follows from the fact that  $e$  is  $\mathcal{O}_Y$ -linear.<sup>(11)</sup> One thus obtains an isomorphism of  $\mathcal{O}_Y(Y)$ -modules:

$$\begin{array}{ccc} \{S\text{-deviations of } u \text{ of order } \leq n\} & \xrightarrow{\sim} & \{Y\text{-deviations of } \Gamma u \text{ of order } \leq n\} \\ d & \longmapsto & \Gamma d. \end{array}$$

Moreover, one sees easily that  $d$  is an  $S$ -derivation of  $u$  if and only if  $\Gamma d$  is a  $Y$ -derivation of  $\Gamma u$ .

Call  $\mathcal{J}_{\Gamma u}$  the kernel of the algebra homomorphism  $(\Gamma u)^{-1}(\mathcal{O}_{Y \times X}) \rightarrow \mathcal{O}_Y$  corresponding to  $\Gamma u$ . Taking account of the preceding lemma, one has obtained:

**Proposition.** *Let  $u : Y \rightarrow X$  be an  $S$ -morphism and  $\Gamma u : Y \rightarrow Y \times X$  its graph. The  $S$ -deviations of  $u$  of order  $\leq n$  identify with the  $Y$ -deviations of  $\Gamma u$  of order  $\leq n$ , which are in bijection with*

$$\text{Hom}_{\mathcal{O}_Y}((\Gamma u)^{-1}(\mathcal{O}_{Y \times X})/\mathcal{J}_{\Gamma u}^{n+1}, \mathcal{O}_Y).$$

**1.3.1.** — <sup>(12)</sup> Return to the case where  $u : S \rightarrow X$  is a section of  $p : X \rightarrow S$ . Then the homomorphism  $\phi : u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_S$  admits a section, which we shall denote simply by  $g \mapsto g \cdot 1$ , so that, with the notation of 1.3, there is an isomorphism of  $\mathcal{O}_S$ -modules:

$$(\star) \quad u^{-1}(\mathcal{O}_X) \simeq \mathcal{O}_S \oplus \mathcal{J}_u,$$

and for every section  $f$  of  $u^{-1}(\mathcal{O}_X)$ ,  $f - \phi(f) \cdot 1$  is a section of  $\mathcal{J}_u$ .

Let  $d$  be an  $S$ -deviation of  $u$  of order  $\leq 1$ , and  $d'$  the  $\mathcal{O}_S$ -morphism  $u^{-1}(\mathcal{O}_X) \rightarrow \mathcal{O}_S$  corresponding to  $d$ . If  $a, b$  are sections of  $u^{-1}(\mathcal{O}_X)$ , one has:

$$\begin{aligned} 0 &= d'((a - \phi(a) \cdot 1)(b - \phi(b) \cdot 1)) \\ &= d'(ab) - \phi(a)d'(b) - \phi(b)d'(a) + \phi(ab)d'(1). \end{aligned}$$

<sup>(11)</sup>**N.D.E.** If  $\lambda, f$  are local sections of  $\mathcal{O}_Y$  and  $\mathcal{O}_X$ , one has  $(\Gamma e_X)(\lambda \otimes f) = \lambda \cdot e(1 \otimes g)$ , and this equals  $e(\lambda \otimes g)$  since  $e$  is  $\mathcal{O}_Y$ -linear.

<sup>(12)</sup>**N.D.E.** This paragraph has been added.

Consequently, one sees that  $d$  is an  $S$ -derivation of  $u$  (cf. 1.2.1 and N.D.E. (2)) if and only if  $d'(1) = 0$ . One therefore obtains:

**Lemma.** *The  $S$ -derivations of  $u$  are exactly the  $S$ -deviations of  $u$  of order 1 which vanish on the unit section of  $\mathcal{O}_X$ ; they correspond to the  $\mathcal{O}_S(S)$ -module*

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}_u/\mathcal{I}_u^2, \mathcal{O}_S),$$

and there is an isomorphism of  $\mathcal{O}_S(S)$ -modules  $\mathrm{Dér}^{\leq 1}(u) \simeq \mathcal{O}_S(S) \oplus \mathrm{Dér}_S(u)$ .

Returning to the general case, one deduces from this, with the notation of 1.3,

**Corollary.** *Let  $u : Y \rightarrow X$  be an  $S$ -morphism and  $\Gamma u : Y \rightarrow Y \times X$  its graph. There is a canonical isomorphism of  $\mathcal{O}_Y(Y)$ -modules*

$$\mathrm{Dér}_S(u) \simeq \mathrm{Dér}_Y(\Gamma u) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_{\Gamma u}/\mathcal{I}_{\Gamma u}^2, \mathcal{O}_Y).$$

**Definition 1.4.** Let  $X$  be an  $S$ -scheme. An  $S$ -differential operator (resp. an  $S$ -differential operator of order  $\leq n$ ) on  $X$  means any  $S$ -deviation (resp. any  $S$ -deviation of order  $\leq n$ ) of the identity morphism of  $X$ .

According to 1.1, an  $S$ -differential operator of order  $\leq n$  is thus an endomorphism of the  $p^{-1}(\mathcal{O}_S)$ -module  $\mathcal{O}_X$  satisfying equalities  $(*_n)$  of 1.1. We shall denote by  $\mathrm{Dif}_{X/S}^n$  the  $\Gamma(\mathcal{O}_S)$ -module<sup>(13)</sup> consisting of the  $S$ -differential operators of order  $\leq n$ , and by  $\mathrm{Dif}_{X/S}$  that consisting of all the  $S$ -differential operators.

As we saw in 1.2, one can compose the  $S$ -deviations of  $\mathrm{id}_X$ , which equips  $\mathrm{Dif}_{X/S}$  with a  $\Gamma(\mathcal{O}_S)$ -algebra structure; we shall say that it is the *algebra of differential operators of  $X/S$* .

Similarly, for every open subset  $V$  of  $X$ , put  $\mathcal{A}_{X/S}(V) = \mathrm{Dif}_{V/S} = \mathrm{Dér}(\mathrm{id}_V)$ ; according to 1.1.3, this defines a sheaf of  $\mathcal{O}_X$ -modules, called the *sheaf of  $S$ -differential operators on  $X$* .<sup>(14)</sup>

**1.4.1.** — As we saw in 1.3, one can interpret the differential operators of  $X/S$  by means of the graph of the identity morphism of  $X$ , that is, of the diagonal morphism  $\Delta = \Delta_{X/S}$  from  $X$  to  $X \times X$ . Let us translate the statements of 1.3 into the present context.

Equip  $\mathcal{O}_{X \times X}$  with the  $\mathrm{pr}_1^{-1}(\mathcal{O}_X)$ -algebra structure defined by  $\mathrm{pr}_1$ , so that  $\Delta^{-1}(\mathcal{O}_{X \times X})$  is equipped with an algebra structure over  $\mathcal{O}_X = \Delta^{-1}\mathrm{pr}_1^{-1}(\mathcal{O}_X)$ . Let  $\mathcal{I}_{X/S}$  be the kernel of the homomorphism

$$\Delta^{-1}(\mathcal{O}_{X \times X}) \rightarrow \mathcal{O}_X$$

adjoint to the homomorphism  $\mathcal{O}_{X \times X} \rightarrow \Delta_*(\mathcal{O}_X)$ , and let  $\mathcal{P}_{X/S}^m$  be the  $\mathcal{O}_X$ -algebra

$$\Delta^{-1}(\mathcal{O}_{X \times X})/\mathcal{I}_{X/S}^{m+1}.$$

If  $V$  is an affine open subset of  $S$  and  $U$  an affine open subset of  $X$  over  $V$ , and if one puts  $k = \Gamma(V, \mathcal{O}_S)$  and  $A = \Gamma(U, \mathcal{O}_X)$ , one therefore has:

$$\Gamma(U, \mathcal{P}_{X/S}^m) = (A \otimes_k A)/I^{m+1},$$

where  $I$  is the ideal generated by the elements  $a \otimes 1 - 1 \otimes a$ , for  $a \in A$ . This being so, according to 1.3 there is an isomorphism of  $\mathcal{O}_X(X)$ -modules:

$$j_X : \mathrm{Dif}_{X/S}^m \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^m, \mathcal{O}_X)$$

which may be defined as follows: if  $d$  belongs to  $\mathrm{Dif}_{X/S}^m$  and  $c$  is a section of  $\mathcal{P}_{X/S}^m$  over  $U$  of the form  $a \otimes b + I^{m+1}$ , one has  $j_X(d)(c) = a \cdot d(b)$ .<sup>(15)</sup>

<sup>(13)</sup>**N.D.E.** In this exposé, the ring  $\Gamma(S, \mathcal{O}_S) = \mathcal{O}_S(S)$  is denoted by  $\Gamma(\mathcal{O}_S)$ .

<sup>(14)</sup>**N.D.E.** Here the original has been modified; it mentioned the sheaf  $U \mapsto \mathrm{Dif}_{X_U/U}$ , where  $U$  ranges over the open subsets of  $S$ ; this is the direct image of  $\mathcal{A}_{X/S}$  under the morphism  $p_X : X \rightarrow S$ .

<sup>(15)</sup>**N.D.E.** Via this isomorphism, the  $X$ -derivations of  $\Delta_{X/S}$  correspond, according to 1.3.1, to the  $S$ -derivations of  $\mathrm{id}_X$ , i.e. to the  $p^{-1}(\mathcal{O}_S)$ -derivations of  $\mathcal{O}_X$ .

**1.4.2.** — Let  $d$  be a differential operator and  $u$  a section of  $X$  over  $S$ . We call the composite  $S$ -deviation

$$S \xrightarrow{u} X \xrightarrow[\text{id}_X]{d} X.$$

the *value of  $d$  at  $u$* .

According to 1.3 and 1.4.1, if  $d$  is a differential operator of order  $\leq m$ , then  $du$  (resp.  $d$ ) is canonically associated to a morphism of  $\mathcal{O}_S$ -modules  $d' : u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1} \rightarrow \mathcal{O}_S$  (resp. a morphism of  $\mathcal{O}_X$ -modules  $d'' : \mathcal{P}_{X/S}^m \rightarrow \mathcal{O}_X$ ).

It is clear that one can construct  $d'$  from  $d''$  in the following way: the square

$$\begin{array}{ccc} X \simeq S \times X & \xrightarrow{u \times X} & X \times X \\ p \downarrow & & \downarrow \text{pr}_1 \\ S & \xrightarrow{u} & X \end{array}$$

is cartesian, which allows one to identify  $X$  with  $S \times_X (X \times X)$ ,  $u$  with  $S \times_X \Delta$ , and hence  $u^*(\mathcal{P}_{X/S}^m)$  with  $u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1}$ . One thus identifies  $u^*(d'')$  with a morphism

$$u^{-1}(\mathcal{O}_X)/\mathcal{I}_u^{m+1} \rightarrow \mathcal{O}_S,$$

which is none other than  $d'$ .

**1.5.** — Put, as usual,  $I_S = \text{Spec } \mathcal{O}_S[T]/(T^2)$ . Let  $\tau : S \rightarrow I_S$  be the zero section and  $\sigma$  the canonical deviation of  $\tau$  which we defined in 1.2.0, i.e. the homomorphism of  $\mathcal{O}_S$ -modules which vanishes on the unit section of  $\mathcal{O}_S[T]/(T^2)$  and sends the class  $t$  of  $T$  modulo  $T^2$  to the unit section of  $\mathcal{O}_S$ .

Let  $X$  be an  $S$ -scheme. To every  $I_S$ -automorphism  $u$  of  $I_S \times X$  inducing the identity on  $X$ , a differential operator  $D_u$  of  $X$  is associated by composition:

$$X \simeq S \times X \xrightarrow{\sigma \times X} I_S \times X \xrightarrow{u} I_S \times X \xrightarrow{\text{pr}_2} X.$$

According to II, 3.14, the map  $u \mapsto D_u$  is an isomorphism of the  $\Gamma(\mathcal{O}_S)$ -Lie algebra

$$\text{Lie}(\underline{\text{Aut}} X) := \text{Lie}(\underline{\text{Aut}} X)(S)$$

onto the  $\Gamma(\mathcal{O}_S)$ -Lie algebra of  $p^{-1}(\mathcal{O}_S)$ -derivations of  $\mathcal{O}_X$ . The inverse isomorphism associates to every derivation  $D$  the automorphism of  $I_S \times X$  corresponding to the automorphism  $a + bt \mapsto a + (Da + b)t$  of  $\mathcal{O}_X[T]/(T^2)$ .

## 2. Invariant differential operators on group schemes

**2.1.** — Let  $G$  be an  $S$ -group scheme; we denote by  $\varepsilon$  or  $\varepsilon_G : S \rightarrow G$  the unit section of  $G$ .

**Definition.** Let  $U(G)$  be the  $\Gamma(\mathcal{O}_S)$ -module of  $S$ -deviations of  $\varepsilon_G$  (or  *$S$ -deviations of the origin*) (cf. 1.1).

If  $d$  and  $e$  are two elements of  $U(G)$ ,  $d \times e$  is an  $S$ -deviation of  $\varepsilon \times \varepsilon : S \simeq S \times S \rightarrow G \times G$ . The image of  $d \times e$  under the multiplication morphism  $m : G \times G \rightarrow G$  (cf. 1.2) will be called the *product of  $d$  and  $e$*  and will be denoted by  $d \cdot e$ .

The  $\Gamma(\mathcal{O}_S)$ -module  $U(G)$  is thus equipped with an associative  $\Gamma(\mathcal{O}_S)$ -algebra structure having  $\varepsilon_G$  as its unit element (1.1). We shall say that  $U(G)$  is the *infinitesimal algebra of  $G$* .<sup>(16)</sup>

As  $T$  ranges over the schemes over  $S$ , the infinitesimal algebra  $U(G_T)$  of the  $T$ -group  $G \times T$  clearly varies contravariantly in  $T$ , so that we shall be able to speak of the *infinitesimal algebra functor*.

<sup>(16)</sup>**N.D.E.** One now says “the algebra of distributions” (at the origin) of  $G$ , cf. [DG70], § II.4, 6.1 and [Ja03], I 7.7.

As  $T$  ranges over the open subsets of  $S$ , one therefore obtains a presheaf  $T \mapsto U(G_T)$  of  $\mathcal{O}_S$ -algebras; moreover, according to 1.1.3, this is a *sheaf*. We shall denote it by  $\mathcal{U}(G)$  and call it the *sheaf of infinitesimal algebras of  $G$* .

The algebra  $U(G)$  is also a covariant functor in  $G$ . Indeed, if  $u : G \rightarrow H$  is a homomorphism of  $S$ -groups and  $d$  an  $S$ -deviation of  $\varepsilon_G$ , the image of  $d$  under  $u$  is an element  $U(u)(d) = ud$  of  $U(H)$ . The map  $U(u) : U(G) \rightarrow U(H)$  thus defined is clearly a homomorphism of  $\Gamma(\mathcal{O}_S)$ -algebras. One similarly defines a homomorphism  $\mathcal{U}(u)$  from  $\mathcal{U}(G)$  to  $\mathcal{U}(H)$ .

**2.2.** — Let  $d$  be an element of  $U(G)$ , i.e. an  $S$ -deviation of the origin of  $G$ . Consider the  $S$ -deviation  $d \times G$  of  $\varepsilon \times G : G \simeq S \times G \rightarrow G \times G$  obtained from  $d$  by base change (1.2.2); the image of  $d \times G$  under the multiplication morphism  $m : G \times G \rightarrow G$  is an  $S$ -deviation of  $m \circ (\varepsilon \times \text{id}_G) = \text{id}_G$ , i.e. an element of  $\text{Dif}_{G/S}$ , which will be denoted by  $d^G$ .

The map  $d \mapsto d^G$  is clearly  $\Gamma(\mathcal{O}_S)$ -linear, and the “commutative” diagram below shows that  $(e \cdot d)^G = d^G \cdot e^G$ .<sup>(17)</sup>

$$\begin{array}{ccccc}
 & & G \times G \times G & \xrightarrow{m \times G} & G \times G \\
 & \nearrow G \times d \times G & & \searrow G \times m & \\
 & \varepsilon \times G \times G & & & \\
 G \times G & \xrightarrow{e \times G} & G \times G & \xrightarrow{d \times G} & G \times G \\
 \nearrow \varepsilon \times G & \searrow m & \nearrow \varepsilon \times G & \searrow m & \\
 G & \xrightarrow{e^G} & G & \xrightarrow{d^G} & G \\
 & \text{id}_G & & \text{id}_G & \\
 & & & & \downarrow m \\
 & & & & G
 \end{array}$$

The commutativity of the two lower triangles follows indeed from the definition of  $d^G$  and  $e^G$ ; on the other hand, the composite  $S$ -deviation of  $e \times G$  and  $G \times d \times G$  is  $(e \times d) \times G$  (cf. 1.2.2), its image under  $m \times G$  is  $(e \cdot d) \times G$ , and the image of the latter under  $m$  is therefore equal to  $(e \cdot d)^G$ .

One thus obtains an anti-homomorphism  $U(G) \rightarrow \text{Dif}_{G/S}$  of  $\Gamma(\mathcal{O}_S)$ -algebras, called *right translation*.<sup>(18)</sup>

If  $\mathcal{A}_{G/S}$  denotes the sheaf of  $S$ -differential operators on  $G$  (cf. 1.4) and  $p$  the structural morphism  $G \rightarrow S$ , one similarly defines a “right translation”:  $\mathcal{U}(G) \rightarrow p_*(\mathcal{A}_{G/S})$ .

**2.3.** — We shall now characterize the differential operators of  $G$  over  $S$  of the form  $d^G$ . Let  $g : S \rightarrow G$  be a section of the structural morphism of  $G$ , and  $g_G$  the right translation of  $G$  by  $g$ , that is, the composite morphism:

$$g_G : G \simeq G \times S \xrightarrow{G \times g} G \times G \xrightarrow{m} G.$$

For every differential operator  $D$  of  $G$  over  $S$ , the composite  $g_G^{-1} D g_G$  (cf. 1.2) is again an  $S$ -deviation of  $\text{id}_G$ , i.e. an element of  $\text{Dif}_{G/S}$ ; we shall write:

$$D^g = g_G^{-1} D g_G.$$

We shall say that  $D$  is *right-invariant* if, for every base change  $t : T \rightarrow S$  and every section  $g : T \rightarrow G \times T$ , one has  $(D_T)^g = D_T$ .

<sup>(17)</sup>**N.D.E.** The original has been corrected by replacing  $d \times G \times G$  in the diagram by  $G \times d \times G$ , so that the composite on the left side of the triangle is  $(e \times d) \times G$ , and the map  $d \mapsto d^G$  is an *anti-isomorphism* from  $U(G)$  onto the *right-invariant* differential operators (cf. 2.3, 2.4 below); on the other hand, defining  ${}^G d$  as the image under  $m$  of  $G \times d$ , one would similarly obtain an isomorphism from  $U(G)$  onto the *left-invariant* differential operators (cf. [DG70], § II.4, Th. 6.5). Sections 2.4 and 2.5 have been corrected accordingly.

<sup>(18)</sup>**N.D.E.** It would be preferable to call it *left operation*. Indeed, let  $d$  be, for example, an  $S$ -derivation of the origin; according to 1.2.1,  $d$  is the composite of the  $S$ -derivation  $(\tau, \partial_t) : S \rightarrow I_S$  and a morphism  $x : I_S \rightarrow G$  such that  $x \circ \tau = \varepsilon$  (i.e.  $x \in \text{Lie}(G/S)(S)$ ), and then  $d^G$  is the derivation of  $\mathcal{O}_G$  sending a local section  $\phi$  to the section  $g \mapsto \partial_t \phi(xg)$ . Moreover, with this terminology one could say that: “left operation commutes with right translations”.

**Lemma.** *For every differential operator  $D$  of  $G$  over  $S$ , the following assertions are equivalent (where  $m$  is the multiplication morphism of  $G$ ):*

(i)  *$D$  is right-invariant.*

(ii) *The following two deviations of  $m$  are equal:  $Dm = m(D \times G)$ .*

(ii)  $\Rightarrow$  (i): since condition (ii) is stable under base change, it suffices to show that (ii) implies the equality  $D^g = D$  for every section  $g : S \rightarrow G$ . Let  $h$  be the morphism  $G \times g : G \times S \rightarrow G \times G$ , so that  $m \circ h$  is the right translation  $g_G$ . The equality  $D^g = D$  is equivalent to the equality  $g_G \circ D = D \circ g_G$ , and the latter follows from the commutative diagram:

$$\begin{array}{ccccc} G & \xleftarrow{m} & G \times G & \xleftarrow{h} & G \\ \text{id}_G \downarrow D & & \text{id}_{(G \times G)} \downarrow D \times G & & \text{id}_G \downarrow D \\ G & \xleftarrow{m} & G \times G & \xleftarrow{h} & G \end{array}$$

(i)  $\Rightarrow$  (ii): take, indeed, for  $t : T \rightarrow S$  the structural morphism  $p : G \rightarrow S$ , and for the section  $g : T \rightarrow G \times T$  the diagonal morphism  $\Delta : G \rightarrow G \times G$ . The right translation

$$\Delta_{G \times G} : G \times G \rightarrow G \times G$$

is then the morphism from  $G \times G$  to  $G \times G$  with components  $m$  and  $\text{pr}_2$ . The equality  $(D_G)^\Delta = D_G$  is then equivalent to the commutativity of the first square of the following diagram:

$$\begin{array}{ccccc} G \times G & \xrightarrow{\Delta_{G \times G}} & G \times G & \xrightarrow{\text{pr}_1} & G \\ \text{id}_{G \times G} \downarrow D_G & & \text{id}_{G \times G} \downarrow D_G & & \text{id}_G \downarrow D \\ G \times G & \xrightarrow{\Delta_{G \times G}} & G \times G & \xrightarrow{\text{pr}_1} & G \end{array}$$

Equality (ii) therefore follows from the fact that  $m = \text{pr}_1 \circ \Delta_{G \times G}$ .

Consider, for example, an element  $d$  of the infinitesimal algebra  $U(G)$ . The squares of the diagram

$$\begin{array}{ccccccc} G \times G & \xlongequal{\quad} & S \times G \times G & \xrightarrow[\varepsilon \times G \times G]{d \times G \times G} & G \times G \times G & \xrightarrow{m \times G} & G \times G \\ m \downarrow & & S \times m \downarrow & & G \times m \downarrow & & m \downarrow \\ G & \xlongequal{\quad} & S \times G & \xrightarrow[\varepsilon \times G]{d \times G} & G \times G & \xrightarrow{m} & G \end{array}$$

are then commutative. Since one has

$$m \circ (d \times G) = d^G \quad \text{and} \quad (m \times G) \circ (d \times G \times G) = d^G \times G,$$

one also has  $d^G \circ m = m \circ (d^G \times G)$ . Thus: *for every  $S$ -deviation  $d$  of the origin,  $d^G$  is a right-invariant differential operator.*

#### Theorem 2.4.

- (i) *The map  $d \mapsto d^G$  is an anti-isomorphism<sup>(19)</sup> from the infinitesimal algebra  $U(G)$  onto the subalgebra  $\text{Dif}_{G/S}^G$  of  $\text{Dif}_{G/S}$  consisting of the right-invariant differential operators.*
- (ii) *Similarly, the map  $d \mapsto {}^G d$  is an isomorphism from  $U(G)$  onto the subalgebra of  $\text{Dif}_{G/S}$  consisting of the left-invariant differential operators.*

<sup>(19)</sup>**N.D.E.** “Isomorphism” has been corrected to “anti-isomorphism”, and assertion (ii) has been added, cf. N.D.E. (17).

Let, indeed,  $D$  be an arbitrary differential operator of  $G$  over  $S$ , and denote by  $D_0$  its *value at the origin*, that is, the composite deviation  $S \xrightarrow{\varepsilon} G \xrightarrow[\text{id}_G]{D} G$ . The right-invariant differential operator  $(D_0)^G$  is then obtained by composition:

$$G \simeq S \times G \xrightarrow{\varepsilon \times G} G \times G \xrightarrow[\text{id}_{G \times G}]{D \times G} G \times G \xrightarrow{m} G.$$

If  $D$  is right-invariant, one has  $Dm = m(D \times G)$ , whence

$$D = Dm(\varepsilon \times G) = m(D \times G)(\varepsilon \times G) = (D_0)^G.$$

In particular, the map  $d \mapsto d^G$  is surjective.

Conversely, let  $d$  be an  $S$ -deviation of the origin. There is then a commutative square

$$\begin{array}{ccc} G \times G & \xleftarrow{d \times G} & G \\ G \times \varepsilon \uparrow & & \uparrow \varepsilon \\ G \times S \simeq G & \xleftarrow{d} & S \end{array}$$

from which it follows that  $d = m(G \times \varepsilon)d = m(d \times G)\varepsilon = (d^G)_0$ . A fortiori, the map  $d \mapsto d^G$  is injective. This proves the theorem.

As  $S$  varies, theorem 2.4 clearly implies that the right translation  $\mathcal{U}(G) \rightarrow p_*(\mathcal{A}f_{G/S})$  is an anti-isomorphism of  $\mathcal{O}_S$ -algebras from  $\mathcal{U}(G)$  onto the sheaf of  $\mathcal{O}_S$ -algebras  $p_*(\mathcal{A}f_{G/S})^G$ , which associates  $\text{Dif}_{G/U}^{GU}$  to every open subset  $U$  of  $S$ .

**Remark 2.4.1.** Consider the commutative diagram

$$\begin{array}{ccc} G & \xleftarrow{\eta} & G \times G \\ \uparrow p\varepsilon & & \uparrow \text{pr}_1 \Delta \\ S & \xleftarrow{p} & G, \end{array}$$

where  $\eta$  denotes the morphism “ $(x, y) \mapsto yx^{-1}$ ”.<sup>(20)</sup> The latter induces morphisms

$$\eta' : \eta^{-1}(\mathcal{O}_G) \rightarrow \mathcal{O}_{G \times G} \quad \text{and} \\ \Delta^{-1}(\eta') : p^{-1}\varepsilon^{-1}(\mathcal{O}_G) \rightarrow \Delta^{-1}(\mathcal{O}_{G \times G}).$$

For every integer  $n \geq 1$ , put  $\mathfrak{p}_{G/S}^n = \varepsilon^{-1}(\mathcal{O}_G)/\mathcal{I}_\varepsilon^{n+1}$  (confer 1.3 and 1.4 for the notation).<sup>(21)</sup> Since the square formed by the morphisms  $\varepsilon, \eta, \Delta$ , and  $p$  is cartesian,  $\Delta^{-1}(\eta')$  induces an isomorphism of  $\mathcal{O}_G$ -modules:

$$p^*(\mathfrak{p}_{G/S}^n) \xrightarrow{\sim} \mathcal{P}_{G/S}^n.$$

The differential operators of  $G$  over  $S$  of order  $\leq n$  therefore correspond bijectively to the morphisms of  $\mathcal{O}_G$ -modules  $p^*(\mathfrak{p}_{G/S}^n) \rightarrow \mathcal{O}_G$ , that is, to the morphisms of  $\mathcal{O}_S$ -modules

$$\mathfrak{p}_{G/S}^n \rightarrow p_*(\mathcal{O}_G).$$

In this bijection, the right-invariant differential operators are associated to the composite arrows

$$\mathfrak{p}_{G/S}^n \rightarrow \mathcal{O}_S \xrightarrow{\text{can.}} p_*(\mathcal{O}_G).$$

One thus recovers the isomorphism of theorem 2.4.

<sup>(20)</sup>**N.D.E.** i.e.  $G$  acts on the left on itself by right translations.

<sup>(21)</sup>**N.D.E.** In what follows, the original has been corrected; it referred to the square formed by the morphisms  $p, p, \eta$ , and  $\text{pr}_1$ , instead of  $\varepsilon, \eta, \Delta$ , and  $p$ .

**2.5.** — <sup>(22)</sup> Let  $\text{Lie}(G)$  be the Lie algebra of  $G$ ; <sup>(23)</sup> we shall define a morphism of  $\Gamma(\mathcal{O}_S)$ -Lie algebras  $\alpha : \text{Lie}(G) \rightarrow U(G)$ .

Let  $s : S \rightarrow I_S$  be the zero section of  $I_S \rightarrow S$ , and  $\sigma$  the deviation of  $s$  defined in 1.2.0. Recall (cf. II, 4.1) that  $\text{Lie}(G)$  is the set of morphisms  $x : I_S \rightarrow G$  such that  $x \circ s = \varepsilon_G$ . Then the composite

$$S \xrightarrow[\sigma]{s} I_S \xrightarrow{x} G$$

is an  $S$ -deviation of  $\varepsilon_G$ , i.e. an element of  $U(G)$ ; with the notation of 1.2 (†), it is denoted by  $\sigma x$ . Moreover, according to 1.2.1, the map  $\alpha : x \mapsto \sigma x$  is an isomorphism of  $\mathcal{O}_S(S)$ -modules from  $\text{Lie}(G)$  onto the submodule  $\text{Dér}(\varepsilon_G)$  of  $U(G)$  consisting of the  $S$ -derivations of  $\varepsilon_G$ . We shall see that  $\alpha$  is a morphism of Lie algebras. <sup>(24)</sup> Let

$$\rho' : U(G) \rightarrow \text{Dif}_{G/S}$$

be the algebra morphism which associates to an  $S$ -deviation  $d$  of  $\varepsilon_G$  the left-invariant differential operator  ${}^G d \in \text{Dif}_{G/S}$ , cf. 2.2, N.D.E. (17).

Let  $\rho : G \rightarrow \underline{\text{Aut}}_S(G)$  be the homomorphism of group functors which associates to an  $S$ -morphism  $g : T \rightarrow G$  the right translation of  $G_T$  by  $g$ , i.e. the morphism:

$$G_T \simeq T \times_T G_T \xrightarrow{G_T \times g} G_T \times_T G_T \xrightarrow{m_T} G_T.$$

Recall also (cf. 1.5 and II, 3.14) that  $\text{Lie}(\underline{\text{Aut}} G) = \underline{\text{Lie}}(\underline{\text{Aut}}_S(G)/S)(S)$  identifies with the infinitesimal automorphisms of  $G$ , i.e. with the automorphisms of  $I_S \times G$  inducing the identity on  $G$ . Since  $\rho$  is a monomorphism, so is the morphism  $\underline{\text{Lie}}(\rho) : \underline{\text{Lie}}(G/S) \rightarrow \underline{\text{Lie}}(\underline{\text{Aut}}_S(G)/S)$  (see, for example, Exp. II, N.D.E. (50)); hence  $\text{Lie}(\rho) : \text{Lie}(G) \rightarrow \text{Lie}(\underline{\text{Aut}} G)$  is injective.

On the other hand, according to 1.5, the map  $\beta$  which associates to every infinitesimal automorphism  $u$  of  $G$  the differential operator  $D_u$  of  $G$ :

$$G \simeq S \times G \xrightarrow{\sigma \times G} I_S \times G \xrightarrow{u} I_S \times G \xrightarrow{\text{pr}_2} G$$

is an isomorphism from  $\text{Lie}(\underline{\text{Aut}} G)$  onto the Lie subalgebra of  $\text{Dif}_{G/S}$  consisting of the  $p^{-1}(\mathcal{O}_S)$ -derivations of  $\mathcal{O}_G$ .

For every  $x \in \text{Lie}(G)$ , there is the following commutative square determining the image of  $x$  under  $\text{Lie}(\rho)$ :

$$\begin{array}{ccc} I_S \times G & \xrightarrow{\text{Lie}(\rho)(x)} & I_S \times G \\ x \times G \downarrow & & \downarrow \text{pr}_2 \\ G \times G & \xrightarrow{m} & G. \end{array}$$

Taking this diagram into account, the image of  $\text{Lie}(\rho)(x)$  under  $\beta$  is the composite deviation

$$G \simeq S \times G \xrightarrow[\sigma \times G]{\sigma \times G} I_S \times G \xrightarrow{G \times x} G \times G \xrightarrow{m} G$$

<sup>(22)</sup>**N.D.E.** In this paragraph the order has been modified, beginning by defining the map  $\alpha : \text{Lie}(G) \rightarrow U(G)$ , and the original has been corrected as indicated in N.D.E. (17).

<sup>(23)</sup>**N.D.E.** In this exposé, if  $G$  (resp.  $X$ ) is an  $S$ -group scheme (resp. an  $S$ -scheme), the “Lie algebra”  $\text{Lie}(G)$  (resp.  $\text{Lie}(\underline{\text{Aut}} X)$ ) denotes, with the notation of Exposé II,  $\underline{\text{Lie}}(G/S)(S)$  (resp.  $\underline{\text{Lie}}(\underline{\text{Aut}}_S(X)/S)(S)$ ); this is a  $\Gamma(\mathcal{O}_S)$ -Lie algebra, according to II, 4.11 and 3.14.

<sup>(24)</sup>**N.D.E.** See also II, 4.11.

which, according to 2.2 N.D.E. (17), is none other than  ${}^G(\sigma x) = \rho'(\alpha(x))$ . One therefore obtains a commutative diagram:

$$\begin{array}{ccc} \mathrm{Lie}(G) & \xrightarrow{\mathrm{Lie}(\rho)} & \mathrm{Lie}(\mathrm{Aut} G) \\ \alpha \downarrow & & \downarrow \beta \\ U(G) & \xrightarrow{\rho'} & \mathrm{Dif}_{G/S} \end{array}$$

where  $\mathrm{Lie}(\rho)$ ,  $\beta$ , and  $\rho'$  are morphisms of Lie algebras. Since  $\rho'$  is injective, it follows that  $\alpha$  too is a morphism of Lie algebras. Consequently, one has obtained:

**Proposition.**  $\alpha$  is an isomorphism of  $\mathcal{O}_S(S)$ -Lie algebras from  $\mathrm{Lie}(G)$  to the Lie algebra of  $S$ -derivations of  $\varepsilon_G$ , itself isomorphic via  $\mathrm{Lie}(\rho)$  to the Lie algebra of left-invariant  $S$ -derivations of  $G$ .<sup>(25)</sup>

### 3. Coalgebras and Cartier duality

**3.1.** — Let  $S$  be a scheme (or, more generally, a ringed space). An  $\mathcal{O}_S$ -coalgebra<sup>(26)</sup> is a pair  $(\mathcal{U}, \Delta_{\mathcal{U}})$  consisting of an  $\mathcal{O}_S$ -module  $\mathcal{U}$  and a morphism of  $\mathcal{O}_S$ -modules  $\Delta_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U}$  (called the *diagonal morphism* or *comultiplication*) such that:

- (i)  $\sigma \circ \Delta_{\mathcal{U}} = \Delta_{\mathcal{U}}$ , where  $\sigma(a \otimes b) = b \otimes a$ .
- (ii) The square

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \mathrm{id}_{\mathcal{U}} \otimes \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} & \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \end{array}$$

is commutative.

- (iii) There exists a morphism of  $\mathcal{O}_S$ -modules  $\varepsilon_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{O}_S$ , called the *augmentation*, such that the composite morphisms

$$\begin{aligned} \mathcal{U} &\xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{O}_S \simeq \mathcal{U} \\ \mathcal{U} &\xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U} \xrightarrow{\varepsilon_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{U} \simeq \mathcal{U} \end{aligned}$$

are the identity morphism of  $\mathcal{U}$ .

If  $\varepsilon_{\mathcal{U}}$  and  $\varepsilon'_{\mathcal{U}}$  are two augmentations, one has  $\varepsilon_{\mathcal{U}} = (\varepsilon_{\mathcal{U}} \otimes \varepsilon'_{\mathcal{U}}) \circ \Delta_{\mathcal{U}} = \varepsilon'_{\mathcal{U}}$ ; the augmentation is therefore uniquely determined by (iii).

If  $(\mathcal{U}, \Delta_{\mathcal{U}})$  and  $(\mathcal{V}, \Delta_{\mathcal{V}})$  are two  $\mathcal{O}_S$ -coalgebras, a *morphism* from the first to the second is a morphism of  $\mathcal{O}_S$ -modules  $f : \mathcal{U} \rightarrow \mathcal{V}$  such that the diagrams

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \Delta_{\mathcal{U}} \downarrow & & \downarrow \Delta_{\mathcal{V}} \\ \mathcal{U} \otimes \mathcal{U} & \xrightarrow{f \otimes f} & \mathcal{V} \otimes \mathcal{V} \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{U} & \xrightarrow{f} & \mathcal{V} \\ \varepsilon_{\mathcal{U}} \searrow & & \swarrow \varepsilon_{\mathcal{V}} \\ & \mathcal{O}_S & \end{array}$$

<sup>(25)</sup>**N.D.E.** There are examples of Lie algebras  $\mathfrak{g}$  over a ring  $A$  such that the map  $\mathfrak{g} \rightarrow U(\mathfrak{g})$  is not injective, cf. [BLie], § I.2, Ex. 9. The result above shows (since  $\alpha$  factors as  $\mathrm{Lie}(G) \rightarrow U(\mathrm{Lie}(G)) \rightarrow U(G)$ ) that this cannot occur for “algebraic” Lie algebras, i.e. those of the form  $\mathrm{Lie}(G)$ , where  $G$  is an  $A$ -group scheme.

<sup>(26)</sup>**N.D.E.** One also says “cogebra”, cf. [BALg], III § 11.1. On the other hand, note that in this exposé (as well as in VII B) one works in the category of *cocommutative* coalgebras (i.e. satisfying condition (i)), which is crucial for defining the product and the notion of coalgebra in groups (cf. 3.1.0 and 3.2).

are commutative. Coalgebra morphisms compose like morphisms of  $\mathcal{O}_S$ -modules, so that we shall be able to speak of the *category of  $\mathcal{O}_S$ -coalgebras*.

**3.1.0.** — <sup>(27)</sup> This category has finite products: the final object is the  $\mathcal{O}_S$ -module  $\mathcal{O}_S$ , the comultiplication being the identity; the product of two coalgebras  $(\mathcal{U}, \Delta_{\mathcal{U}})$  and  $(\mathcal{V}, \Delta_{\mathcal{V}})$  is the tensor product  $\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{V}$ , the comultiplication being the composite morphism

$$\begin{aligned} \mathcal{U} \otimes \mathcal{V} &\xrightarrow{\Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{V} \\ &\xrightarrow{\text{id}_{\mathcal{U}} \otimes \sigma \otimes \text{id}_{\mathcal{V}}} \mathcal{U} \otimes \mathcal{V} \otimes \mathcal{U} \otimes \mathcal{V} \end{aligned}$$

where  $\sigma(a \otimes b) = b \otimes a$ ; the canonical projections from  $\mathcal{U} \otimes \mathcal{V}$  to the factors  $\mathcal{U}$  and  $\mathcal{V}$  are the morphisms  $\text{id}_{\mathcal{U}} \otimes \varepsilon_{\mathcal{V}}$  and  $\varepsilon_{\mathcal{U}} \otimes \text{id}_{\mathcal{V}}$ ,<sup>(28)</sup> and the “diagonal morphism”  $\mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$  (corresponding to the pair of morphisms  $(\text{id}_{\mathcal{U}}, \text{id}_{\mathcal{U}})$ ) is none other than the comultiplication  $\Delta_{\mathcal{U}}$ .

**3.1.1.** — Let  $\mathcal{A}$  be a commutative  $\mathcal{O}_S$ -algebra, *locally free and of finite type* as an  $\mathcal{O}_S$ -module. If we put

$$\mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}, \mathcal{O}_S),$$

the canonical morphism  $\varphi$  from  $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}^*$  to  $(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A})^*$  is invertible. If  $m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$  is the morphism defining the multiplication of  $\mathcal{A}$ , one obtains by composition a diagonal morphism

$$\Delta_{\mathcal{A}^*} : \mathcal{A}^* \xrightarrow{m^*} (\mathcal{A} \otimes \mathcal{A})^* \xrightarrow{\varphi^{-1}} \mathcal{A}^* \otimes \mathcal{A}^*.$$

This diagonal morphism clearly makes  $\mathcal{A}^*$  a  $\mathcal{O}_S$ -coalgebra having as augmentation the transpose of the morphism  $\mathcal{O}_S \rightarrow \mathcal{A}$  defined by the unit section of  $\mathcal{A}$ . Moreover, it is clear that:

*The functor  $\mathcal{A} \mapsto \mathcal{A}^*$  is an anti-equivalence from the category of  $\mathcal{O}_S$ -algebras which are locally free and of finite type as  $\mathcal{O}_S$ -modules onto the category of  $\mathcal{O}_S$ -coalgebras locally free and of finite type as  $\mathcal{O}_S$ -modules.*

**3.1.2.** — To every  $\mathcal{O}_S$ -coalgebra  $\mathcal{U}$  there is canonically associated an  $S$ -functor

$$\text{Spec}^* \mathcal{U} : (\mathbf{Sch}/S)^{\circ} \rightarrow (\mathbf{Ens}).$$

Observe, indeed, that for every  $S$ -scheme  $q : T \rightarrow S$ ,  $q^*(\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{U})$  identifies with  $q^*(\mathcal{U}) \otimes_{\mathcal{O}_T} q^*(\mathcal{U})$ , so that  $q^*(\Delta_{\mathcal{U}})$  makes  $\mathcal{U}_T = q^*(\mathcal{U})$  an  $\mathcal{O}_T$ -coalgebra; we may therefore put, by definition and with an obvious abuse of notation:<sup>(29)</sup>

$$(\text{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1 \text{ and } \Delta_{\mathcal{U}_T}(x) = x \otimes x\}.$$

The sections  $x$  of  $\mathcal{U}_T$  clearly correspond to the morphisms of  $\mathcal{O}_T$ -modules  $\xi : \mathcal{O}_T \rightarrow \mathcal{U}_T$ ; the conditions  $\varepsilon(x) = 1$  and  $\Delta(x) = x \otimes x$  express simply that  $\xi$  is a coalgebra morphism. One therefore also has:

$$(\text{Spec}^* \mathcal{U})(T) = \text{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{U}_T).$$

In particular, one has the following proposition:<sup>(30)</sup>

<sup>(27)</sup>**N.D.E.** The numbering 3.1.0 has been added for later references.

<sup>(28)</sup>**N.D.E.** What follows has been added. Recall also that, to show that  $\mathcal{U} \otimes \mathcal{V}$  is indeed the product of  $\mathcal{U}$  and  $\mathcal{V}$  in the category of cocommutative  $\mathcal{O}_S$ -cogbras, one verifies that if one has an arbitrary  $\mathcal{O}_S$ -cogebra  $\mathcal{E}$  and cogebra morphisms  $f : \mathcal{E} \rightarrow \mathcal{U}$  and  $g : \mathcal{E} \rightarrow \mathcal{V}$ , then every cogebra morphism  $\phi : \mathcal{E} \rightarrow \mathcal{U} \otimes \mathcal{V}$  such that  $\text{pr}_{\mathcal{U}} \circ \phi = f$  and  $\text{pr}_{\mathcal{V}} \circ \phi = g$  is necessarily equal to  $(f \otimes g) \circ \Delta_{\mathcal{E}}$ , and that the latter is a cogebra morphism if and only if it equals  $(g \otimes f) \circ \Delta_{\mathcal{E}}$ .

<sup>(29)</sup>**N.D.E.** For every  $x \otimes y \in \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \Gamma(T, \mathcal{U}_T)$ , its image in  $\Gamma(T, \mathcal{U}_T \otimes_{\mathcal{O}_T} \mathcal{U}_T)$  is still denoted by  $x \otimes y$ .

<sup>(30)</sup>**N.D.E.** The numbering 3.1.2.1 has been added for later references. Observe, on the other hand, that the  $S$ -functor  $\text{Spec}^* \mathcal{U}$  is a sheaf for the Zariski topology (and even for the (fpqc) topology if  $\mathcal{U}$  is a quasi-coherent  $\mathcal{O}_S$ -module).

**Proposition 3.1.2.1.** *Let  $\mathcal{A}$  be a commutative  $\mathcal{O}_S$ -algebra which is locally free of finite type as an  $\mathcal{O}_S$ -module. Then the  $S$ -functor  $\mathrm{Spec}^* \mathcal{A}^*$  is represented by  $\mathrm{Spec} \mathcal{A}$ .*

Indeed, for every  $S$ -scheme  $T$ , there are canonical isomorphisms:

$$\begin{aligned} (\mathrm{Spec}^* \mathcal{A}^*)(T) &= \mathrm{Hom}_{\mathcal{O}_T\text{-coalg.}}(\mathcal{O}_T, \mathcal{A}_T^*) \\ &\simeq \mathrm{Hom}_{\mathcal{O}_T\text{-alg.}}(\mathcal{A}_T, \mathcal{O}_T) \simeq (\mathrm{Spec} \mathcal{A})(T). \end{aligned}$$

**3.2.** — An  $\mathcal{O}_S$ -coalgebra in groups, that is, a group in the category of  $\mathcal{O}_S$ -coalgebras, consists in giving a  $\mathcal{O}_S$ -coalgebra  $(\mathcal{U}, \Delta_{\mathcal{U}})$  and three morphisms of  $\mathcal{O}_S$ -coalgebras  $m_{\mathcal{U}} : \mathcal{U} \otimes \mathcal{U} \rightarrow \mathcal{U}$ ,  $\eta_{\mathcal{U}} : \mathcal{O}_S \rightarrow \mathcal{U}$ , and  $c_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U}$  satisfying conditions (ii)\*, (iii)\*, and (vi) below; on the other hand, the fact that  $m_{\mathcal{U}}$  is a cogeбра morphism is expressed by the commutativity of diagrams (iv) and (v) below:

(iv)

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ \Delta_{\mathcal{U}} \otimes \Delta_{\mathcal{U}} \downarrow & & \downarrow \Delta_{\mathcal{U}} \\ \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & & \\ \mathrm{id}_{\mathcal{U}} \otimes \sigma \otimes \mathrm{id}_{\mathcal{U}} \downarrow & & \\ \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}} \otimes m_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U} \end{array}$$

(v)

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \\ \searrow \varepsilon_{\mathcal{U}} \otimes \varepsilon_{\mathcal{U}} & & \swarrow \varepsilon_{\mathcal{U}} \\ & \mathcal{O}_S & \end{array}$$

(ii)\* The square

$$\begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} \otimes \mathcal{U} & \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes m_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U} \\ m_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}} \downarrow & & \downarrow m_{\mathcal{U}} \\ \mathcal{U} \otimes \mathcal{U} & \xrightarrow{m_{\mathcal{U}}} & \mathcal{U} \end{array}$$

is commutative.

(iii)\* The two composites below equal the identity morphism of  $\mathcal{U}$ :

$$\begin{aligned} \mathcal{U} &\simeq \mathcal{U} \otimes \mathcal{O}_S \xrightarrow{\mathrm{id}_{\mathcal{U}} \otimes \eta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U} \\ \mathcal{U} &\simeq \mathcal{O}_S \otimes \mathcal{U} \xrightarrow{\eta_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}. \end{aligned}$$

(vi) The composite morphism below is equal to  $\eta_{\mathcal{U}} \circ \varepsilon_{\mathcal{U}}$ :

$$\mathcal{U} \xrightarrow{\Delta_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{c_{\mathcal{U}} \otimes \mathrm{id}_{\mathcal{U}}} \mathcal{U} \otimes \mathcal{U} \xrightarrow{m_{\mathcal{U}}} \mathcal{U}.$$

**3.2.1.** — The morphisms  $\eta_{\mathcal{U}}$  and  $c_{\mathcal{U}}$  are uniquely determined by  $m_{\mathcal{U}}$ . On the other hand, conditions (ii)\* and (iii)\* express simply that  $m_{\mathcal{U}}$  makes  $\mathcal{U}$  an  $\mathcal{O}_S$ -algebra having as unit section the image under  $\eta_{\mathcal{U}}$  of the unit section of  $\mathcal{O}_S$ . Condition (iv) also expresses that the diagonal morphism  $\Delta_{\mathcal{U}}$  is compatible with multiplication; and indeed,  $\Delta_{\mathcal{U}} : \mathcal{U} \rightarrow \mathcal{U} \otimes \mathcal{U}$  must be a homomorphism of coalgebras in groups, which also implies the commutativity of the triangle

(v\*)

$$\begin{array}{ccc} & \mathcal{O}_S & \\ \eta_{\mathcal{U}} \swarrow & & \searrow \eta_{\mathcal{U}} \otimes \eta_{\mathcal{U}} \\ \mathcal{U} & \xrightarrow{\Delta_{\mathcal{U}}} & \mathcal{U} \otimes \mathcal{U}. \end{array}$$

On the other hand, as in every category, the antipode  $c_{\mathcal{U}}$  is an isomorphism from  $\mathcal{U}$  onto the “opposite” group object;<sup>(31)</sup> in particular,  $c_{\mathcal{U}}$  induces an algebra isomorphism from  $\mathcal{U}$  onto the opposite algebra  $\mathcal{U}^\circ$ .

**3.2.2.** — Since the functor  $\mathcal{U} \mapsto \mathrm{Spec}^* \mathcal{U}$  commutes with finite products, it transforms a coalgebra in groups into an  $S$ -group functor; and indeed, for every  $S$ -scheme  $T$ , the elements  $x \in \Gamma(T, \mathcal{U}_T)$  belonging to  $(\mathrm{Spec}^* \mathcal{U})(T)$  form a group for the multiplication of the algebra  $\Gamma(T, \mathcal{U}_T)$ ; the inverse of  $x$  is none other than  $c_{\mathcal{U}}(x)$ . According to 3.1.2.1, one has:

**Scholium 3.2.2.1.** <sup>(32)</sup> *Let  $\mathcal{U}$  be a  $\mathcal{O}_S$ -coalgebra in groups, finite and locally free as an  $\mathcal{O}_S$ -module. Then the  $S$ -group functor  $\mathrm{Spec}^* \mathcal{U}$  is represented by the finite and locally free  $S$ -group  $\mathrm{Spec} \mathcal{U}^*$ .*

**Remark 3.2.2.2.** Let  $\mathcal{L}$  be a  $\mathcal{O}_S$ -Lie algebra and  $\mathcal{U}(\mathcal{L})$  the *enveloping algebra* of  $\mathcal{L}$ , that is, the sheaf on  $S$  associated to the presheaf assigning to every open subset  $V$  the enveloping algebra  $U(\Gamma(V, \mathcal{L}))$  of the Lie algebra  $\Gamma(V, \mathcal{L})$ .

Every homomorphism from  $\mathcal{L}$  to the Lie algebra underlying an associative  $\mathcal{O}_S$ -algebra factors in one and only one way through the canonical morphism from  $\mathcal{L}$  to  $\mathcal{U}(\mathcal{L})$ ; furthermore, this universal property implies, besides the functoriality of  $\mathcal{U}(\mathcal{L})$  in  $\mathcal{L}$ , that the enveloping algebra of a product of Lie algebras identifies with the tensor product of the enveloping algebras.

In particular, the diagonal morphism  $\delta : \mathcal{L} \rightarrow \mathcal{L} \times \mathcal{L}$  induces an algebra homomorphism  $\Delta : \mathcal{U}(\mathcal{L}) \rightarrow \mathcal{U}(\mathcal{L} \times \mathcal{L}) \simeq \mathcal{U}(\mathcal{L}) \otimes \mathcal{U}(\mathcal{L})$ . The zero morphism  $\mathcal{L} \rightarrow 0$  induces an algebra homomorphism  $\varepsilon : \mathcal{U}(\mathcal{L}) \rightarrow \mathcal{U}(0) \simeq \mathcal{O}_S$ . The isomorphism  $x \mapsto -x$  from  $\mathcal{L}$  onto the opposite Lie algebra  $\mathcal{L}^\circ$  induces an anti-isomorphism  $c$  of the algebra  $\mathcal{U}(\mathcal{L})$ . One then verifies easily that the multiplication  $m$  of the algebra  $\mathcal{U}(\mathcal{L})$  makes  $(\mathcal{U}(\mathcal{L}), \Delta)$  a  $\mathcal{O}_S$ -coalgebra in groups having  $\varepsilon$  as augmentation and  $c$  as antipode.<sup>(33)</sup>

**3.2.3.** — <sup>(34)</sup> Let  $\mathcal{U}$  be a  $\mathcal{O}_S$ -coalgebra in groups. We shall see that the  $S$ -group functor  $G = \mathrm{Spec}^* \mathcal{U}$  is *very good*, in the sense of II, 4.6 and 4.10.

Let  $\mathcal{M}$  be a free  $\mathcal{O}_S$ -module of rank  $r$ , and let  $T \rightarrow S$  be an  $S$ -scheme. Since  $I_T(\mathcal{M}) = \mathrm{Spec}(\mathcal{O}_T \oplus \mathcal{M}_T)$ , so that  $\pi : I_T(\mathcal{M}) \rightarrow T$  is affine, one has

$$\pi_*(\mathcal{U}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} \pi_*(\mathcal{O}_{I_T(\mathcal{M})}) = \mathcal{U}_T \otimes_{\mathcal{O}_T} (\mathcal{O}_T \oplus \mathcal{M}_T),$$

and therefore

$$(1) \quad \Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} (\mathcal{O}(T) \oplus \Gamma(T, \mathcal{M}_T)).$$

Let  $(d_1, \dots, d_r)$  be a basis of  $\mathcal{M}$ . Then an element  $u_0 + \sum_i u_i d_i$  of  $\Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})})$  belongs to  $G(I_T(\mathcal{M}))$  if and only if one has:

$$1 = \varepsilon\left(u_0 + \sum_i u_i d_i\right) = \varepsilon(u_0) + \sum_i \varepsilon(u_i) d_i,$$

and

$$\begin{aligned} \left(u_0 + \sum_i u_i d_i\right) \otimes \left(u_0 + \sum_i u_i d_i\right) &= \Delta\left(u_0 + \sum_i u_i d_i\right) \\ &= \Delta(u_0) + \sum_i \Delta(u_i) d_i, \end{aligned}$$

<sup>(31)</sup>**N.D.E.** i.e. equipped with the multiplication  $m'_{\mathcal{U}} = m_{\mathcal{U}} \circ \sigma$ .

<sup>(32)</sup>**N.D.E.** This scholium, implicit in the original, has been added.

<sup>(33)</sup>**N.D.E.** The  $S$ -group functor  $\mathrm{Spec}^* \mathcal{U}(\mathcal{L})$  is not representable in general, but we shall see below (5.5) that if  $S$  is a scheme of characteristic  $p$ , if  $\mathcal{L}$  is finite locally free over  $\mathcal{O}_S$ , and if  $\mathcal{U}_p(\mathcal{L})$  is its *restricted enveloping algebra* (cf. 5.3), then  $\mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$  is represented by a finite and locally free  $S$ -group.

<sup>(34)</sup>**N.D.E.** This paragraph has been expanded.

that is:

$$(2) \quad \begin{cases} \varepsilon(u_0) = 1, & \Delta u_0 = u_0 \otimes u_0, & \text{i.e. } u_0 \in G(T), \\ \varepsilon(u_i) = 0, & \Delta(u_i) = u_i \otimes u_0 + u_0 \otimes u_i, & i = 1, \dots, r. \end{cases}$$

Moreover, the morphism  $G(I_T(\mathcal{M})) \rightarrow G(T)$  corresponding to the zero section of  $I_T(\mathcal{M}) \rightarrow T$  is given by:  $u_0 + \sum_i u_i d_i \mapsto u_0$ . From this, combined with (1) and (2), one deduces that if  $\mathcal{N}$  is a second free  $\mathcal{O}_S$ -module of finite rank, the diagram of sets

$$\begin{array}{ccc} G(I_T(\mathcal{M} \oplus \mathcal{N})) & \longrightarrow & G(I_T(\mathcal{N})) \\ \downarrow & & \downarrow \\ G(I_T(\mathcal{M})) & \longrightarrow & G(T) \end{array}$$

is cartesian, i.e.  $G$  satisfies condition (E) of II, 3.5.

Denote by  $\text{Prim } \Gamma(T, \mathcal{U}_T)$  the sub- $\mathcal{O}(T)$ -module of  $\Gamma(T, \mathcal{U}_T)$  consisting of *primitive elements*, i.e. elements  $u$  satisfying (with the abuse of notation pointed out in 3.1.2):

$$\Delta u = u \otimes 1 + 1 \otimes u, \quad \varepsilon(u) = 0.$$

<sup>(35)</sup> Since  $(\underline{\text{Lie}}G)(T)$  is the set of elements  $u_0 + ud \in G(I_T)$  over the unit element  $u_0 = 1$  of  $G(T)$ , one obtains an isomorphism of  $\mathcal{O}(T)$ -modules, functorial in  $T$ :<sup>(36)</sup>

$$(\underline{\text{Lie}}G)(T) \simeq \text{Prim } \Gamma(T, \mathcal{U}_T).$$

On the other hand, one deduces from (1) that

$$\text{Prim } \Gamma(I_T(\mathcal{M}), \mathcal{U}_{I_T(\mathcal{M})}) \simeq \text{Prim } \Gamma(T, \mathcal{U}_T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})),$$

and it follows that the natural morphism of  $\mathcal{O}(I_T(\mathcal{M}))$ -modules:

$$(\underline{\text{Lie}}G)(T) \otimes_{\mathcal{O}(T)} \mathcal{O}(I_T(\mathcal{M})) \rightarrow (\underline{\text{Lie}}G)(I_T(\mathcal{M}))$$

is an isomorphism, i.e.  $\underline{\text{Lie}}G$  is a good  $\mathbf{O}_S$ -module (cf. II, Def. 4.4).

Thus  $G$  is a good  $S$ -group functor (cf. II, Def. 4.6), and according to II, 4.7.2,  $\underline{\text{Lie}}G$  is equipped with a  $\mathbf{O}_S$ -bilinear “Lie bracket” satisfying the Jacobi identity. It remains to show that  $G$  is very good, i.e. that the “bracket” on  $(\underline{\text{Lie}}G)(T)$  does indeed satisfy  $[u, u] = 0$  for every  $u \in (\underline{\text{Lie}}G)(T)$  (cf. II, 4.10).

Let  $u, v$  be two elements of  $(\underline{\text{Lie}}G)(T)$ , i.e. two primitive elements of  $\Gamma(T, \mathcal{U}_T)$ . Put  $I = \text{Spec } \mathcal{O}_S[d]/(d^2)$  and  $I' = \text{Spec } \mathcal{O}_S[d']/(d'^2)$ . Since the composition law of  $G(I \times I')$  is induced by the multiplication of the algebra  $\mathcal{U}_{I \times I'}$ , one has in  $G(I \times I')$  the equality:

$$\begin{aligned} & (1 + ud)(1 + vd')(1 + ud)^{-1}(1 + vd')^{-1} \\ &= (1 + ud)(1 + vd')(1 - ud)(1 - vd) \\ &= 1 + (uv - vu)dd'. \end{aligned}$$

According to the description of the bracket  $[u, v]$  given before Prop. 4.8 of Exp. II, one obtains that

$$[u, v] = uv - vu,$$

where the right-hand term is the commutator of  $u$  and  $v$  in the algebra  $\Gamma(T, \mathcal{U}_T)$ , whence  $[u, u] = 0$ . One has therefore obtained the following proposition:<sup>(37)</sup>

**Proposition.** *Let  $\mathcal{U}$  be a  $\mathcal{O}_S$ -coalgebra in groups. The  $S$ -group functor  $G = \text{Spec}^* \mathcal{U}$  is very good, and there is an isomorphism  $\underline{\text{Lie}}G \simeq \text{Prim } \mathbf{W}(\mathcal{U})$  of  $\mathbf{O}_S$ -Lie algebras, where  $\text{Prim } \mathbf{W}(\mathcal{U})$  denotes the functor which associates to every  $T \rightarrow S$  the  $\mathcal{O}(T)$ -Lie algebra consisting of the primitive elements of  $\mathbf{W}(\mathcal{U})(T) = \Gamma(T, \mathcal{U}_T)$ .*

<sup>(35)</sup>**N.D.E.** Note that the second condition is a consequence of the first, for the latter implies that  $u = (\text{id} \otimes \varepsilon)\Delta(u) = u + \varepsilon(u)$ , whence  $\varepsilon(u) = 0$ .

<sup>(36)</sup>**N.D.E.** The  $\mathbf{O}_S$ -module structure on  $\underline{\text{Lie}}G$  is defined in II, Prop. 3.6.

<sup>(37)</sup>**N.D.E.** This proposition, which summarizes the preceding discussion, has been added.

**3.3.** — Finally, suppose that  $\mathcal{U}$  is a  $\mathcal{O}_S$ -coalgebra in commutative groups, that is, that the triangle

$$(i^*) \quad \begin{array}{ccc} \mathcal{U} \otimes \mathcal{U} & \xrightarrow{\sigma} & \mathcal{U} \otimes \mathcal{U} \\ & \searrow m_{\mathcal{U}} \quad \swarrow m_{\mathcal{U}} & \\ & \mathcal{U} & \end{array}$$

is commutative, or again that  $m_{\mathcal{U}}$  makes  $\mathcal{U}$  a commutative  $\mathcal{O}_S$ -algebra. Conditions (i), (ii), (iii), (iv), (v), (vi), (i)\*, (ii)\*, (iii)\*, and (v)\* then also mean that  $\mathcal{U}$  is a cogroup in the category of commutative  $\mathcal{O}_S$ -algebras. Thus, if in addition  $\mathcal{U}$  is a quasi-coherent  $\mathcal{O}_S$ -module, the affine  $S$ -scheme  $\text{Spec } \mathcal{U}$  is a commutative  $S$ -group scheme.

In this case, since the diagonal morphism  $\Delta'$  of  $\mathcal{O}_S[T, T^{-1}]$  sends  $T$  to  $T \otimes T$ , the morphisms of  $S$ -groups from  $\text{Spec } \mathcal{U}$  to  $\mathbf{G}_{m,S}$  (I 4.3.2) correspond bijectively to the morphisms of unital  $\mathcal{O}_S$ -algebras

$$\varphi : \mathcal{O}_S[T, T^{-1}] \longrightarrow \mathcal{U}$$

such that  $(\varphi \otimes \varphi) \circ \Delta' = \Delta_{\mathcal{U}} \circ \varphi$  (in this case,  $\varepsilon_{\mathcal{U}} \circ \varphi$  is the neutral element of  $\mathbf{G}_{m,S}(S)$ , i.e. the augmentation  $\varepsilon'$ ). Such a morphism  $\varphi$  is determined by the image  $\varphi(T)$ , which must be an invertible element  $x$  of  $\mathcal{U}$  satisfying  $\Delta_{\mathcal{U}} x = x \otimes x$  and  $\varepsilon_{\mathcal{U}}(x) = \varepsilon'(T) = 1$ . One therefore has:

$$\text{Hom}_{S\text{-gr.}}(\text{Spec } \mathcal{U}, \mathbf{G}_{m,S}) \simeq (\text{Spec}^* \mathcal{U})(S)$$

and since this formula remains valid after every base change, this gives:

$$\text{Spec}^* \mathcal{U} \simeq \underline{\text{Hom}}_{S\text{-gr.}}(\text{Spec } \mathcal{U}, \mathbf{G}_{m,S}).$$

One has therefore obtained the

**Proposition 3.3.0.** *If  $\mathcal{U}$  is a  $\mathcal{O}_S$ -coalgebra in commutative groups, quasi-coherent as an  $\mathcal{O}_S$ -module, then the affine  $S$ -scheme  $G = \text{Spec } \mathcal{U}$  is a commutative  $S$ -group scheme, and there is an isomorphism of  $S$ -group functors  $\text{Spec}^* \mathcal{U} \simeq \underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$ .*

If one assumes, in addition, that  $\mathcal{U}$  is a locally free  $\mathcal{O}_S$ -module of finite type, then, according to 3.1.2.1, the  $S$ -group functor  $\text{Spec}^* \mathcal{U}$  is represented by  $\text{Spec } \mathcal{U}^*$ . One therefore obtains the

**Proposition 3.3.1.** (Cartier duality). *The functor*

$$\mathcal{A}(G) \mapsto \mathcal{A}(G)^* = \mathcal{H}om_{\mathcal{O}_S\text{-Mod.}}(\mathcal{A}(G), \mathcal{O}_S)$$

*induces a duality\* of the category of commutative, finite and locally free  $S$ -group schemes; it associates to  $G$  the  $S$ -group  $\underline{\text{Hom}}_{S\text{-gr.}}(G, \mathbf{G}_{m,S})$ .*

#### 4. “Frobeniusseries”

Let  $p$  be a fixed prime number and  $(\mathbf{Sch}/\mathbf{F}_p)$  the category of schemes of characteristic  $p$ , that is, schemes over the prime field  $\mathbf{F}_p$ . Following the general conventions of this seminar, we identify  $(\mathbf{Sch}/\mathbf{F}_p)$  with a subcategory of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$  by means of the functor  $h$  of I 1.1. We likewise use the isomorphism from  $\text{Hom}(h_X, \mathbf{F})$  onto  $\mathbf{F}(X)$  defined in I 1.1 to identify these two sets whenever  $X$  is an  $\mathbf{F}_p$ -scheme and  $\mathbf{F}$  an object of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ .

\*A duality of a category  $\mathcal{C}$  is a pair  $(D, \varphi)$  consisting of a contravariant functor  $D$  from  $\mathcal{C}$  to  $\mathcal{C}$  and a functorial isomorphism  $\varphi : \text{Id}_{\mathcal{C}} \longrightarrow DD$  such that the isomorphisms  $\varphi D : D \longrightarrow DDD$  and  $D\varphi^{-1} : DDD \longrightarrow D$  are inverse to one another.\*

(38) **N.D.E.**  $D\varphi$  has been corrected to  $D\varphi^{-1}$ .

**Notation 4.0.** <sup>(39)</sup> If  $T$  is an  $\mathbf{F}_p$ -scheme, a  $T$ -functor is a morphism  $q : \mathbf{F} \rightarrow T$  of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$  with target  $T$ ; for every  $T$ -scheme  $r : X \rightarrow T$ , the set of  $T$ -morphisms  $X \rightarrow \mathbf{F}$ , i.e.  $\mathbf{F}_p$ -morphisms  $s : X \rightarrow \mathbf{F}$  such that  $q \circ s = r$ , will then be denoted by  $q(r)$ ,  $q(X/T)$ ,  $\mathbf{F}(r)$ , or  $\mathbf{F}(X/T)$  (or even  $\mathbf{F}(X)$  when no confusion will be possible with  $\text{Hom}(h_X, \mathbf{F})$ ).

**4.1.** — For every scheme  $S$  of characteristic  $p$ , we denote by  $\text{fr}(S)$ , or simply  $\text{fr}$ , the endomorphism of  $S$  inducing the identity on the topological space underlying  $S$  and associating  $x^p$  to a section  $x$  of  $\mathcal{O}_S$  over an open subset  $U$ .

Then the map  $\text{fr} : S \mapsto \text{fr}(S)$  is an *endomorphism of the identity functor* of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ ,<sup>(40)</sup> which implies the following results. Let  $\mathbf{E}$  be an  $\mathbf{F}_p$ -functor, that is, an object of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ ; the map associating to every  $\mathbf{F}_p$ -scheme  $S$  the endomorphism  $\mathbf{E}(\text{fr}(S))$  of  $\mathbf{E}(S)$  is a functorial endomorphism of  $\mathbf{E}$  which we shall denote by  $\text{fr}(\mathbf{E})$  or  $\text{fr}$ ; this notation is compatible with the identification of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$  with a subcategory of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$ . Moreover, the map  $\mathbf{E} \mapsto \text{fr}(\mathbf{E})$  is an *endomorphism of the identity functor* of  $(\widehat{\mathbf{Sch}/\mathbf{F}_p})$  (which we shall again denote by  $\text{fr}$ ).<sup>(41)</sup>

For every  $\mathbf{F}_p$ -scheme  $S$  and every  $S$ -functor  $q : X \rightarrow S$ , we denote by  $X^{(p/S)}$  or  $X^{(p)}$  the inverse image of  $X$  by the base change  $\text{fr}(S)$ :

$$\begin{array}{ccc} X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ \downarrow & & \downarrow q \\ S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

The commutative square

$$\begin{array}{ccc} X & \xrightarrow{\text{fr}(X)} & X \\ \downarrow q & & \downarrow q \\ S & \xrightarrow{\text{fr}(S)} & S \end{array}$$

then induces an  $S$ -morphism denoted by  $\text{Fr}(X/S)$  (or simply  $\text{Fr}$ ) from  $X$  to  $X^{(p/S)}$  such that  $\text{fr}(X) = \text{pr}_X \circ \text{Fr}(X/S)$ :

$$\begin{array}{ccccc} X & & \xrightarrow{\text{fr}(X)} & & X \\ & \searrow \text{Fr}(X/S) & & & \downarrow q \\ & & X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ & \downarrow q & \downarrow & & \downarrow q \\ & & S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

We shall say that  $\text{Fr}(X/S)$  is the *Frobenius morphism of  $X$  relative to  $S$* ; it is clear that the map  $\text{Fr} : X \mapsto \text{Fr}(X/S)$  is a functorial homomorphism.

<sup>(39)</sup>**N.D.E.** The numbering 4.0 has been added for later references.

<sup>(40)</sup>**N.D.E.** i.e. for every morphism of  $\mathbf{F}_p$ -schemes  $f : Y \rightarrow X$ , the diagram below is commutative:

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \text{fr}(Y) \downarrow & & \downarrow \text{fr}(X) \\ Y & \xrightarrow{f} & X. \end{array}$$

<sup>(41)</sup>**N.D.E.** One says that  $\text{fr}(X)$  is the “absolute” Frobenius morphism of  $X$ , to distinguish it from the “relative” Frobenius morphism  $\text{Fr}(X/S)$  introduced below.

(42) Let  $r : T \rightarrow S$  be an  $S$ -scheme. For every  $\phi \in X(r) = \text{Hom}_S(T, X)$  (cf. 4.0), there is a commutative diagram:

$$\begin{array}{ccccc} X & \xrightarrow{\text{Fr}(X/S)} & X^{(p/S)} & \xrightarrow{\text{pr}_X} & X \\ \uparrow \phi & \searrow q & \downarrow q^{(p/S)} & & \downarrow q \\ T & \xrightarrow{r} & S & \xrightarrow{\text{fr}(S)} & S. \end{array}$$

According to the definition of  $X^{(p/S)}$  as a fibered product,  $\text{pr}_X$  induces a bijection:

$$X^{(p/S)}(r) = \text{Hom}_S(T, X^{(p/S)}) \xrightarrow{\sim} \text{Hom}_S(T, X) = X(\text{fr}(S) \circ r).$$

On the other hand,  $r \circ \text{fr}(T) = \text{fr}(S) \circ r$ , since  $\text{fr}$  is an endomorphism of the identity functor. It follows that the map  $\text{Fr}(X/S)(r) : X(r) \rightarrow X^{(p/S)}(r)$  may be characterized by the commutativity of the following square:

$$\begin{array}{ccc} X(r) & \xrightarrow{\text{Fr}(X/S)(r)} & X^{(p/S)}(r) \\ \downarrow X(\text{fr}(T)) & & \downarrow \wr \\ X(r \circ \text{fr}(T)) & \xlongequal{\quad} & X(\text{fr}(S) \circ r). \end{array}$$

For example, if  $X$  is the subscheme of  $S$  defined by a quasi-coherent ideal  $\mathcal{I}$ , then  $X^{(p)}$  is the subscheme of  $S$  defined by the ideal  $\mathcal{I}^{[p]}$  generated by the  $p$ -th powers of sections of  $\mathcal{I}$ ; furthermore,  $\text{Fr}(X/S)$  is then the canonical immersion from  $\text{Spec}(\mathcal{O}_X/\mathcal{I})$  into  $\text{Spec}(\mathcal{O}/\mathcal{I}^{[p]})$ .

**4.1.1.1.** — (43) Let  $t : T \rightarrow S$  be a base change and  $X_T = X \times_{q,t} T$ . Consider the inverse image of  $X_T$  under  $\text{fr}(T)$ :

$$\begin{array}{ccccc} (X_T)^{(p/T)} & \longrightarrow & X_T & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow q \\ T & \xrightarrow{\text{fr}(T)} & T & \xrightarrow{t} & S. \end{array}$$

Since  $t \circ \text{fr}(T) = \text{fr}(S) \circ t$ ,  $(X_T)^{(p/T)}$  identifies with the inverse image of  $X^{(p/S)}$  under  $t$ ; in other words, there is a canonical isomorphism:

$$X_T^{(p/T)} \xrightarrow{\sim} (X^{(p/S)})_T.$$

It is clear that, under this identification,  $\text{Fr}(X_T/T)$  identifies with the inverse image  $\text{Fr}(X/S)_T$  of  $\text{Fr}(X/S)$ .

**4.1.1.1.1.** — In particular, if  $S$  is the spectrum of the prime field  $\mathbf{F}_p$ ,  $X^{(p/S)}$  is equal to  $X$  and  $\text{Fr}(X/S)$  to  $\text{fr}(X)$ . Consequently,  $X_T^{(p/T)}$  identifies with  $X_T$  and  $\text{Fr}(X_T/T)$  with  $\text{fr}(X)_T$ .

For example, if  $E$  is a set and  $E_T$  the constant  $T$ -scheme of type  $E$ , one has  $E_T^{(p/T)} \simeq E_T$  and  $\text{Fr}(E_T/T) \simeq \text{id}_{E_T}$ .

**4.1.1.2.** — The functor  $X \mapsto X^{(p/S)}$  clearly commutes with products; it therefore transforms an  $S$ -group  $G$  into an  $S$ -group  $G^{(p/S)}$ ; moreover, since  $\text{Fr}$  is a functorial homomorphism,

$$\text{Fr}(G/S) : G \rightarrow G^{(p/S)}$$

is a homomorphism of  $S$ -groups. We shall denote its kernel by  $_{\text{Fr}}G$ .

(42) **N.D.E.** The original has been expanded in what follows.

(43) **N.D.E.** The original has been expanded in what follows.

If  $r : T \rightarrow S$  is a scheme over  $S$ , it follows from diagram (†) of 4.1 that the value of  $\mathrm{Fr}G$  at  $r$  is the kernel of the homomorphism

$$G(\mathrm{fr}(T)) : G(r) \rightarrow G(r \circ \mathrm{fr}(T)).$$

Now, when  $T$  is the scheme  $I_R$  of dual numbers over an  $S$ -scheme  $R$ ,  $\mathrm{fr}(I_R)$  factors as follows:

$$I_R \xrightarrow{\mathrm{can.}} R \xrightarrow{\mathrm{fr}(R)} R \xrightarrow{s} I_R,$$

where  $s$  is the zero section. It follows that  $(\mathrm{Fr}G)(I_R)$  contains the kernel  $\underline{\mathrm{Lie}}(G/S)(R)$  of the morphism  $G(s) : G(I_R) \rightarrow G(R)$ , and that one therefore has:  $\underline{\mathrm{Lie}}(G/S) = \underline{\mathrm{Lie}}(\mathrm{Fr}G/S)$ .

**4.1.3.** — More generally, for every  $S$ -functor  $X$ , we define the  $S$ -functor  $X^{(p^n)}$  by induction on  $n$  using the formulas:

$$X^{(p)} = X^{(p/S)} \quad \text{and} \quad X^{(p^n)} = (X^{(p^{n-1})})^{(p)}.$$

Similarly,  $\mathrm{Fr}^n(X/S)$  or  $\mathrm{Fr}^n$  denote the composite functorial homomorphism

$$\begin{aligned} X &\xrightarrow{\mathrm{Fr}(X/S)} X^{(p)} \xrightarrow{\mathrm{Fr}(X^{(p)}/S)} X^{(p^2)} \\ &\rightarrow \dots \rightarrow X^{(p^{n-1})} \xrightarrow{\mathrm{Fr}(X^{(p^{n-1})}/S)} X^{(p^n)}. \end{aligned}$$

Note that, according to 4.1.1,  $\mathrm{Fr}(X^{(p)}/S)$  coincides with  $\mathrm{Fr}(X/S)^{(p)}$ , i.e. the following diagram is commutative:

$$\begin{array}{ccc} X^{(p)} & \xrightarrow{\quad} & X \\ \mathrm{Fr}(X^{(p)}/S) \downarrow & & \downarrow \mathrm{Fr}(X/S) \\ X^{(p^2)} & \xrightarrow{\quad} & X^{(p)}. \end{array}$$

If  $G$  is an  $S$ -group functor,  $G^{(p^n)}$  is one also, and  $\mathrm{Fr}^n(G/S)$  is a homomorphism of  $S$ -group functors.

**Definition.** We shall denote by  $\mathrm{Fr}^n G$  the kernel of  $\mathrm{Fr}^n(G/S)$  and shall say that  $G$  is of *height*  $\leq n$  if  $\mathrm{Fr}^n(G/S)$  is zero, that is, if  $\mathrm{Fr}^n G = G$ .

**Lemma.** The group subfunctor  $\mathrm{Fr}^n G$  of  $G$  is characteristic, i.e. for every  $S$ -scheme  $T$ , every endomorphism  $\phi$  of the  $T$ -group functor  $G_T$  induces an endomorphism of  $(\mathrm{Fr}^n G)_T$ .

Indeed, since the construction of  $G^{(p^n)}$  and of  $\mathrm{Fr}^n(G/S)$  commutes with base changes according to 4.1.1, one may suppose  $T = S$ ; in this case, the assertion follows from the fact that  $\mathrm{Fr}^n(G/S)$  is a functorial homomorphism.

**4.1.4.** — Here are a few examples.

a) First consider an “abstract” abelian group  $M$  and the diagonalizable group  $G = D_S(M)$  of type  $M$  (I 4.4): for every  $S$ -scheme  $T$ ,  $G(T)$  is therefore the abelian group

$$\mathrm{Hom}_{(\mathrm{Ab})}(M, \Gamma(T, \mathcal{O}_T)^\times).$$

Since  $G$  is the inverse image of the diagonalizable group  $D(M)$  over  $\mathbf{F}_p$ ,  $G^{(p)}$  identifies with  $G$ , and  $\mathrm{Fr}(G/S)(T)$  identifies with the endomorphism  $x \mapsto x^p$  of  $G(T)$  (4.1.1). In particular, when  $M$  is equal to  $\mathbf{Z}$ , one has  $D_S(M) = \mathbf{G}_{m,S}$ , so that:

$\mathrm{Fr} \mathbf{G}_{m,S}$  is the  $S$ -group  $\mu_{p,S}$  which associates to every  $S$ -scheme  $T$  the group of  $p$ -th roots of unity in  $\Gamma(T, \mathcal{O}_T)^*$ .

b) Now consider a scheme  $S$  of characteristic  $p$  and a sheaf of modules  $\mathcal{E}$  on  $S$ . According to I 4.6.2, there is a canonical isomorphism

$$\mathbf{W}(\mathcal{E})^{(p)} \simeq \mathbf{W}(\mathcal{E}^{(p)}),$$

where  $\mathcal{E}^{(p)}$  is the inverse image of  $\mathcal{E}$  under  $\mathrm{fr}(S)$ . For every  $S$ -scheme  $\pi : T \rightarrow S$ , the map  $\mathrm{Fr}(\mathbf{W}(\mathcal{E}))(\pi)$  is determined, according to 4.1 (†), by the commutative triangle

$$\begin{array}{ccc} \Gamma(T, \pi^* \mathrm{fr}(S)^* \mathcal{E}) & \xrightarrow[\mathrm{can.}]{\sim} & \Gamma(T, \mathrm{fr}(T)^* \pi^* \mathcal{E}) \\ & \nwarrow \mathrm{Fr}(\mathbf{W}(\mathcal{E})/S)(\pi) \quad \nearrow f' & \\ & \Gamma(T, \pi^* \mathcal{E}) & \end{array},$$

where  $f'$  is the map induced by  $\mathrm{fr}(T)$ .

In particular, if  $\mathcal{E}$  is equal to  $\mathcal{O}_S$ ,  $\mathbf{W}(\mathcal{E})$  identifies with the additive group  $\mathbf{G}_{a,S}$ . In this case, one has  $\mathcal{E}^{(p)} = \mathcal{E} = \mathcal{O}_S$ , and the Frobenius morphism  $\mathrm{Fr}(\mathbf{G}_{a,S}/S)$  sends  $x \in \Gamma(T, \mathcal{O}_T)$  to  $x^p$ . Thus:

$\mathrm{Fr} \mathbf{G}_{a,S}$  is the  $S$ -group  $\alpha_{p,S}$  which associates to every  $S$ -scheme  $T$  the group:  $\{x \in \Gamma(T, \mathcal{O}_T) \mid x^p = 0\}$ .

c) One would see similarly that, for every quasi-coherent  $\mathcal{O}_S$ -algebra  $\mathcal{A}$ ,  $(\mathrm{Spec} \mathcal{A})^{(p)}$  identifies with the spectrum  $\mathrm{Spec} \mathcal{A}^{(p)}$  of the inverse image of  $\mathcal{A}$  under  $\mathrm{fr}(S)$ . If  $\pi$  denotes the endomorphism  $x \mapsto x^p$  of the sheaf of rings  $\mathcal{O}_S$ , one therefore has

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_{\pi} \mathcal{O}_S$$

(44) and  $\mathrm{Fr}((\mathrm{Spec} \mathcal{A})/S)$  is induced by the morphism of  $\mathcal{O}_S$ -algebras  $\mathcal{A} \otimes_{\pi} \mathcal{O}_S \rightarrow \mathcal{A}$  defined by  $a \otimes_{\pi} x \mapsto a^p x$ .

Finally, for every quasi-coherent  $\mathcal{O}_S$ -module  $\mathcal{E}$ , there are canonical isomorphisms

$$\mathbf{V}(\mathcal{E})^{(p)} \simeq \mathbf{V}(\mathcal{E}^{(p)}) \quad \text{and} \quad S(\mathcal{E})^{(p)} \simeq S(\mathcal{E}^{(p)}),$$

where  $S(\mathcal{E})$  denotes the symmetric algebra of the  $\mathcal{O}_S$ -module  $\mathcal{E}$ .

d) Let  $\mathcal{U}$  be a  $\mathcal{O}_S$ -coalgebra (3.1), and  $T$  a scheme of characteristic  $p$ . If  $\mathcal{U}^{(p/S)}$  or  $\mathcal{U}^{(p)}$  denote the inverse image of the coalgebra  $\mathcal{U}$  under  $\mathrm{fr}(S)$ , there is, as in b), a canonical isomorphism:

$$(\mathrm{Spec}^* \mathcal{U})^{(p)} \simeq \mathrm{Spec}^* \mathcal{U}^{(p)}.$$

If  $\mathcal{U}$  is a coalgebra in groups, the value of  $\mathrm{Fr}(\mathrm{Spec}^* \mathcal{U})$ , i.e. of the kernel of the Frobenius morphism  $\mathrm{Spec}^* \mathcal{U} \rightarrow (\mathrm{Spec}^* \mathcal{U})^{(p)}$ , at an  $S$ -scheme  $T$  is therefore the set of elements  $\gamma$  of

$$(\mathrm{Spec}^* \mathcal{U})(T) = \{x \in \Gamma(T, \mathcal{U}_T) \mid \varepsilon_{\mathcal{U}_T}(x) = 1, \Delta_{\mathcal{U}_T} x = x \otimes x\}$$

such that the image in  $\Gamma(T, \mathcal{U}_T \otimes_{\mathrm{fr}(T)} \mathcal{O}_T)$  of the element  $\gamma \otimes_{\mathrm{fr}(T)} 1$  of  $\Gamma(T, \mathcal{U}_T) \otimes_{\mathrm{fr}(T)} \mathcal{O}(T)$  is equal to 1.

**4.2.** — (45) We shall now deal with a construction close to the preceding one: let  $S$  be a scheme of characteristic  $p$ ,  $X$  an  $S$ -scheme, and  $X_S^p$  the product in the category  $(\mathbf{Sch}/S)$  of  $p$  copies of  $X$ .

We then denote by  $U^p(X)$  the open subscheme of  $X_S^p$  which is the union of the products  $U_S^p$ , as  $U$  ranges over the affine open subsets of  $X$ . A point  $x$  of  $X_S^p$  therefore belongs to  $U^p(X)$  if and only if the projections  $\mathrm{pr}_i x$  of  $x$  onto the factors of  $X_S^p$  belong to the same affine open subset of  $X$ . For example, if every finite subset of  $X$  is contained in an affine open subset, one has  $U^p(X) = X_S^p$ .

The symmetric group  $\mathcal{S}_p$  of order  $p$  acts on  $X_S^p$  by permutation of the factors and leaves the open subset  $U^p(X)$  stable. We shall call the quotient of  $X_S^p$  by  $\mathcal{S}_p$  in the category of ringed spaces the  $p$ -fold symmetric product of  $X$ , and denote it by  $\Sigma^p X$ . Let  $q(X)$ , or simply  $q$ , be the canonical projection  $X_S^p \rightarrow \Sigma^p X$ .

Then  $q$  maps  $U^p(X)$  onto an open subset  $V^p(X)$  of the symmetric product, which may be described as follows (cf. V 4.1). The structure sheaf of  $\Sigma^p X$  induces on  $V^p(X)$  a scheme

(44) **N.D.E.**  $\mathcal{A} \otimes_{\pi} \mathcal{O}_S$  denotes the  $\mathcal{O}_S$ -algebra obtained by the extension of scalars  $\pi : \mathcal{O}_S \rightarrow \mathcal{O}_S$ , i.e. one has:  $ax \otimes_{\pi} 1 = a \otimes_{\pi} x^p$ , and  $x \cdot (a \otimes_{\pi} 1) = a \otimes_{\pi} x$ .

(45) **N.D.E.** For the contents of nos. 4.2 and 4.3, one may also refer to [DG70], § IV.3, nos. 4–6.

structure; the morphism  $q'(X) : U^p(X) \rightarrow V^p(X)$  induced by  $q(X)$  is affine and even integral; as  $U$  ranges over the affine open subsets of  $X$  which project into a varying affine open subset  $V$  of  $S$ , the  $\Sigma^p U$  form an affine covering of  $V^p(X)$ ; if  $R$  denotes the affine algebra of  $V$  and  $A$  that of  $U$ ,  $\Sigma^p U$  has as affine algebra the subalgebra  $\Sigma^p A$  of  $\bigotimes_R^p A$  consisting of symmetric tensors.

Now consider the diagonal morphism  $\delta$  from  $X$  to  $U^p(X)$ . If  $V = \text{Spec } R$  is an affine open subset of  $S$  and  $U = \text{Spec } A$  an affine open subset of  $X$  over  $V$ , the restriction of  $\delta$  to  $U$  is defined by the algebra morphism

$$\eta : \bigotimes_R^p A \longrightarrow A, \quad a_1 \otimes \cdots \otimes a_p \longmapsto a_1 a_2 \cdots a_p.$$

One therefore has, if  $N$  is the symmetrization operator:

$$\begin{aligned} \eta(N(a_1 \otimes \cdots \otimes a_p)) &= \eta\left(\sum_{\sigma \in \mathcal{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}\right) \\ &= p! a_1 \cdots a_p = 0. \end{aligned}$$

In other words,  $\eta$  vanishes on the subspace  $N(\bigotimes_R^p A)$  of  $\Sigma^p A$  consisting of symmetrized tensors. Moreover, if  $f$  is a symmetric tensor, one clearly has  $N(fa) = fN(a)$ , which shows that  $N(\bigotimes_R^p A)$  is an ideal of  $\Sigma^p A$ . From now on we shall write

$$U^{[p/S]} = \text{Spec}\left(\Sigma^p A / N\left(\bigotimes_R^p A\right)\right);$$

this is a closed subscheme of  $\Sigma^p(U) = V^p(U)$ . The union of the  $U^{[p/S]}$ , as  $U$  ranges over the affine open subsets of  $X$  which project into a varying affine open subset  $V$  of  $S$ , is a closed subscheme of  $V^p(X)$ , denoted by  $X^{[p/S]}$ .

Moreover, if  $i(X)$  denotes the inclusion of  $X^{[p/S]}$  in  $V^p(X)$ , we have just seen that  $q'(X) \circ \delta$  factors through  $X^{[p/S]}$ , whence a morphism  $F^{[p]}(X/S) : X \rightarrow X^{[p/S]}$ ; <sup>(46)</sup>

$$\begin{array}{ccc} X_S^p & \supset & U^p(X) \xleftarrow{\delta(X)} X \\ q(X) \downarrow & & \downarrow q'(X) \quad \downarrow F^{[p]}(X/S) \\ \Sigma^p(X) & \supset & V^p(X) \xleftarrow{i(X)} X^{[p/S]}. \end{array}$$

It is clear that  $X^{[p/S]}$  is functorial in  $X$  and that the map  $F^{[p]} : X \mapsto F^{[p]}(X/S)$  is a functorial homomorphism.

**4.2.1.** — The schemes  $X^{[p/S]}$  and  $X^{(p/S)}$  are clearly related: let  $V$  be an affine open subset of  $S$  with affine ring  $R$ , and  $U$  an affine open subset of  $X$  over  $V$ ; let  $A$  be the affine algebra of  $U$ . If  $\pi$  denotes the endomorphism  $x \mapsto x^p$  of  $R$ , then  $U^{(p/S)}$  has  $A \otimes_\pi R$  as affine algebra. One verifies, moreover, that the map

$$a \otimes_\pi \lambda \longmapsto \left( \lambda a \otimes \cdots \otimes a \quad \text{mod } N\left(\bigotimes_R^p A\right) \right)$$

defines a morphism of  $R$ -algebras from  $A \otimes_\pi R$  to  $\Sigma^p A / N(\bigotimes_R^p A)$ , and the latter induces a morphism  $\varphi(U) : U^{[p/S]} \rightarrow U^{(p/S)}$  such that  $\varphi(U) \circ F^{[p]}(U/S) = \text{Fr}(U/S)$ .

<sup>(46)</sup>**N.D.E.** In the original, this morphism (resp. the relative Frobenius morphism) was denoted by  $F'$  (resp.  $F$ ).

“Gluing the pieces together”, one then obtains a commutative triangle

$$\begin{array}{ccc} & X & \\ F^{[p]}(X/S) \swarrow & & \searrow \text{Fr}(X/S) \\ X^{[p/S]} & \xrightarrow{\varphi(X)} & X^{(p/S)}. \end{array}$$

For example, if  $X$  is the subscheme of  $S$  defined by a quasi-coherent ideal  $\mathcal{I}$ ,  $F^{[p]}(X/S)$  identifies with the identity morphism of  $X$ , so that  $\varphi(X)$  is the canonical immersion from  $\text{Spec}(\mathcal{O}_S/\mathcal{I})$  into  $\text{Spec}(\mathcal{O}_S/\mathcal{I}^{\{p\}})$ . One thus sees that  $\varphi(X)$  is not an isomorphism in general.

However, when  $M$  is a free  $R$ -module, it is clear that the map

$$\begin{aligned} M \otimes_{\pi} R &\longrightarrow \Sigma^p M / N \left( \bigotimes_R^p M \right), \\ m \otimes_{\pi} \lambda &\longmapsto \left( \lambda m \otimes \cdots \otimes m \mod N \left( \bigotimes_R^p M \right) \right) \end{aligned}$$

is bijective; this map therefore remains bijective when  $M$  is  $R$ -flat, because every flat module is a filtered inductive limit of free modules (Lazard<sup>\*(47)</sup>). It follows that

$\varphi(X) : X^{[p/S]} \longrightarrow X^{(p/S)}$  is an isomorphism if  $X$  is a flat  $S$ -scheme.

**4.3.** — Finally, consider an  $S$ -scheme in abelian groups  $G$ . Then the composite morphism  $\mu(G) : U^p(G) \hookrightarrow G_S^p \rightarrow G$ , which is defined by multiplication, factors through  $V^p(G)$ , i.e. there exists a morphism  $\nu(G) : V^p(G) \rightarrow G$  such that  $\nu(G) \circ q'(G) = \mu(G)$ , so that one has the following commutative diagram:

$$\begin{array}{ccccc} G & \xleftarrow{\mu(G)} & U^p(G) & \xleftarrow{\delta(G)} & G \\ & \searrow \nu(G) & \downarrow q'(G) & & \downarrow F^{[p]}(G/S) \\ & & V^p(G) & \xleftarrow{i(G)} & G^{[p/S]} \\ & & & \searrow \varphi(G) & \downarrow \text{Fr}(G/S) \\ & & & & G^{(p/S)} \end{array}$$

When  $G$  is  $S$ -flat,  $\varphi(G)$  is an isomorphism and one can define a morphism (called *Verschiebung*)

$$\text{Ver}(G/S) : G^{(p/S)} \longrightarrow G$$

by means of the formula  $\text{Ver}(G/S) = \nu(G) \circ i(G) \circ \varphi(G)^{-1}$ . As  $G$  ranges over the flat  $S$ -schemes in abelian groups, the map  $\text{Ver} : G \mapsto \text{Ver}(G/S)$  is clearly a functorial homomorphism; consequently,  $\text{Ver}(G/S)$  is a *group homomorphism*. Finally, for every  $S$ -scheme  $T$ , the composite map

$$G(T) \xrightarrow{\delta(G)(T)} U^p(G)(T) \xrightarrow{\mu(G)(T)} G(T)$$

sends  $x \in G(T)$  to  $p \cdot x$ . We may write  $p \cdot \text{id}_G$  instead of  $\mu(G) \circ \delta(G)$ , thus obtaining the classical formula:

$$(*) \quad \text{Ver}(G/S) \circ \text{Fr}(G/S) = p \cdot \text{id}_G.$$

**Examples 4.3.1.** (a) When  $G$  is a constant  $S$ -scheme in abelian groups, we know that  $\text{Fr}(G/S)$  identifies with the identity morphism of  $G$  (cf. 4.1.1.1). One therefore has  $\text{Ver}(G/S) = p \text{id}_G$ .

\*D. Lazard, *C. R. Acad. Sc. Paris* **258**, 1964, pp. 6313–6316.

(47)**N.D.E.** See also: D. Lazard, *Bull. Soc. Math. France* **97** (1969), 81–128, or: [BA]g, X § 1.6, Th. 1.

(b) When  $G$  is the diagonalizable  $S$ -group of type  $M$ ,  $\mathrm{Fr}(G/S)$  is equal to  $p \mathrm{id}_G$  according to 4.1.4 (a); one then sees easily that  $\mathrm{Ver}(G/S)$  is the identity morphism of  $G$ .

(c) When  $\mathcal{E}$  is a flat  $\mathcal{O}_S$ -module and  $G$  is the  $S$ -group  $\mathbf{V}(\mathcal{E})$ , the morphism  $\mathrm{Ver}(G/S)$  is zero, as is  $p \cdot \mathrm{id}_G$ . We shall see in Exposé VII B that a commutative algebraic group  $G$  over a field  $k$  is “unipotent” if and only if the composite homomorphism

$$G^{(p^n)} \xrightarrow{\mathrm{Ver}(G^{(p^{n-1})}/S)} G^{(p^{n-1})} \longrightarrow \cdots \longrightarrow G^{(p)} \xrightarrow{\mathrm{Ver}(G/S)} G$$

is zero for some  $n$  (one has put  $G^{(p^n)} = (G^{(p^{n-1})})^{(p)}$ , cf. 4.1.3).<sup>(48)</sup>

**4.3.2.** — Since the map  $\mathrm{Ver} : G \mapsto \mathrm{Ver}(G/S)$  is a functorial homomorphism as  $G$  ranges over the flat  $S$ -schemes in commutative groups, the square

$$\begin{array}{ccc} G^{(p)} & \xrightarrow{\mathrm{Ver}(G/S)} & G \\ \mathrm{Fr}(G/S)^{(p)} \downarrow & & \downarrow \mathrm{Fr}(G/S) \\ G^{(p^2)} & \xrightarrow{\mathrm{Ver}(G^{(p)}/S)} & G^{(p)} \end{array}$$

is commutative, where  $\mathrm{Fr}(G/S)^{(p)}$  denotes the inverse image of  $\mathrm{Fr}(G/S)$  by the base change  $\mathrm{fr}(S)$ . According to 4.1.1, one has  $\mathrm{Fr}(G/S)^{(p)} = \mathrm{Fr}(G^{(p)}/S)$ , hence, according to 4.3 (\*) applied to  $G^{(p)}$ , one obtains:

$$\begin{aligned} (**) \quad \mathrm{Fr}(G/S) \circ \mathrm{Ver}(G/S) &= \mathrm{Ver}(G^{(p)}/S) \circ \mathrm{Fr}(G^{(p)}/S) \\ &= p \cdot \mathrm{id}_{G^{(p)}}. \end{aligned}$$

**4.3.3.** — Finally, suppose that  $G$  is a commutative, finite and locally free  $S$ -group; let  $\mathcal{A}$  be the affine  $\mathcal{O}_S$ -algebra of  $G$ , and  $\pi$  the endomorphism of the sheaf of rings  $\mathcal{O}_S$  sending a section  $x$  of  $\mathcal{O}_S$  to  $x^p$ .<sup>(49)</sup> Denote by  $\Sigma^p \mathcal{A}$  the subalgebra of  $\bigotimes_{\mathcal{O}_S}^p \mathcal{A}$  consisting of sections invariant under the action of the symmetric group, and by  $i(\mathcal{A})$  the inclusion of  $\Sigma^p \mathcal{A}$  into the tensor product. Let  $\Delta^p(\mathcal{A}) : \mathcal{A} \rightarrow \bigotimes_{\mathcal{O}_S}^p \mathcal{A}$  be the morphism obtained by iterating the diagonal morphism of the coalgebra  $\mathcal{A}$  (it corresponds to the multiplication morphism from  $U^p(G) = G_S^p$  to  $G$ ); according to the beginning of paragraph 4.3,  $\Delta^p(\mathcal{A})$  factors through  $\Sigma^p \mathcal{A}$ , i.e. induces a morphism

$$a(\mathcal{A}) : \mathcal{A} \rightarrow \Sigma^p \mathcal{A}$$

such that  $i(\mathcal{A}) \circ a(\mathcal{A}) = \Delta^p(\mathcal{A})$ .

On the other hand, let  $S^p(\mathcal{A})$  be the degree  $p$  component of the symmetric algebra of  $\mathcal{A}$ , and  $q(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^p \mathcal{A} \rightarrow S^p(\mathcal{A})$  the canonical projection. The multiplication  $m^p(\mathcal{A}) : \bigotimes_{\mathcal{O}_S}^p \mathcal{A} \rightarrow \mathcal{A}$  factors through  $S^p(\mathcal{A})$ , i.e. induces a map

$$b(\mathcal{A}) : S^p(\mathcal{A}) \rightarrow \mathcal{A}$$

such that  $b(\mathcal{A}) \circ q(\mathcal{A}) = m^p(\mathcal{A})$ .

Since  $\Sigma^p \mathcal{A}$  is the affine algebra of  $V^p(\mathcal{A})$ , according to the beginning of 4.3 again, the composite morphism  $i(G) \circ \varphi(G)^{-1}$  induces an algebra homomorphism

$$r(\mathcal{A}) : \Sigma^p \mathcal{A} \rightarrow \mathcal{A} \otimes_{\pi} \mathcal{O}_S;$$

this homomorphism vanishes on sections of the form

$$\sum_{\sigma \in \mathcal{S}_p} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(p)}$$

<sup>(48)</sup>**N.D.E.** Since this does not appear explicitly in VII B, we refer to [DG70], § IV.3, Prop. 4.11.

<sup>(49)</sup>**N.D.E.** The order has been modified by introducing first the objects occurring in the diagram which is to follow.

and sends  $a \otimes \cdots \otimes a$  to  $a \otimes_\pi 1$ . Similarly,  $j(\mathcal{A})$  is the morphism of  $\mathcal{O}_S$ -modules  $a \otimes_\pi 1 \mapsto q(a \otimes \cdots \otimes a)$ . One therefore obtains the commutative diagram:

$$(\mathcal{A}) \quad \begin{array}{ccccc} \mathcal{A} & \xrightarrow{\Delta^p(\mathcal{A})} & \bigotimes_{\mathcal{O}_S}^p \mathcal{A} & \xrightarrow{m^p(\mathcal{A})} & \mathcal{A} \\ & \searrow a(\mathcal{A}) & \nearrow i(\mathcal{A}) & \searrow q(\mathcal{A}) & \nearrow b(\mathcal{A}) \\ & \Sigma^p \mathcal{A} & & S^p(\mathcal{A}) & \\ & \searrow r(\mathcal{A}) & & \nearrow j(\mathcal{A}) & \\ & \mathcal{A} \otimes_\pi \mathcal{O}_S & & & \end{array} .$$

The composite  $r(\mathcal{A}) \circ a(\mathcal{A})$  is associated to the Verschiebung morphism  $\text{Ver}(G/S)$ , while  $b(\mathcal{A}) \circ j(\mathcal{A})$  is associated to the Frobenius morphism  $\text{Fr}(G/S)$ .

The commutative diagram  $(\mathcal{A})$  above is self-dual; let, indeed,  $D$  be the functor associating to every  $\mathcal{O}_S$ -module  $\mathcal{M}$  the dual  $\mathcal{O}_S$ -module  $\mathcal{H}om_{\mathcal{O}_S}(\mathcal{M}, \mathcal{O}_S)$ ; it is clear that the image of diagram  $(\mathcal{A})$  under the functor  $D$  is none other than diagram  $(D\mathcal{A})$ , the morphisms  $Dr(\mathcal{A})$ ,  $Da(\mathcal{A})$ ,  $Dj(\mathcal{A})$ , and  $Db(\mathcal{A})$  identifying respectively with  $j(D\mathcal{A})$ ,  $b(D\mathcal{A})$ ,  $r(D\mathcal{A})$ , and  $a(D\mathcal{A})$ . According to 3.3.1, one therefore sees that:

*In the category of commutative, finite and locally free  $S$ -groups, Cartier duality interchanges the Frobenius morphism and Verschiebung.*<sup>(50)</sup>

## 5. Lie $p$ -algebras

Let us first recall a few results from the Séminaire Sophus Lie.<sup>(51)</sup>

**5.1.** — Let  $p$  be a prime number,  $R$  a commutative ring of characteristic  $p$ , and  $A$  an associative, but not necessarily commutative,  $R$ -algebra. If  $a$  and  $b$  are two elements of  $A$ , we put  $[a, b] = ab - ba$  and  $ab = L_a(b) = R_b(a)$ . One then has:

$$\begin{aligned} (\text{ad } x^p)(y) &= [x^p, y] = (L_x^p - R_x^p)(y) \\ &= (L_x - R_x)^p(y) = (\text{ad } x)^p(y) \end{aligned}$$

whence the *first Jacobson formula*:

$$(i) \quad \text{ad}(x^p) = (\text{ad } x)^p.$$

If  $a_1, \dots, a_p$  are  $p$  arbitrary elements of  $A$ , then, denoting by  $N$  the symmetrization operator (cf. 4.2), one has the equalities:

$$\begin{aligned} (*) \quad N(a_1 \otimes \cdots \otimes a_p) &= \sum_{\sigma} a_{\sigma(1)} \cdots a_{\sigma(p)} \\ &= \sum_{\tau} [a_{\tau(1)} [a_{\tau(2)} [\cdots [a_{\tau(p-1)}, a_p] \cdots ]]] \end{aligned}$$

where  $\sigma$  ranges over the permutations of  $p$  letters and  $\tau$  over those of  $(p-1)$  letters. Indeed, the last term is equal to

$$\sum_{\tau} \sum_{r=0}^{p-1} \sum_{i_1 < \cdots < i_r} (-1)^s a_{\tau(i_1)} a_{\tau(i_2)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots a_{\tau(j_1)}$$

where  $\tau$  ranges over the permutations of  $p-1$  letters,  $i_1, \dots, i_r$  over the strictly increasing sequences of integers in the interval  $[1, p-1]$ , and where  $j_1, \dots, j_s$  denotes the strictly

<sup>(50)</sup>**N.D.E.** See also [DG70], § IV.3, 4.9.

<sup>(51)</sup>**N.D.E.** Cf. P. Cartier, *Exemples d'hyperalgèbres*, Sémin. Sophus Lie 1955/56, Exp. 3 (accessible on the Numdam site: <http://www.numdam.org>).

increasing sequence whose values are the integers of  $[1, p-1]$  different from  $i_1, \dots, i_r$ . For a fixed value of  $r$ , the sum of the terms  $(-1)^s a_{\tau(i_1)} \cdots a_{\tau(i_r)} a_p a_{\tau(j_s)} \cdots a_{\tau(j_1)}$  is clearly equal to

$$(-1)^s \binom{p-1}{s} \sum_{\rho} a_{\rho(1)} \cdots a_{\rho(r)} a_p a_{\rho(r+1)} \cdots a_{\rho(p-1)}$$

where  $\rho$  ranges over the permutations of  $p-1$  letters. Now  $(-1)^s \binom{p-1}{s} = 1$  in  $\mathbf{F}_p$ , since in  $\mathbf{F}_p[x]$  ( $x$  an indeterminate) one has:  $(x-1)^p = x^p - 1 = (x-1)(x^{p-1} + \cdots + 1)$ , and hence  $(x-1)^{p-1} = x^{p-1} + \cdots + 1$ . This proves  $(*)$ .

On the other hand, if  $x_0$  and  $x_1$  are two elements of  $A$ , one has

$$(x_0 + x_1)^p = x_0^p + x_1^p + \sum x_{z(1)} x_{z(2)} \cdots x_{z(p)},$$

where  $z$  ranges over the nonconstant maps from  $[1, p]$  to  $\{0, 1\}$ . From this one obtains

$$(x_0 + x_1)^p = x_0^p + x_1^p + \sum_{0 < r < p} \frac{1}{r!(p-r)!} N(\underbrace{x_0, \dots, x_0}_r, \underbrace{x_1, \dots, x_1}_{p-r}).$$

(52) Now, according to  $(*)$ , one has:

$$\begin{aligned} & N(\underbrace{x_0, \dots, x_0}_r, \underbrace{x_1, \dots, x_1}_{p-r}) \\ &= r!(p-1-r)! \sum_t [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots ]]] \end{aligned}$$

where  $t$  ranges over the maps  $[1, p-1] \rightarrow \{0, 1\}$  taking the value 0  $r$  times. One deduces from this the *second Jacobson formula*:

$$(ii) \quad (x_0 + x_1)^p = x_0^p + x_1^p - \sum_{0 < r < p} \sum_t \frac{1}{r} [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots ]]]$$

where  $t$  ranges over the maps  $[1, p-1] \rightarrow \{0, 1\}$  taking the value 0  $r$  times.

**5.2.** — Now let  $\mathfrak{g}$  be an  $R$ -Lie algebra. One says that a map  $x \mapsto x^{(p)}$  from  $\mathfrak{g}$  to  $\mathfrak{g}$  makes  $\mathfrak{g}$  a *Lie  $p$ -algebra over  $R$*  if the following conditions hold:

- (0)  $(\lambda x)^{(p)} = \lambda^p \cdot x^{(p)}$ , for  $\lambda \in R$ ,  $x \in \mathfrak{g}$ .
- (i)  $\text{ad } x^{(p)} = (\text{ad } x)^p$ , for  $x \in \mathfrak{g}$ .
- (ii)

$$(x_0 + x_1)^{(p)} = x_0^{(p)} + x_1^{(p)} - \sum_{0 < r < p} \sum_t \frac{1}{r} [x_{t(1)} [x_{t(2)} [\cdots [x_{t(p-1)}, x_1] \cdots ]]]$$

where  $t$  ranges over the maps  $[1, p-1] \rightarrow \{0, 1\}$  taking the value 0  $r$  times ( $x_0, x_1 \in \mathfrak{g}$ ).

The map  $x \mapsto x^{(p)}$  will then be called the “symbolic  $p$ -th power”.

For example, if  $A$  is an associative  $R$ -algebra, we saw in 5.1 that one obtains a Lie  $p$ -algebra, which will be denoted by  $A_{\text{Lie}}$ , by taking the  $R$ -module underlying  $A$  and putting, for  $x, y \in A$ ,

$$[x, y] = xy - yx \quad \text{and} \quad x^{(p)} = x^p.$$

We shall say that  $A_{\text{Lie}}$  is the *Lie  $p$ -algebra underlying  $A$* .

In what follows we shall chiefly consider Lie  $p$ -subalgebras of  $p$ -algebras of the form  $A_{\text{Lie}}$ ; here is an example: let  $S$  be a scheme of characteristic  $p > 0$ , and  $X$  an  $S$ -scheme. Recall that a derivation of  $X$  over  $S$  is an endomorphism  $D$  of the sheaf of abelian groups  $\mathcal{O}_X$  such that

$$D(\lambda \cdot s) = \lambda \cdot D(s) \quad \text{and} \quad D(st) = (Ds)t + s(Dt)$$

(52) **N.D.E.** The following explanation, taken from [DG70], § II.7, 3.2, has been inserted.

as  $\lambda$  and  $s, t$  range over sections of  $\mathcal{O}_S$  and  $\mathcal{O}_X$  over open subsets such that the formulas make sense. The Leibniz formula

$$D^n(st) = \sum_{i=0}^n \binom{n}{i} (D^i s)(D^{n-i} t)$$

shows that  $D^p$  is again a derivation of  $X$  over  $S$ , taking account of the equality  $\binom{p}{i} \equiv 0 \pmod{p}$  for  $i \neq 0, p$ . It follows that:

*The algebra  $\text{Dér}_{X/S}$  of derivations of  $X$  over  $S$  is a Lie  $p$ -subalgebra of the  $\Gamma(S, \mathcal{O}_S)$ -algebra of differential operators of  $X$  over  $S$ .*

**5.2.1.** — If  $\mathfrak{g}$  and  $\mathfrak{h}$  are two Lie  $p$ -algebras, a homomorphism  $h : \mathfrak{g} \rightarrow \mathfrak{h}$  is an  $R$ -linear map from  $\mathfrak{g}$  to  $\mathfrak{h}$  such that  $h([x, y]) = [h(x), h(y)]$  and  $h(x^{(p)}) = h(x)^{(p)}$  if  $x, y \in \mathfrak{g}$ . The composite map of two homomorphisms is again a homomorphism, so that we shall be able to speak of the category of Lie  $p$ -algebras over  $R$ .

If  $(X, \mathcal{R})$  is a ringed space, we shall say that an  $\mathcal{R}$ -module  $\mathfrak{g}$  is equipped with a Lie  $p$ -algebra structure over  $\mathcal{R}$  if, for every open subset  $U$ ,  $\Gamma(U, \mathfrak{g})$  is equipped with a Lie  $p$ -algebra structure over  $\Gamma(U, \mathcal{R})$  and the restrictions are homomorphisms.

**5.3.** — We are now interested in the functor left adjoint to the functor  $A \mapsto A_{\text{Lie}}$  of 5.2. Let  $\mathfrak{g}$  be a Lie  $p$ -algebra over the ring  $R$  of characteristic  $p$ ,  $U(\mathfrak{g})$  the enveloping algebra of the Lie algebra underlying  $\mathfrak{g}$  (cf. [BLie], I § 2.1), and  $i_{\mathfrak{g}}$  (or simply  $i$ ) the canonical map  $\mathfrak{g} \rightarrow U(\mathfrak{g})$ .

Let  $A$  be an associative unital  $R$ -algebra. One knows that, for every homomorphism of Lie algebras  $\phi : \mathfrak{g} \rightarrow A_{\text{Lie}}$ , there exists a unique homomorphism of unital  $R$ -algebras  $\psi : U(\mathfrak{g}) \rightarrow A$  such that  $\psi \circ i = \phi$ . Furthermore,  $\phi$  is a homomorphism of Lie  $p$ -algebras if and only if  $\psi$  vanishes on the elements  $i(x)^p - i(x^{(p)})$ , as  $x$  ranges over  $\mathfrak{g}$ .

**Definition.** One denotes by  $U_p^R(\mathfrak{g})$ , or simply  $U_p(\mathfrak{g})$ , the quotient of  $U(\mathfrak{g})$  by the two-sided ideal generated by the elements  $i(x)^p - i(x^{(p)})$ , and by  $j_{\mathfrak{g}}$  (or simply  $j$ ) the map  $\mathfrak{g} \rightarrow U_p(\mathfrak{g})$  which is the composite of  $i : \mathfrak{g} \rightarrow U(\mathfrak{g})$  and the canonical map  $U(\mathfrak{g}) \rightarrow U_p(\mathfrak{g})$ . One says that  $U_p(\mathfrak{g})$  is the *restricted enveloping algebra* of  $\mathfrak{g}$ .

According to what precedes, one has the

**Proposition.** *For every associative unital  $R$ -algebra and every morphism of Lie  $p$ -algebras  $\phi : \mathfrak{g} \rightarrow A_{\text{Lie}}$ , there exists a unique homomorphism of unital algebras  $\psi : U_p(\mathfrak{g}) \rightarrow A$  such that  $\psi \circ j = \phi$ . In other words, the functor  $\mathfrak{g} \mapsto U_p(\mathfrak{g})$  is left adjoint to the forgetful functor  $A \mapsto A_{\text{Lie}}$ .*

**5.3.1.** — With the notation of 5.3, now put  $\beta(x) = i(x)^p - i(x^{(p)})$ . For every element  $y$  of  $\mathfrak{g}$ , one has, according to 5.1 (i) and 5.2 (i):

$$\begin{aligned} \beta(x)i(y) &= i(y)\beta(x) + [\beta(x), i(y)] \\ &= i(y)\beta(x) + (\text{ad } i(x))^p i(y) - i((\text{ad } x)^p y) \\ &= i(y)\beta(x), \end{aligned}$$

so that  $\beta(x)$  belongs to the center of  $U(\mathfrak{g})$ ; in particular, the left ideal generated by the elements  $\beta(x)$  is already two-sided.

On the other hand, it is clear that  $\beta(\lambda x) = \lambda^p \beta(x)$ , for  $\lambda \in R$ , and it follows from 5.1 (ii) and 5.2 (ii) that, for  $x, y \in \mathfrak{g}$ ,

$$\beta(x + y) = \beta(x) + \beta(y).$$

In particular, if  $(x_{\alpha})$  is a family of generators of the  $R$ -module  $\mathfrak{g}$ , the left ideal generated by the elements  $\beta(x)$  is already generated by the  $\beta(x_{\alpha})$ .

**Proposition 5.3.2.** <sup>(53)</sup> *Let  $\mathfrak{g}$  be an  $R$ -Lie algebra whose underlying  $R$ -module is free with basis  $(x_\alpha)$ . Then the Lie  $p$ -algebra structures on  $\mathfrak{g}$  correspond bijectively to the families  $(y_\alpha)$  of  $\mathfrak{g}$  such that  $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$ .*

Indeed, if  $\mathfrak{g}$  is equipped with a Lie  $p$ -algebra structure  $x \mapsto x^{(p)}$ , then according to 5.2 (i) and (0), (ii), the  $y_\alpha = x_\alpha^{(p)}$  satisfy  $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$  and determine the Lie  $p$ -algebra structure.

Let us prove the converse. Since  $\mathfrak{g}$  is a free  $R$ -module, the canonical map  $i : \mathfrak{g} \rightarrow U(\mathfrak{g})$  is injective, according to the Poincaré–Birkhoff–Witt theorem (cf. [BLie], I § 2.7), so one may identify  $\mathfrak{g}$  with a sub- $R$ -module of  $U(\mathfrak{g})$ . Suppose that  $(y_\alpha)$  is a family of elements of  $\mathfrak{g}$  such that  $\text{ad } y_\alpha = (\text{ad } x_\alpha)^p$ . Let  $\pi$  be the map  $r \mapsto r^p$  from  $R$  to  $R$ , and let  $\mathfrak{g} \otimes_\pi R$  be the  $R$ -Lie algebra obtained by the extension of scalars  $\pi : R \rightarrow R$ .<sup>(54)</sup>

There then exists an  $R$ -linear map  $\gamma$  from  $\mathfrak{g} \otimes_\pi R$  to  $U(\mathfrak{g})$  sending  $x_\alpha \otimes_\pi 1$  to  $x_\alpha^p - y_\alpha$ ; moreover, since one has, for every  $x \in \mathfrak{g}$ ,

$$(\text{ad } x_\alpha^p)(x) = (\text{ad } x_\alpha)^p(x) = (\text{ad } y_\alpha)(x),$$

$\gamma$  maps  $\mathfrak{g} \otimes_\pi R$  into the center of  $U(\mathfrak{g})$ . Put, for every  $x \in \mathfrak{g}$ :

$$x^{(p)} = x^p - \gamma(x \otimes_\pi 1).$$

Then, for every  $\alpha$ , one has  $x_\alpha^{(p)} = y_\alpha$ . If  $x = \sum \lambda_\alpha x_\alpha$ , one deduces from 5.1 (ii) (by induction on the number of indices  $\alpha$  such that  $\lambda_\alpha \neq 0$ ) that

$$x^p - \sum_\alpha \lambda_\alpha^p x_\alpha^p \in \mathfrak{g};$$

denoting this element by  $z$ , one then has  $x^{(p)} = \sum \lambda_\alpha^p y_\alpha + z$  and hence  $x^{(p)} \in \mathfrak{g}$ .

It is clear that the map  $x \mapsto x^{(p)}$  satisfies  $(\lambda x)^{(p)} = \lambda^p x^{(p)}$ . Moreover, since  $\gamma(x \otimes_\pi 1)$  is central,  $\text{ad } x^{(p)} = \text{ad } x^p$  and hence, according to the first Jacobson formula (5.1 (i)), one has

$$\text{ad } x^{(p)} = (\text{ad } x)^p.$$

Finally, according to the second Jacobson formula (5.1 (ii)), the map  $x \mapsto x^{(p)}$  satisfies condition (ii) of 5.2. It therefore makes  $\mathfrak{g}$  a Lie  $p$ -algebra. This proves the proposition.

**Proposition 5.3.3.** *Let  $\mathfrak{g}$  be a Lie  $p$ -algebra over  $R$  whose underlying module is free with basis  $(x_\alpha)$ . Then the map  $j : \mathfrak{g} \rightarrow U_p(\mathfrak{g})$  is injective and, if one puts  $z_\alpha = j(x_\alpha)$ ,  $U_p(\mathfrak{g})$  has as basis the monomials*

$$\prod_\alpha z_\alpha^{n_\alpha} \quad \text{where } 0 \leq n_\alpha < p,$$

(the  $n_\alpha$  are assumed to be zero except for finitely many of them; the basis is assumed totally ordered and the products taken in increasing order).

Indeed, let us identify  $\mathfrak{g}$  with a submodule of the enveloping algebra  $U(\mathfrak{g})$  by means of the canonical map  $i$ . For every family  $n = (n_\alpha)$  of natural integers, zero except for finitely many of them, put

$$|n| = \sum_\alpha n_\alpha \quad \text{and} \quad x^n = \prod_\alpha x_\alpha^{n_\alpha}.$$

Writing  $n_\alpha = m_\alpha + p\ell_\alpha$ , with  $0 \leq m_\alpha < p$ , also put

$$T_n = \prod_\alpha x^{m_\alpha} \beta(x_\alpha)^{\ell_\alpha}$$

where  $\beta(x) = x^p - x^{(p)}$  is the map  $\mathfrak{g} \rightarrow U(\mathfrak{g})$  defined in 5.3.1. For every  $r \in \mathbb{N}$ , denote by  $U^r$  the sub- $R$ -module of  $U(\mathfrak{g})$  generated by the  $x^n$  such that  $|n| \leq r$ . Since the graded ring

<sup>(53)</sup>**N.D.E.** In this paragraph the order has been modified, stating the result first and then expanding the proof.

<sup>(54)</sup>**N.D.E.** i.e.  $xr \otimes_\pi 1 = x \otimes_\pi r^p$  and  $r \cdot (x \otimes_\pi 1) = x \otimes_\pi r$ , for  $x \in \mathfrak{g}$ ,  $r \in R$ .

$\bigoplus_r U^r/U^{r-1}$  is commutative (cf. [BLie], I § 2.6), one sees that, for every  $n$ :

$$T_n - \prod_{\alpha} x^{n_{\alpha}} \in U^{|n|-1}.$$

For every  $s \in \mathbf{N}$ , the  $x^n$  such that  $|n| = s$  form, according to the Poincaré–Birkhoff–Witt theorem (loc. cit., § 2.7), a basis of  $U^s/U^{s-1}$ , and therefore the same is true of the  $T_n$  such that  $|n| = s$ .

Thus, as  $s = |n|$  varies, the  $T_n$  form a basis of  $U(\mathfrak{g})$ . Now the kernel  $J$  of the canonical map  $U(\mathfrak{g}) \rightarrow U_p(\mathfrak{g})$  is the left ideal of  $U(\mathfrak{g})$  generated by the central elements  $\beta(x_{\alpha})$  (5.3.1). Consequently, the  $T_n$  such that  $\ell = (\ell_{\alpha}) \neq 0$  form a basis of  $J$ , and the  $T_n$  such that  $n_{\alpha} < p$  for every  $\alpha$  form a basis of  $U_p(\mathfrak{g}) = U(\mathfrak{g})/J$ .

**5.3.3 bis.** — Let  $\mathfrak{g}$  be a Lie  $p$ -algebra over  $R$  and  $f : R \rightarrow R'$  an extension of the base ring. I say that *there exists one and only one Lie  $p$ -algebra structure on the  $R'$ -module  $R' \otimes_R \mathfrak{g}$  such that*

$$(*) \quad [\lambda \otimes x, \mu \otimes y] = \lambda \mu \otimes [x, y] \quad \text{and} \quad (\lambda \otimes x)^{(p)} = \lambda^p \otimes x^{(p)}.$$

It will follow, in particular, that the functor  $\mathfrak{g} \mapsto R' \otimes_R \mathfrak{g}$  is left adjoint to the functor “restriction of scalars from  $R'$  to  $R$ ”.

The uniqueness of the Lie  $p$ -algebra structure defined by  $(*)$  being clear, let us prove existence. When  $\mathfrak{g}$  is free with basis  $(x_{\alpha})$ , there exists, according to 5.3.2, one and only one Lie  $p$ -algebra structure on the Lie algebra  $R' \otimes_R \mathfrak{g}$  such that

$$(1 \otimes x_{\alpha})^{(p)} = 1 \otimes x_{\alpha}^{(p)};$$

this structure is the one we seek.

When  $\mathfrak{g}$  is an arbitrary Lie  $p$ -algebra, there exists a Lie  $p$ -algebra  $L_0$ , free (as an  $R$ -module), and a surjective homomorphism  $q_0 : L_0 \rightarrow \mathfrak{g}$ ; it suffices, for example, to take for  $L_0$  the Lie  $p$ -algebra  $R \otimes_{\mathbf{F}_p} \mathfrak{g}$ , where  $\mathbf{F}_p$  denotes the prime field of characteristic  $p$ , and for  $q_0$  the homomorphism  $\lambda \otimes x \mapsto \lambda x$  ( $\mathfrak{g}$  is free over  $\mathbf{F}_p$ !). The kernel of  $q_0$  is then a  $p$ -ideal of  $L_0$ , that is, an ideal of the Lie algebra  $L_0$  stable under the endomorphism  $x \mapsto x^{(p)}$ ; there is therefore also a Lie  $p$ -algebra  $L_1$ , free (as an  $R$ -module), and a homomorphism  $q_1 : L_1 \rightarrow L_0$  whose image is  $\text{Ker } q_0$ , whence the exact sequence:

$$L_1 \xrightarrow{q_1} L_0 \xrightarrow{q_0} \mathfrak{g} \rightarrow 0.$$

One deduces from this an exact sequence of  $R'$ -Lie algebras

$$R' \otimes_R L_1 \xrightarrow{R' \otimes_R q_1} R' \otimes_R L_0 \xrightarrow{R' \otimes_R q_0} R' \otimes_R \mathfrak{g} \rightarrow 0.$$

Since  $R' \otimes_R q_1$  is manifestly a homomorphism of Lie  $p$ -algebras, the kernel of  $R' \otimes_R q_0$  is a  $p$ -ideal, so that the symbolic  $p$ -th power operation of  $R' \otimes_R L_0$  induces, by passage to the quotient, a map from  $R' \otimes_R \mathfrak{g}$  to  $R' \otimes_R \mathfrak{g}$  (use formula (ii) of 5.2.); the latter equips  $R' \otimes_R \mathfrak{g}$  with the desired Lie  $p$ -algebra structure.

**5.3.4.** — The canonical map  $j_{\mathfrak{g}} : \mathfrak{g} \rightarrow U_p(\mathfrak{g})$  induces, for every extension  $R \rightarrow R'$  of the base ring, a homomorphism

$$R' \otimes_R j_{\mathfrak{g}} : R' \otimes_R \mathfrak{g} \rightarrow R' \otimes_R U_p(\mathfrak{g}),$$

whence a homomorphism

$$h : U_p(R' \otimes_R \mathfrak{g}) \rightarrow R' \otimes_R U_p(\mathfrak{g})$$

such that  $h \circ j_{R' \otimes_R \mathfrak{g}} = R' \otimes_R j_{\mathfrak{g}}$ . It clearly follows from the universal properties of  $R' \otimes_R \mathfrak{g}$  and of the restricted enveloping algebra that  $h$  is an isomorphism, which will allow us to identify  $U_p(R' \otimes_R \mathfrak{g})$  with  $R' \otimes_R U_p(\mathfrak{g})$ .

In particular, if  $r$  is an element of  $R$  and  $R'$  is the localized ring  $R_r$ , one sees that  $\mathfrak{g}_r = R_r \otimes_R \mathfrak{g}$  is canonically equipped with a Lie  $p$ -algebra structure over  $R_r$ , so that the sheaf  $\tilde{\mathfrak{g}}$  on  $\text{Spec } R$  is a quasi-coherent Lie  $p$ -algebra on  $\text{Spec } R$ . Moreover, the restricted

enveloping algebra  $U_p^{Rr}(\mathfrak{g}_r)$  identifies with  $U_p^R(\mathfrak{g})_r$ , so that the sheaf associated to the presheaf  $V \mapsto U_p(\Gamma(V, \mathfrak{g}))$  is quasi-coherent.

**Definition.** More generally, if  $S$  is a scheme of characteristic  $p$  and  $\mathcal{G}$  a quasi-coherent Lie  $p$ -algebra over  $\mathcal{O}_S$ , the sheaf associated to the presheaf  $V \mapsto U_p(\Gamma(V, \mathcal{G}))$  is quasi-coherent; it will be denoted by  $\mathcal{U}_p(\mathcal{G})$  and called the *restricted enveloping algebra of  $\mathcal{G}$* . If  $V$  is affine,  $U_p(\Gamma(V, \mathcal{G}))$  identifies with  $\Gamma(V, \mathcal{U}_p(\mathcal{G}))$ .

**5.4.** — The universal character of  $U_p(\mathfrak{g})$  implies that  $U_p(\mathfrak{g})$  is functorial in  $\mathfrak{g}$ : every homomorphism of Lie  $p$ -algebras  $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$  induces one and only one homomorphism of unital algebras  $U_p(\phi)$  such that  $j_{\mathfrak{h}} \circ \phi = U_p(\phi) \circ j_{\mathfrak{g}}$ . Here are a few examples:

a) If  $\mathfrak{h} = 0$ ,  $U_p(\mathfrak{h})$  identifies with the base ring and  $U_p(\phi)$  is an algebra homomorphism  $\varepsilon_{\mathfrak{g}} : U_p(\mathfrak{g}) \rightarrow R$  called the *augmentation*.

b) Now take for  $\mathfrak{h}$  the algebra  $\mathfrak{g}^\circ$  opposite to  $\mathfrak{g}$ , i.e.  $\mathfrak{g}^\circ$  has the same underlying module as  $\mathfrak{g}$ , the same symbolic  $p$ -th power, the bracket of two elements in  $\mathfrak{g}^\circ$  being the negative of the bracket in  $\mathfrak{g}$ . It is clear that we may identify  $U_p(\mathfrak{g}^\circ)$  with the algebra opposite to  $U_p(\mathfrak{g})$ . Moreover, the isomorphism  $x \mapsto -x$  from  $\mathfrak{g}$  to  $\mathfrak{g}^\circ$  induces an isomorphism  $c_{\mathfrak{g}}$  from  $U_p(\mathfrak{g})$  to  $U_p(\mathfrak{g}^\circ) \simeq U_p(\mathfrak{g})^\circ$ . One says that  $c_{\mathfrak{g}}$  is the *antipode* of  $U_p(\mathfrak{g})$ .

c) Finally, let  $\mathfrak{f}$  and  $\mathfrak{g}$  be two Lie  $p$ -algebras and  $\mathfrak{h}$  the *product Lie  $p$ -algebra*  $\mathfrak{f} \times \mathfrak{g}$  having as underlying  $R$ -module the direct product  $\mathfrak{f} \times \mathfrak{g}$ , the bracket and symbolic  $p$ -th power being defined by the formulas

$$[(x, y), (x', y')] = ([x, x'], [y, y']) \quad \text{and} \quad (x, y)^{(p)} = (x^{(p)}, y^{(p)}).$$

If  $h_1 : \mathfrak{f} \rightarrow \mathfrak{k}$  and  $h_2 : \mathfrak{g} \rightarrow \mathfrak{k}$  are two homomorphisms of Lie  $p$ -algebras such that  $[h_1(x), h_2(y)] = 0$  for every  $x$  of  $\mathfrak{f}$  and every  $y$  of  $\mathfrak{g}$ , the map  $h_1 + h_2 : (x, y) \mapsto h_1(x) + h_2(y)$  is a homomorphism of Lie  $p$ -algebras; conversely, every homomorphism from  $\mathfrak{f} \times \mathfrak{g}$  to  $\mathfrak{k}$  is of this type, which allows one to characterize  $\mathfrak{f} \times \mathfrak{g}$  as the solution to a universal problem. For example, the maps

$$h_1 : x \mapsto i_{\mathfrak{f}}(x) \otimes 1 \quad \text{and} \quad h_2 : y \mapsto 1 \otimes i_{\mathfrak{g}}(y)$$

induce a homomorphism  $h_1 + h_2$  from  $\mathfrak{f} \times \mathfrak{g}$  to the Lie  $p$ -algebra underlying  $U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g})$ . It follows from the universal characters of  $\mathfrak{f} \times \mathfrak{g}$  and of the restricted enveloping algebras that  $h_1 + h_2$  extends to an isomorphism:

$$\varphi : U_p(\mathfrak{f} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{f}) \otimes U_p(\mathfrak{g}).$$

**Definition.** If  $\mathfrak{f} = \mathfrak{g}$ , the diagonal map  $\delta : x \mapsto (x, x)$  from  $\mathfrak{g}$  to  $\mathfrak{g} \times \mathfrak{g}$  induces a homomorphism from  $U_p(\mathfrak{g})$  to  $U_p(\mathfrak{g} \times \mathfrak{g})$ . We shall denote by  $\Delta_{\mathfrak{g}}$  the composite of this homomorphism with the isomorphism  $\varphi : U_p(\mathfrak{g} \times \mathfrak{g}) \xrightarrow{\sim} U_p(\mathfrak{g}) \otimes U_p(\mathfrak{g})$ .<sup>(55)</sup>

One then sees easily that  $\Delta_{\mathfrak{g}}$  and the multiplication of the algebra  $U_p(\mathfrak{g})$  make  $U_p(\mathfrak{g})$  an  $R$ -coalgebra in groups (cf. 3.2) having  $\varepsilon_{\mathfrak{g}}$  as augmentation and  $c_{\mathfrak{g}}$  as antipode.

**5.5.** — <sup>(56)</sup> Now let  $S$  be a scheme of characteristic  $p$ . First, if  $\mathcal{U}$  is a  $\mathcal{O}_S$ -coalgebra in groups and  $G$  the  $S$ -group functor  $\text{Spec}^* \mathcal{U}$ , we saw (3.2.3) that, for every  $T \rightarrow S$ ,  $(\underline{\text{Lie}} G)(T)$  is the Lie subalgebra of  $\Gamma(T, \mathcal{U}_T)$  consisting of primitive elements. Now, if  $x$  is such an element, one has  $\Delta(x^p) = x^p \otimes 1 + 1 \otimes x^p$  (since  $\binom{p}{i} = 0 \pmod p$  for  $0 < i < p$ ), i.e.  $x^p$  is again a primitive element. It follows, according to 5.1 and 5.2, that the map  $x \mapsto x^p$  equips  $(\underline{\text{Lie}} G)(T)$  with an  $\mathcal{O}(T)$ -Lie  $p$ -algebra structure.

<sup>(55)</sup>**N.D.E.** i.e.  $\Delta_{\mathfrak{g}}(x) = x \otimes 1 + 1 \otimes x$  for every  $x \in \mathfrak{g}$ ; in particular, the comultiplication  $\Delta_{\mathfrak{g}}$  is indeed cocommutative. . .

<sup>(56)</sup>**N.D.E.** Section 5.4.1 of the original has been transformed into this § 5.5: on the one hand, proposition 5.5.1 brings together the results of Section 5 and proposition 3.2.3, and contains lemma 7.3 of the original; on the other hand, the proof of 5.5.3 (ii) takes up and expands that of the implication (i)  $\Rightarrow$  (ii) in theorem 7.4 below.

Now let  $\mathcal{L}$  be a  $\mathcal{O}_S$ -Lie  $p$ -algebra, quasi-coherent over  $\mathcal{O}_S$ . As  $V$  ranges over the open subsets of  $S$ , the structures of coalgebras in groups previously defined on the sets  $U_p(\Gamma(V, \mathcal{L}))$  induce on the associated sheaf, i.e. on the restricted enveloping algebra  $\mathcal{U}_p(\mathcal{L})$ , a structure of  $\mathcal{O}_S$ -coalgebra in groups. Moreover, for every  $S$ -scheme  $T$ , there is an isomorphism  $\mathcal{U}_p(\mathcal{L}_T) \xrightarrow{\sim} \mathcal{U}_p(\mathcal{L})_T$ .

Denote by  $\text{Prim } \mathcal{U}_p(\mathcal{L})$  the subpresheaf of  $\mathcal{U}_p(\mathcal{L})$  associating to every open subset  $V$  the set of primitive elements of  $\mathcal{U}_p(\mathcal{L})(V)$ ; one sees easily that this is a sheaf. As  $V$  ranges over the open subsets of  $S$ , the composite maps

$$\Gamma(V, \mathcal{L}) \xrightarrow{j} \text{Prim } U_p(\Gamma(V, \mathcal{L})) \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})(V)$$

define a morphism  $\mathcal{L} \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})$ , which we shall still denote by  $j$  or  $j_{\mathcal{L}}$ , and the latter again defines a morphism of  $\mathbf{O}_S$ -Lie  $p$ -algebras  $\mathbf{W}(\mathcal{L}) \longrightarrow \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L}))$  (cf. 3.2.3).

**Proposition 5.5.1.** *Let  $\mathcal{L}$  be a  $\mathcal{O}_S$ -Lie  $p$ -algebra, locally free as an  $\mathcal{O}_S$ -module. Then  $j_{\mathcal{L}}$  induces an isomorphism of  $\mathbf{O}_S$ -Lie  $p$ -algebras:*

$$\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L})).$$

PROOF. Let  $T$  be an  $S$ -scheme; taking account of the identification  $\mathcal{U}_p(\mathcal{L}_T) = \mathcal{U}_p(\mathcal{L})_T$ , one must show that the map  $\Gamma(T, \mathcal{L}_T) \longrightarrow \text{Prim } \Gamma(T, \mathcal{U}_p(\mathcal{L}_T))$  is bijective. Replacing  $S$  by  $T$ , one is reduced to the case where  $T = S$ , and it then suffices to show that the sheaf morphism  $j_{\mathcal{L}} : \mathcal{L} \longrightarrow \text{Prim } \mathcal{U}_p(\mathcal{L})$  is an isomorphism. This question being local on  $S$ , we may suppose that  $S$  is affine with ring  $R$  and that  $\mathcal{L}$  is the sheaf associated to an  $R$ -Lie  $p$ -algebra  $L$  with basis  $(x_{\alpha})$ . As in 5.3.3, denote by  $z_{\alpha}$  the image of  $x_{\alpha}$  in  $U = U_p(L)$ , and, for every family  $n = (n_{\alpha})$  of integers between 0 and  $p - 1$ , zero except for finitely many of them, denote by  $z^{(n)}$  the product

$$\prod_{\alpha} \frac{z_{\alpha}^{n_{\alpha}}}{n_{\alpha}!}$$

(the basis  $(x_{\alpha})$  is assumed totally ordered and the products taken in increasing order).

Since  $\Delta(z_{\alpha}) = z_{\alpha} \otimes 1 + 1 \otimes z_{\alpha}$ , one sees easily that

$$\Delta(z^{(n)}) = \sum_r z^{(n-r)} \otimes z^{(r)}$$

the sum being taken over the (finite!) set of  $r$  such that  $0 \leq r_{\alpha} \leq n_{\alpha}$  for every  $\alpha$ . Since the  $z^{(n)}$  (resp. the  $z^{(n)} \otimes z^{(m)}$ ) form a basis of  $U$  (resp. of  $U \otimes U$ ), one deduces that an element  $u$  of  $U$  satisfies  $\Delta(u) = u \otimes 1 + 1 \otimes u$  if and only if  $u$  is a linear combination of the  $z_{\alpha}$ . This proves 5.5.1.  $\square$

**Remark 5.5.2.** Recall (cf. 3.2.2 and 3.2.3) that the  $S$ -group functor  $G = \text{Spec}^* \mathcal{U}_p(\mathcal{L})$  is very good and that  $\underline{\text{Lie}}(G) = \text{Prim } \mathbf{W}(\mathcal{U}_p(\mathcal{L}))$ . The preceding proposition therefore means that  $j_{\mathcal{L}}$  induces an isomorphism  $\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \underline{\text{Lie}}(G)$ .

If one assumes, in addition, that  $\mathcal{L}$  is a locally free  $\mathcal{O}_S$ -module of finite rank, then  $\mathcal{U}_p(\mathcal{L})$  is finite locally free over  $\mathcal{O}_S$ , according to 5.3.3, hence  $\text{Spec}^* \mathcal{U}_p(\mathcal{L})$  is represented by the  $S$ -group  $G_p(\mathcal{L}) = \text{Spec } \mathcal{U}_p(\mathcal{L})^*$  (cf. 3.2.2.1), and one obtains the following more precise proposition:

**Proposition 5.5.3.** *Let  $\mathcal{L}$  be a  $\mathcal{O}_S$ -Lie  $p$ -algebra, locally free of finite rank as an  $\mathcal{O}_S$ -module, let  $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$ , and let  $G = G_p(\mathcal{L})$  be the affine  $S$ -group  $\text{Spec } \mathcal{A}$ .*

- (i)  *$j_{\mathcal{L}}$  induces an isomorphism  $\mathbf{W}(\mathcal{L}) \xrightarrow{\sim} \underline{\text{Lie}}(G)$  of  $\mathbf{O}_S$ -Lie  $p$ -algebras.*
- (ii) *Let  $\mathcal{I}$  be the augmentation ideal of  $\mathcal{A}$ , and  $\omega_G = \mathcal{I}/\mathcal{I}^2$  (cf. II, 4.11.4). Then  $\omega_G$  identifies with  $\mathcal{L}^* = \mathcal{H}om_{\mathcal{O}_S}(\mathcal{L}, \mathcal{O}_S)$ , hence is a locally free  $\mathcal{O}_S$ -module of finite rank (and one has  $\omega_{G/S}^* = \mathcal{L}$ ).*

PROOF. Since (i) follows from 5.5.2, let us prove (ii). Denote by  $\eta_{\mathcal{U}}$  and  $\varepsilon_{\mathcal{U}}$  the unit section and augmentation of  $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ , by  $\eta_{\mathcal{A}}$  and  $\varepsilon_{\mathcal{A}}$  those of  $\mathcal{A}$ , and put  $\mathcal{J} = \text{Ker } \varepsilon_{\mathcal{U}}$ . One then has:

$$(1) \quad \mathcal{U} = \eta_{\mathcal{U}}(\mathcal{O}_S) \oplus \mathcal{J}.$$

Let  $\delta$  be the morphism defined by the diagram below, where  $\tau$  and  $\pi$  denote the inclusion and projection deduced from decomposition (1):

$$\begin{array}{ccc} \mathcal{J} & \xrightarrow{\delta} & \mathcal{J} \otimes \mathcal{J} \\ \tau \downarrow & & \uparrow \pi \\ \mathcal{U} & \xrightarrow{\Delta} & \mathcal{U} \otimes \mathcal{U} \end{array}$$

then there is an exact sequence:

$$(*) \quad 0 \longrightarrow \mathcal{L}^* \xrightarrow{j_{\mathcal{L}}} \mathcal{J} \xrightarrow{\delta} \mathcal{J} \otimes \mathcal{J}.$$

Moreover, according to 5.3.3, the  $\mathcal{O}_S$ -module  $\mathcal{J}/\mathcal{L}$  is locally free and, according to 5.5.1, sequence  $(*)$  remains exact after every base change. Thus, according to [BAC], II § 3, prop. 6,  $\delta$  induces an isomorphism from  $\mathcal{J}/\mathcal{L}$  onto a submodule  $\mathcal{D}$  locally a direct factor of  $\mathcal{J} \otimes \mathcal{J}$ . It follows that  $(*)$  gives by duality the exact sequence:

$$(**) \quad 0 \longleftarrow \mathcal{L} \xleftarrow{t_{j_{\mathcal{L}}}} \mathcal{J} \xleftarrow{t_{\delta}} \mathcal{J} \otimes \mathcal{J}.$$

Now decomposition (1) corresponds by duality to the decomposition:

$$(2) \quad \mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{J}$$

and the transpose of  $\Delta$  is the multiplication  $m_{\mathcal{A}} : \mathcal{A} \otimes \mathcal{A} \longrightarrow \mathcal{A}$ . Since  $\mathcal{J}$  is an ideal of  $\mathcal{A}$ ,  $m_{\mathcal{A}}$  sends  $\mathcal{J} \otimes \mathcal{J}$  into  $\mathcal{J}$ ; more precisely, taking account of decomposition (2), there is a commutative square

$$\begin{array}{ccc} \mathcal{J} & \xleftarrow{m'} & \mathcal{J} \otimes \mathcal{J} \\ \uparrow t_{\tau} & & \downarrow t_{\pi} \\ \mathcal{A} & \xleftarrow{m_{\mathcal{A}}} & \mathcal{A} \otimes \mathcal{A} \end{array}$$

which shows that the restriction  $m'$  of  $m_{\mathcal{A}}$  to  $\mathcal{J} \otimes \mathcal{J}$  is the transpose of  $\delta$ . Exact sequence  $(**)$  then gives  $\mathcal{J}/\mathcal{J}^2 \simeq \mathcal{L}^*$ , and the proposition follows.  $\square$

## 6. Lie $p$ -algebra of an $S$ -group scheme

Let  $S$  be a scheme of characteristic  $p > 0$ . In paragraph 5.5 we associated to every quasi-coherent  $\mathcal{O}_S$ -Lie  $p$ -algebra  $\mathcal{L}$  an  $S$ -group functor  $G_p(\mathcal{L}) = \text{Spec}^* \mathcal{U}_p(\mathcal{L})$ . We shall now see that, for every  $S$ -group scheme  $G$ , the  $\mathbf{O}_S$ -Lie algebra  $\underline{\text{Lie}}(G/S)$  defined in II 4.11 is naturally equipped with a  $\mathbf{O}_S$ -Lie  $p$ -algebra structure.

**6.1.** — First identify  $\underline{\text{Lie}}(G/S)(S)$  and  $\underline{\text{Lie}}(\text{Aut } G/S)(S)$  respectively with Lie subalgebras of  $U(G)$  and  $\text{Dif}_{G/S}$  by means of the injections  $\alpha$  and  $\beta$  of 2.5;  $\underline{\text{Lie}}(\text{Aut } G/S)(S)$  is therefore identified with the  $\Gamma(\mathcal{O}_S)$ -Lie algebra of  $S$ -derivations of  $\mathcal{O}_G$ . According to 5.2, the latter is a Lie  $p$ -subalgebra of  $\text{Dif}_{G/S}$ .

On the other hand, the image of  $L = \underline{\text{Lie}}(G/S)(S)$  under the injective algebra morphism  $\ell : U(G) \longrightarrow \text{Dif}_{G/S}$ ,  $d \mapsto {}^G d$ , consists of the left-invariant derivations (cf. 2.2, N.D.E. (17), 2.4 and 2.5). If  $x$  belongs to  $L$ ,  $\ell(x)^p$  is none other than  $\ell(x^p)$ , according to loc. cit. Since  $\ell(x)^p$  is again a derivation, one sees that  $x^p$  belongs to  $\underline{\text{Lie}}(G/S)(S)$ . Thus:<sup>(57)</sup>

<sup>(57)</sup>N.D.E. One can also show directly (without using  $\text{Dif}_{G/S}$ ) that the Lie algebra of derivations of  $G$  at the origin (isomorphic to  $\underline{\text{Lie}}(G/S)(S)$  according to 2.5) is stable under raising to the power  $p$  in  $U(G)$ : this is done in 6.2 below.

$\underline{\text{Lie}}(G/S)(S)$  is a Lie  $p$ -subalgebra of the infinitesimal algebra  $U(G)$ .

**6.1.1.** — Let  $\phi : G \rightarrow H$  be a homomorphism of  $S$ -group schemes. It is clear that the homomorphisms  $\underline{\text{Lie}}(\phi/S)(S)$  and  $U(\phi)$  are compatible with the identifications of  $\underline{\text{Lie}}(G/S)(S)$  and  $\underline{\text{Lie}}(H/S)(S)$  with Lie  $p$ -subalgebras of  $U(G)$  and  $U(H)$ . Since  $U(\phi)$  is an algebra homomorphism, one therefore sees that  $\underline{\text{Lie}}(\phi/S)(S)$  is a homomorphism of Lie  $p$ -algebras.

Similarly, if  $s : T \rightarrow S$  is a base change, the map from  $\underline{\text{Lie}}(G/S)(S)$  to  $\underline{\text{Lie}}(G/S)(T)$  induced by  $s$  is a homomorphism of Lie  $p$ -algebras. One may express this by saying that the functor  $\underline{\text{Lie}}(G/S)$  is equipped with a  $\mathbf{O}_S$ -Lie  $p$ -algebra structure. In particular, as  $T$  ranges over the open subsets of  $S$ , one sees that the sheaf  $\mathcal{L}\text{ie}(G/S)$  is equipped with a  $\mathcal{O}_S$ -Lie  $p$ -algebra structure.

**6.2.** — Following an idea of Demazure, we shall now generalize what precedes to certain  $S$ -group functors which are not necessarily representable. For this purpose, we shall first give another definition of the symbolic  $p$ -th power in the Lie algebra of an  $S$ -group scheme  $G$ .

Let  $D$  be a derivation of  $G$  at the origin;<sup>(58)</sup> according to 1.2.1,  $D$  is the composite of the  $S$ -derivation  $\delta = (\tau, \partial_t)$  of the zero section  $\tau : S \rightarrow I_S$  and a morphism  $x : I_S \rightarrow G$  such that  $x \circ \tau = \varepsilon$  (i.e.  $x \in \underline{\text{Lie}}(G/S)(S)$ ). According to the definition which we gave in 2.1,  $D^p$  is the following composite deviation:

$$\begin{aligned} S &\simeq \underbrace{S \times S \times \cdots \times S}_p \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \\ &\xrightarrow{x \times \cdots \times x} G \times \cdots \times G \xrightarrow{m^{(p)}} G \end{aligned}$$

where  $m^{(p)}$  is the morphism induced by the multiplication  $m : G \times G \rightarrow G$ . Since  $I_S \times \cdots \times I_S$  is affine over  $S$  and has as affine algebra  $\mathcal{B} = \mathcal{O}_S[d_1, \dots, d_p]/(d_1^2, \dots, d_p^2)$ , the deviation  $\delta \times \cdots \times \delta$  is defined by the morphism of  $\mathcal{O}_S$ -modules

$$\phi : \mathcal{B} \rightarrow \mathcal{O}_S$$

sending the monomial  $d_1 d_2 \cdots d_p$  to 1 and the other monomials  $d_{i_1} \cdots d_{i_r}$ , for  $0 \leq r < p$ , to 0. On the other hand, if  $\text{pr}_i$  denotes the projection from  $I_S^p$  onto the  $i$ -th factor and  $x_i$  is the image in  $G(I_S^p)$  of  $x$  under  $G(\text{pr}_i)$ , then the composite morphism  $m^{(p)} \circ (x \times \cdots \times x)$  is none other than the product  $x_1 x_2 \cdots x_p$ . Consequently,  $D^p$  is also the following composite deviation:

$$S \xrightarrow{\delta \times \cdots \times \delta} I_S \times \cdots \times I_S \xrightarrow{x_1 x_2 \cdots x_p} G.$$

This description allows us to prove again that  $D^p$  is a derivation of  $G$  at the origin. Indeed, since  $G$  is a very good group (II 4.11), the images  $G(\text{pr}_1)(x)$  and  $G(\text{pr}_2)(x)$  of  $x$  in  $G(I_S \times I_S)$  commute with one another. It follows that the elements  $x_i$  of  $G(I_S^p)$  commute pairwise, and therefore, for every permutation  $\gamma$  of the factors of  $I_S^p$ , one has  $(x_1 \cdots x_p) \circ \gamma = x_1 \cdots x_p$ ; it follows that  $x_1 \cdots x_p$  factors through the canonical projection from  $I_S^p$  to the symmetric product  $\Sigma^p I_S$  (cf. 4.2).

The symmetric product  $\Sigma^p I_S$  has as affine algebra the subalgebra  $\mathcal{A}$  of  $\mathcal{B}$  having as basis over  $\mathcal{O}_S$  the elementary symmetric functions  $1 = \sigma_0, \sigma_1, \dots, \sigma_p$  of  $d_1, \dots, d_p$ . Denote by  $\kappa$  the inclusion  $\mathcal{A} \hookrightarrow \mathcal{B}$  and by  $\pi$  the morphism of  $\mathcal{O}_S$ -algebras  $\mathcal{A} \rightarrow \mathcal{O}_S[t]/(t^2)$  vanishing on  $\sigma_1, \dots, \sigma_{p-1}$  and sending  $\sigma_p$  to  $t$ ; then one has  $\phi \circ \kappa = \partial_t \circ \pi$  (recall that  $\partial_t : \mathcal{O}_S[t]/(t^2) \rightarrow \mathcal{O}_S$  is the morphism of  $\mathcal{O}_S$ -modules vanishing on 1 and sending  $t$  to 1). Consequently, denoting

<sup>(58)</sup>**N.D.E.** The original has been expanded in what follows.

by  $i$  the closed immersion  $I_S \hookrightarrow \Sigma^p I_S$  defined by  $\pi$ , there is a commutative diagram:

$$\begin{array}{ccccc}
 & & D^p & & \\
 & \curvearrowright & & \curvearrowright & \\
 S & \xrightarrow{\delta \times \dots \times \delta} & I_S^p & \xrightarrow{x_1 \dots x_p} & G \\
 \downarrow \delta & & \downarrow \text{can.} & & \parallel \\
 I_S & \xrightarrow{i} & \Sigma^p I_S & \longrightarrow & G \\
 & \curvearrowright & & \curvearrowright & \\
 & & y & & 
 \end{array}$$

which shows that  $D^p$  is of the form  $y \circ \delta$ , and thus is indeed a derivation of  $G$  at the origin.

**6.3.** — Let  $\mathcal{S}_p$  be the symmetric group of order  $p$ , and  $I_S^p \times \mathcal{S}_p$  the direct sum of a family of copies of  $I_S^p$  indexed by  $\mathcal{S}_p$ . We denote by  $\pi : I_S^p \times \mathcal{S}_p \rightarrow I_S^p$  the canonical projection, and by

$$\mu : I_S^p \times \mathcal{S}_p \rightarrow I_S^p$$

the morphism defining the action of  $\mathcal{S}_p$  on  $I_S^p$  (i.e. if  $\tau$  is an element of  $\mathcal{S}_p$ , the restriction of  $\mu$  to  $I_S^p \times \tau$  has  $\text{pr}_{\tau(j)}$  as  $j$ -th component). This being so, we put forward the following definition:

**Definition.** A functor  $X : (\mathbf{Sch}/S)^\circ \rightarrow (\mathbf{Ens})$  satisfies condition (F) if:

- a)  $X$  transforms finite direct sums into direct products,
- b) for every  $S$ -scheme  $T$ , the sequence below is exact:

$$X(T \times \Sigma^p I_S) \longrightarrow X(T \times I_S^p) \xrightarrow[X(\text{id}_T \times \mu)]{X(\text{id}_T \times \pi)} X(T \times I_S^p \times \mathcal{S}_p).$$

Every  $S$ -scheme satisfies (F); if  $\mathcal{F}$  is an  $\mathcal{O}_S$ -module,  $\mathbf{W}(\mathcal{F})$  satisfies (F); every projective limit of functors satisfying (F) also satisfies (F); if  $Y$  satisfies (F) and  $X$  is an arbitrary  $S$ -functor,  $\underline{\text{Hom}}_S(X, Y)$  satisfies (F).

Let  $X$  be a *very good group* (II 4.10) satisfying condition (F). Denoting by  $x : I_S \rightarrow X$  a morphism extending the unit section of  $X$  and using again the notation of 6.2, one sees as above that  $x_1 \dots x_p : I_S^p \rightarrow X$  factors through  $\Sigma^p I_S$ :

$$\begin{array}{ccc}
 I_S^p & \xrightarrow{x_1 \dots x_p} & X \\
 \searrow \text{can.} & & \nearrow \Sigma^p(x) \\
 & \Sigma^p I_S & 
 \end{array}$$

and defines by composition a morphism

$$x^{(p)} : I_S \xrightarrow{i} \Sigma^p I_S \xrightarrow{\Sigma^p(x)} X$$

which we shall call the *symbolic  $p$ -th power* of  $x$ .

The endomorphism  $x \mapsto x^{(p)}$  of  $\underline{\text{Lie}}(G/S)(S)$  is clearly compatible with base changes and functorial in  $G$ . It would be interesting to know for which  $G$  this endomorphism makes  $\underline{\text{Lie}}(G/S)(S)$  a Lie  $p$ -algebra.

**6.4.** — The last definition of symbolic  $p$ -th power, which we have just given, is particularly well suited to calculation. Here are a few examples:

**6.4.1.** — Let  $M$  be an “abstract” abelian group and  $D_S(M)$  the diagonalizable  $S$ -group of type  $M$  (I 4.4.2). For every  $S$ -scheme  $T$ , one therefore has

$$D_S(M)(T) = \text{Hom}_{(\text{Ab})}(M, \mathcal{O}(T)^\times).$$

Let  $x$  be an element of  $\underline{\mathrm{Lie}}(D_S(M)/S)(S)$ , that is, a homomorphism of abelian groups

$$M \xrightarrow{x} \Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$$

of the form  $m \mapsto 1 + d\xi(m)$ , where  $\xi \in \mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$ . With the notation of 6.2 and 6.3, the product  $x_1 \cdots x_p$  associates to an element  $m$  of  $M$  the expression

$$(1 + d_1\xi(m)) \cdots (1 + d_p\xi(m))$$

that is,  $1 + \sigma_1\xi(m) + \sigma_2\xi(m)^2 + \cdots + \sigma_p\xi(m)^p$ .

This expression does indeed belong to  $\mathcal{O}(\Sigma^p I_S)$ . Projecting this into  $\mathcal{O}(S)[d]/(d^2)$  by annihilating  $\sigma_1, \dots, \sigma_{p-1}$  and sending  $\sigma_p$  to  $d$ , one sees that  $x^{(p)}$  is the following homomorphism from  $M$  to  $\Gamma(S, \mathcal{O}_S + d\mathcal{O}_S)^\times$ :

$$m \mapsto 1 + d\xi(m)^p.$$

To summarize, if one identifies  $\underline{\mathrm{Lie}}(D_S(M)/S)(S)$  with  $\mathrm{Hom}_{(\mathrm{Ab})}(M, \mathcal{O}(S))$  as in II 5.1, the symbolic  $p$ -th power associates to  $\xi$  the homomorphism  $\xi^{(p)} : m \mapsto \xi(m)^p$ .

**6.4.2.** — Let  $\mathcal{F}$  be an  $\mathcal{O}_S$ -module and  $G$  the  $S$ -functor in abelian groups  $\mathbf{W}(\mathcal{F})$  (cf. I, 4.6). Let  $y$  be an element of  $\mathbf{W}(\mathcal{F})(S) = \Gamma(S, \mathcal{F})$  and  $y'$  its image in  $\mathbf{W}(\mathcal{F})(I_S)$  under  $\mathbf{W}(\mathcal{F})(I_S \rightarrow S)$ .

One knows (cf. II, 4.4.2 and 4.5.1) that the map  $y \mapsto dy'$  is an isomorphism of  $\mathcal{O}(S)$ -modules from  $\mathbf{W}(\mathcal{F})(S)$  onto  $\underline{\mathrm{Lie}}(\mathbf{W}(\mathcal{F})/S)(S)$ . If one puts  $x = dy'$ , the quantity  $x_i$  of 6.2 is none other than  $d_i y''$ , where  $y''$  denotes the canonical image of  $y'^{(59)}$  in  $\mathbf{W}(\mathcal{F})(I_S^p)$ . Consequently, the product  $x_1 \cdots x_p$  is here equal to

$$x_1 + \cdots + x_p = (d_1 + \cdots + d_p)y'' = \sigma_1 y''$$

and belongs to  $\mathbf{W}(\mathcal{F})(\Sigma^p I_S)$ . Since the homomorphism  $\mathcal{O}(\Sigma^p I_S) \rightarrow \mathcal{O}(I_S)$  defining the morphism  $i$  of 6.1 annihilates  $\sigma_1$ , one sees that  $x^{(p)}$  is zero. Thus:

*For every  $\mathcal{O}_S$ -module  $\mathcal{F}$ , the operation  $x \mapsto x^{(p)}$  in  $\underline{\mathrm{Lie}}\mathbf{W}(\mathcal{F})$  is zero.*

**6.4.3.** — Let  $X$  be an  $S$ -scheme,  $G$  the  $S$ -group functor  $\underline{\mathrm{Aut}}_S X$ , and  $D$  an  $S$ -derivation of the structure sheaf  $\mathcal{O}_X$ . According to 1.5,  $D$  may be identified with an  $I_S$ -automorphism  $x$  of  $X_{I_S}$  inducing the identity on  $X$ , which may be described as follows. If  $f$  is a section of  $\mathcal{O}_S[d]/(d^2)$  of the form  $a + bd$ , put  $D_{I_S} f = Da + (Db)d$ ; in other words,  $D_{I_S}$  is deduced from  $D$  by the base change  $I_S \rightarrow S$ ; then the automorphism in question of  $X_{I_S}$  is associated to the endomorphism  $f \mapsto f + (D_{I_S} f)d = a + (D(a) + b)d$  of  $\mathcal{O}_S[d]/(d^2)$ .

Similarly, let  $D_{I_S^p}$  be the differential operator of  $X_{I_S^p}$  deduced from  $D$  by the base change  $I_S^p \rightarrow S$ . With the notation of 6.2, the automorphism  $x_i$  of  $X_{I_S^p}$  is then associated to the endomorphism  $f \mapsto f + (D_{I_S^p} f)d_i$  of  $\mathcal{O}_S[d_1, \dots, d_p]/(d_1^2, \dots, d_p^2)$ . The product  $x_1 \cdots x_p$  is therefore associated to the endomorphism

$$(1 + d_1 D_{I_S^p})(1 + d_2 D_{I_S^p}) \cdots (1 + d_p D_{I_S^p})$$

that is, to  $1 + \sigma_1 D_{I_S^p} + \sigma_2 D_{I_S^p}^2 + \cdots + \sigma_p D_{I_S^p}^p$ . The coefficient of  $\sigma_p$  is  $D_{I_S^p}^p$ , which means that the Lie algebra isomorphism  $\mathrm{Dér}_S(\mathcal{O}_X) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S X)$ ,  $D \mapsto x$  (cf. 1.5), is also an isomorphism of Lie  $p$ -algebras.

**6.4.4.** — Using the same method, one sees that, for every  $\mathcal{O}_S$ -module  $\mathcal{F}$ , the isomorphism

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathbf{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})/S)(S) \xrightarrow{\sim} (\underline{\mathrm{End}}_{\mathbf{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F}))(S)$$

of Lie algebras (cf. II 4.8) is also an isomorphism of Lie  $p$ -algebras.

**6.4.5.** — <sup>(60)</sup> Let  $\mathcal{U}$  be a  $\mathcal{O}_S$ -coalgebra in groups and  $G$  the  $S$ -group functor  $\mathrm{Spec}^* \mathcal{U}$ ; suppose that  $G$  is *representable*. In this case, for every  $T \rightarrow S$ , two Lie  $p$ -algebra structures have

<sup>(59)</sup>**N.D.E.**  $x$  has been corrected to  $y'$ .

<sup>(60)</sup>**N.D.E.** The numbering 6.4.5 has been added, and the original expanded.

been defined in 5.5 and 6.1.1 on  $L(T) = \underline{\mathrm{Lie}}(G)(T)$ . Since one has a commutative diagram

$$\begin{array}{ccc} L(T) & \xrightarrow{\tau} & \Gamma(T, \mathcal{U}_T) \\ \alpha \downarrow & \searrow i & \uparrow \psi \\ U(G_T) & \xleftarrow{\phi} & U(L(T)) \end{array}$$

where  $U(L(T))$  is the enveloping algebra of  $L(T)$  and  $\phi, \psi$  the algebra morphisms induced by  $\alpha, \tau$ , one sees that the two Lie  $p$ -algebra structures coincide: if one identifies  $x \in L(T)$  with its image in  $U(G_T)$  (resp.  $\Gamma(T, \mathcal{U}_T)$ ), then  $x^{(p)}$  is the image of the element  $x^p$  of  $U(L(T))$  under  $\phi$  (resp.  $\psi$ ).

## 7. Radicial groups of height 1

<sup>(61)</sup> Let  $S$  be a scheme of characteristic  $p > 0$ . We shall say that a  $\mathcal{O}_S$ -algebra  $\mathcal{A}$  (resp. a  $\mathcal{O}_S$ -Lie  $p$ -algebra  $\mathcal{L}$ ) is *finite locally free* if the  $\mathcal{O}_S$ -module underlying  $\mathcal{A}$  (resp.  $\mathcal{L}$ ) is locally free and of finite type. If  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra, we know (cf. 5.5.2) that the  $S$ -group functor  $\mathrm{Spec}^* \mathcal{U}_p(\mathcal{L})$  is represented by a finite and locally free  $S$ -group scheme  $G_p(\mathcal{L})$ . We shall see that this  $S$ -group scheme is the solution to a universal problem (7.2), and we shall characterize the  $S$ -group schemes of the form  $G_p(\mathcal{L})$  (7.4).

**Definition 7.0.** <sup>(62)</sup> Let  $H = \mathrm{Spec} \mathcal{A}$  be a finite locally free  $S$ -group scheme. One says that  $H$  is *infinitesimal* if the unit section  $\varepsilon_H : S \rightarrow H$  is a homeomorphism, which is equivalent to saying that the augmentation ideal of  $\mathcal{A}$  is locally nilpotent.

**7.1.** — <sup>(63)</sup> Let  $\mathcal{L}$  be a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra and  $G_p(\mathcal{L})$  the affine  $S$ -group  $\mathrm{Spec} \mathcal{U}_p(\mathcal{L})$ . According to 5.5, one knows that  $\mathcal{L}$  identifies with  $\mathcal{L}\mathrm{ie} G_p(\mathcal{L})$ . Now consider a very good  $S$ -group functor  $G$  satisfying condition (F) of 6.3, and let  $\phi : G_p(\mathcal{L}) \rightarrow G$  be a morphism of  $S$ -group functors. According to 6.3, the morphism of  $\mathcal{O}_S$ -Lie algebras  $\mathcal{L}\mathrm{ie} \phi : \mathcal{L}\mathrm{ie} G_p(\mathcal{L}) \rightarrow \mathcal{L}\mathrm{ie} G$  is compatible with symbolic  $p$ -th power. If we denote by  $\mathrm{Hom}_p(\mathcal{L}, \mathcal{L}\mathrm{ie} G)$  the set of morphisms of  $\mathcal{O}_S$ -Lie algebras compatible with symbolic  $p$ -th power, there is therefore a map

$$\mathcal{L}\mathrm{ie} : \mathrm{Hom}_{S\text{-Gr.}}(G_p(\mathcal{L}), G) \rightarrow \mathrm{Hom}_p(\mathcal{L}, \mathcal{L}\mathrm{ie} G), \quad \phi \mapsto \mathcal{L}\mathrm{ie} \phi.$$

**Theorem 7.2.** *If  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra, the map*

$$\mathrm{Hom}_{S\text{-gr.}}(G_p(\mathcal{L}), G) \rightarrow \mathrm{Hom}_p(\mathcal{L}, \mathcal{L}\mathrm{ie} G)$$

*is bijective in each of the following cases:*

- (i)  $G$  is an  $S$ -group scheme;
- (ii)  $G$  is of the form  $\underline{\mathrm{Aut}}_S X$ , where  $X$  is an  $S$ -scheme;
- (iii)  $G$  is of one of the forms  $\mathbf{W}(\mathcal{F})$  or  $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod}} \mathbf{W}(\mathcal{F})$ , where  $\mathcal{F}$  denotes a quasi-coherent  $\mathcal{O}_S$ -module.

The proof of the theorem rests on the following lemma:

**Lemma.** *If  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra, the  $S$ -group  $G = G_p(\mathcal{L})$  is annihilated by the Frobenius morphism  $\mathrm{Fr} : G \rightarrow G^{(p)}$ . In particular,  $G$  is infinitesimal.*

<sup>(61)</sup>**N.D.E.** For the results of this section, one may also refer to [DG70], § II.7, nos. 3–4.

<sup>(62)</sup>**N.D.E.** This definition (cf. [DG70], § II.4, 7.1), which will be used in 7.2.1, has been added.

<sup>(63)</sup>**N.D.E.** The original has been simplified, taking account of the additions made in 5.5.

<sup>(64)</sup> Let, indeed,  $\mathcal{U}$  be the restricted enveloping algebra of  $\mathcal{L}$ ,  $\mathcal{A} = \mathcal{U}^*$  the affine algebra of  $G$ , and  $\mathcal{J} = \text{Ker } \varepsilon_{\mathcal{A}}$  the augmentation ideal of  $\mathcal{A}$ . One has

$$(1) \quad \mathcal{A} = \mathcal{J} \oplus \eta_{\mathcal{A}}(\mathcal{O}_S),$$

where  $\eta_{\mathcal{A}}$  denotes the unit section of  $\mathcal{A}$ , and since  $\varepsilon_{\mathcal{A}}$  (resp.  $\eta_{\mathcal{A}}$ ) is the transpose of  $\eta_{\mathcal{U}}$  (resp.  $\varepsilon_{\mathcal{U}}$ ), this decomposition corresponds by duality to the decomposition

$$(2) \quad \mathcal{U} = \mathcal{J} \oplus \eta_{\mathcal{U}}(\mathcal{O}_S),$$

where  $\mathcal{J}$  is the augmentation ideal of  $\mathcal{U}$ ; one therefore has  $\mathcal{J} = \mathcal{J}^*$ .

Denote by  $\pi$  the endomorphism  $x \mapsto x^p$  of  $\mathcal{O}_S$ . We must show that the morphism  $\text{Fr} : G \rightarrow G^{(p)}$  factors through the unit section of  $G^{(p)}$ , which is equivalent to saying (cf. 4.1.4 (c)) that the morphism  $\Phi : a \otimes_{\pi} x \mapsto a^p x$  from  $\mathcal{J} \otimes_{\pi} \mathcal{O}_S$  to  $\mathcal{A}$  is zero. Since  $\mathcal{A}$  is finite locally free over  $\mathcal{O}_S$ , it suffices to see that the transpose morphism  ${}^t\Phi$  is zero.

Now  $\Phi$  is none other than the following composite

$$\mathcal{J} \otimes_{\pi} \mathcal{O}_S \xrightarrow{\tau} \mathcal{A} \otimes_{\pi} \mathcal{O}_S \xrightarrow{j(\mathcal{A})} S^p \mathcal{A} \xrightarrow{b(\mathcal{A})} \mathcal{A},$$

where  $\tau$  is deduced from the inclusion  $\mathcal{J} \hookrightarrow \mathcal{A}$ , and  $b(\mathcal{A})$  and  $j(\mathcal{A})$  are defined as in 4.3.3 (i.e.  $b(\mathcal{A})$  is induced by the multiplication of  $\mathcal{A}$  and  $j(\mathcal{A})$  sends  $a \otimes_{\pi} 1$  to the image of  $a \otimes \cdots \otimes a$  in  $S^p \mathcal{A}$ ). Since the dual  $\mathcal{O}_S$ -module of  $S^p \mathcal{A}$  is none other than the submodule  $\Sigma^p \mathcal{U}$  of  $\bigotimes^p \mathcal{U}$  consisting of sections invariant under the action of the symmetric group of order  $p$ , one sees that  ${}^t\Phi$  is the following composite morphism:

$$\mathcal{U} \xrightarrow{a(\mathcal{U})} \Sigma^p \mathcal{U} \xrightarrow{r(\mathcal{U})} \mathcal{U} \otimes_{\pi} \mathcal{O}_S \xrightarrow{q} \mathcal{J} \otimes_{\pi} \mathcal{O}_S,$$

where  $q$  is deduced from the projection  $\mathcal{U} \rightarrow \mathcal{J}$  with kernel  $\eta_{\mathcal{U}}(\mathcal{O}_S)$ ,  $a(\mathcal{U})$  is induced by the comultiplication of  $\mathcal{U}$ , and  $r(\mathcal{U})$  vanishes on symmetrized tensors and sends a section  $x \otimes \cdots \otimes x$  to  $x \otimes_{\pi} 1$  (confer 4.3.3).

It is clear that  ${}^t\Phi \circ \eta_{\mathcal{U}} = 0$  and hence, according to (2), it remains to see that  ${}^t\Phi$  annihilates the augmentation ideal  $\mathcal{J}$ . Since  ${}^t\Phi$  is an algebra morphism and the ideal  $\mathcal{J}$  is generated by  $\mathcal{L}$  (identified with its image in  $\mathcal{U}$ ), it suffices to see that  ${}^t\Phi(x) = 0$  for every section  $x$  of  $\mathcal{L}$ . Now  $-a(\mathcal{U})(x) = (p-1)! a(\mathcal{U})(x)$  is the symmetrization of  $x \otimes 1 \otimes \cdots \otimes 1$ , hence its image under  $r(\mathcal{U})$  is zero. This proves the first assertion of the lemma.

The second follows from it. Indeed, since every local section of  $\mathcal{J}$  has zero  $p$ -th power and  $\mathcal{J}$  is an  $\mathcal{O}_S$ -module of finite type,  $\mathcal{J}$  is locally nilpotent (explicitly, if  $V$  is an affine open subset of  $S$  such that  $I = \Gamma(V, \mathcal{J})$  is generated by  $r$  elements, then  $I^{r(p-1)+1} = 0$ ), whence  $G_{\text{red}} = S_{\text{red}}$ , and hence the unit section  $\varepsilon_G : S \rightarrow G$  is a homeomorphism.

**7.2.1.** — <sup>(65)</sup> We shall first prove assertion (ii) of theorem 7.2. Let  $\pi : X \rightarrow S$  be an  $S$ -scheme. First consider an arbitrary infinitesimal  $S$ -group  $H$ . Morphisms  $\phi$  from  $H$  to  $\underline{\text{Aut}} X$  correspond bijectively to left actions  $\mu : H \times X \rightarrow X$  of  $H$  on  $X$ . For such an action, if  $\varepsilon$  is the unit section of  $H$ , the composite morphism

$$X \simeq S \times X \xrightarrow{\varepsilon \times X} H \times X \xrightarrow{\mu} X$$

must be the identity. Since  $(H \times X)_{\text{red}}$  identifies with  $X_{\text{red}}$ , one sees that  $\mu$  must induce the identity on the associated reduced schemes. In particular,  $\mu$  induces an action of  $H$  on every open subset of  $X$ , and one therefore obtains, for every open subset  $U$  of  $X$  affine over  $S$ , a morphism of associative unital algebras:

$$\mathcal{A}(U) \rightarrow \mathcal{A}(H) \otimes \mathcal{A}(U)$$

making  $\mathcal{A}(U)$  a left  $\mathcal{A}(H)$ -comodule, in such a way that the restriction maps  $\mathcal{A}(U) \rightarrow \mathcal{A}(U')$ , for  $U' \subset U$ , are comodule morphisms. Conversely, every datum of this type comes from a unique left action  $\mu : H \times X \rightarrow X$ . On the other hand, one has the following lemma:

<sup>(64)</sup>**N.D.E.** The original has been expanded in what follows. For another proof, see [DG70], § II.7, 3.9.

<sup>(65)</sup>**N.D.E.** The original has been expanded in what follows.

**Lemma.** *Let  $X = \operatorname{Spec} \mathcal{C}$  be an affine  $S$ -scheme,  $H = \operatorname{Spec} \mathcal{A}$  an infinitesimal  $S$ -group, and  $\mathcal{U} = \mathcal{A}^* = \mathcal{H}om_{\mathcal{O}_S}(\mathcal{A}, \mathcal{O}_S)$ . The left actions of  $H$  on  $X$  correspond bijectively to the representations of the algebra  $\mathcal{U}$  in the  $\mathcal{O}_S$ -module  $\mathcal{C}$  such that one has:*

$$\begin{aligned} (a) \quad & u(1_{\mathcal{C}}) = \varepsilon(u) \cdot 1_{\mathcal{C}} \\ (b) \quad & u(xy) = \sum_i v_i(x)w_i(y) \quad \text{if } \Delta u = \sum_i v_i \otimes w_i. \end{aligned}$$

(In the formulas above,  $u$  denotes an arbitrary section of  $\mathcal{U}$  over an affine open subset  $V$  of  $S$ , and  $x$  and  $y$  sections of  $\mathcal{C}$  over  $V$ ;  $1_{\mathcal{C}}$  denotes the unit section of  $\mathcal{C}$ , and  $\varepsilon$  and  $\Delta$  the augmentation and diagonal morphism of  $\mathcal{U}$ .) Indeed, a left action  $\mu$  of  $H$  on  $X$  is defined by a morphism of associative unital algebras:

$$\lambda : \mathcal{C} \longrightarrow \mathcal{A} \otimes \mathcal{C}$$

making  $\mathcal{C}$  a left  $\mathcal{A}$ -comodule. We shall denote by  $\alpha$  the composite morphism

$$\mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\mathcal{U} \otimes \lambda} \mathcal{U} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{C} \xrightarrow{\gamma \otimes \mathcal{C}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{C} \simeq \mathcal{C}$$

where  $\gamma$  is the “contraction” from  $\mathcal{A}^* \otimes_{\mathcal{O}_S} \mathcal{A}$  to  $\mathcal{O}_S$ . Since  $\mathcal{A}$  is finite locally free over  $\mathcal{O}_S$ , one knows that the map  $\lambda \mapsto (\gamma \otimes \mathcal{C})(\mathcal{U} \otimes \lambda)$  is a bijection from  $\operatorname{Hom}_{\mathcal{O}_S}(\mathcal{C}, \mathcal{A} \otimes \mathcal{C})$  onto  $\operatorname{Hom}_{\mathcal{O}_S}(\mathcal{U} \otimes \mathcal{C}, \mathcal{C})$ . Moreover, one sees easily that the condition that  $\lambda$  define a left  $\mathcal{A}$ -comodule structure (resp. be a morphism of associative unital algebras) is equivalent, by duality, to the condition that  $\alpha$  be a representation of  $\mathcal{U}$  in  $\mathcal{C}$  (resp. that  $\alpha$  satisfy conditions (a) and (b)). This proves the lemma.

Moreover, it is clear that, for every representation of  $\mathcal{U}$  in the  $\mathcal{O}_S$ -module  $\mathcal{C}$ , the sections  $u$  of  $\mathcal{U}$  satisfying conditions (a) and (b) of the lemma form a subalgebra of  $\mathcal{U}$ .

In the particular case  $H = G_p(\mathcal{L})$  which interests us, these conditions will therefore hold for all sections  $u$  of  $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$  if they are true for the sections  $u$  of  $\mathcal{L}$  (identifying  $\mathcal{L}$  with its image in  $\mathcal{U}_p(\mathcal{L})$ ). Now, if  $u$  is a section of  $\mathcal{L}$ , conditions (a) and (b) mean simply that  $u(1_{\mathcal{C}}) = 0$  and  $u(xy) = u(x)y + xu(y)$ , i.e. that  $\alpha(u)$  is a  $\mathcal{O}_S$ -derivation of  $\mathcal{C} = \mathcal{A}(X)$ . Assertion (ii) of 7.2 follows. Indeed, every morphism  $\phi$  from  $G_p = G_p(\mathcal{L})$  to  $\underline{\operatorname{Aut}} X$  defines a morphism of Lie  $p$ -algebras  $\mathcal{L}ie \phi$  from  $\mathcal{L} = \mathcal{L}ie G_p$  to  $\pi_*(\mathcal{D}er_{\mathcal{O}_S} \mathcal{O}_X)$ , and conversely every datum of this type comes, according to what precedes, from a unique action  $\mu : G \times X \longrightarrow X$ .

**7.2.2.** — Let us now show how assertion (i) of theorem 7.2 follows from (ii). Let  $G$  be an  $S$ -group scheme. If  $T$  is an  $S$ -scheme and  $x$  an element of  $G(T)$ , we denote by  $\ell_x^T$  (resp.  $r_x^T$ ) the left (resp. right) translation of  $G_T$  defined by  $x$ . The maps  $\ell^T : x \mapsto \ell_x^T$  therefore determine a homomorphism  $\ell$  from  $G$  to  $\underline{\operatorname{Aut}} G$ . Let, on the other hand,  $f$  be a  $T$ -automorphism of  $G_T$ ; one then defines  ${}^x f$  to be equal to  $(r_x^T)^{-1} f r_x^T$ , i.e. for every  $T' \longrightarrow T$  and  $g \in G(T')$ ,  $({}^x f)(g) = f(gx)x^{-1}$ . In this way  $G$  acts on the left on the  $S$ -group functor  $\underline{\operatorname{Aut}} G$ , hence also on the functors  $T \mapsto \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), \underline{\operatorname{Aut}} G_T)$  and  $T \mapsto \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(\underline{\operatorname{Aut}} G_T/T))$ . On the other hand, the morphism  $\ell_T : G_T \longrightarrow \underline{\operatorname{Aut}} G_T$  identifies  $G_T$  with the group of automorphisms of the  $T$ -scheme  $G_T$  commuting with right translations, and the derived morphism  $\mathcal{L}ie(\ell_T)$  identifies  $\mathcal{L}ie G_T$  with the Lie  $p$ -algebra of  $\mathcal{O}_T$ -derivations of  $\mathcal{O}_{G_T}$  commuting with right translations (cf. II, 4.11.1); they therefore induce commutative squares

$$\begin{array}{ccc} \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), G_T) & \xrightarrow{\mathcal{L}ie} & \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(G_T/T)) \\ \ell_T \downarrow & & \downarrow \mathcal{L}ie \ell_T \\ \operatorname{Hom}_{T\text{-Gr.}}(G_p(\mathcal{L}_T), \underline{\operatorname{Aut}} G_T) & \xrightarrow{\mathcal{L}ie} & \operatorname{Hom}_p(\mathcal{L}_T, \mathcal{L}ie(\underline{\operatorname{Aut}} G_T/T)). \end{array}$$

The images of the two vertical arrows are the subfunctors consisting of the invariants under the action of the  $S$ -group  $G$ . Since the lower horizontal arrow is invertible according to 7.2.1 and is compatible with the action of  $G$ , the upper horizontal arrow is invertible also. This proves 7.2 (i).

**7.2.3.** — Now consider the case where  $G = \underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})$ .<sup>(66)</sup> Put  $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ ,  $\mathcal{A} = \mathcal{U}^*$ , and  $H = G_p(\mathcal{L}) = \text{Spec } \mathcal{A}$ . Since  $H$  is affine over  $S$ , according to VI B 11.6.1 a morphism of  $S$ -groups from  $H$  to  $\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}} \mathbf{W}(\mathcal{F})$  is the same thing as a right  $\mathcal{A}$ -comodule structure

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}.$$

Moreover, since  $\mathcal{A}$  is finite locally free over  $\mathcal{O}_S$ , this is equivalent to giving a representation

$$\alpha : \mathcal{U} \longrightarrow \text{End}_{\mathcal{O}_S}(\mathcal{F})$$

of  $\mathcal{U}$  in  $\mathcal{F}$ . Finally, according to the universal property of  $\mathcal{U} = \mathcal{U}_p(\mathcal{L})$ , giving such a morphism  $\alpha$  is equivalent to giving its restriction  $\rho$  to  $\mathcal{L}$  (identified with its image in  $\mathcal{U}$ ), which is a morphism of Lie  $p$ -algebras from  $\mathcal{L}$  to  $\text{End}_{\mathcal{O}_S}(\mathcal{F})$ .

<sup>(67)</sup> Finally, consider the case where  $G = \mathbf{W}(\mathcal{F})$ , keeping the preceding notation. First, giving a morphism of  $S$ -functors  $\phi : H \longrightarrow \mathbf{W}(\mathcal{F})$  is equivalent to giving an element  $\theta$  of  $\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H)$ , and since  $H$  is finite locally free over  $S$ , one has:

$$\Gamma(H, \mathcal{F} \otimes \mathcal{O}_H) = \Gamma(S, \mathcal{F} \otimes \mathcal{A}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{U}, \mathcal{F}).$$

The condition that  $\phi$  be a group morphism is then expressed by the fact that  $\theta$ , considered as a morphism of  $\mathcal{O}_S$ -modules  $\mathcal{U} \longrightarrow \mathcal{F}$ , vanishes on  $1_{\mathcal{U}}$  and on  $\mathcal{J} / \mathcal{J}^2$ , where  $\mathcal{J}$  is the augmentation ideal of  $\mathcal{U}$ , whence

$$(1) \quad \text{Hom}_{S\text{-gr.}}(H, \mathbf{W}(\mathcal{F})) = \text{Hom}_{\mathcal{O}_S}(\mathcal{J} / \mathcal{J}^2, \mathcal{F}).$$

On the other hand, consider the quasi-coherent sheaf  $[\mathcal{L}, \mathcal{L}]$ , the image of the morphism  $\mathcal{L} \otimes \mathcal{L} \longrightarrow \mathcal{L}$ ,  $x \otimes y \mapsto [x, y]$ ; for every affine open subset  $V$  of  $S$ , one has  $[\mathcal{L}, \mathcal{L}](V) = [\mathcal{L}(V), \mathcal{L}(V)]$ . Then there is an exact sequence

$$(\dagger) \quad 0 \longrightarrow [\mathcal{L}, \mathcal{L}] \longrightarrow \mathcal{L} \xrightarrow{\pi} \mathcal{J} / \mathcal{J}^2 \longrightarrow 0,$$

where  $\pi$  is the composite of the inclusion  $\mathcal{L} \hookrightarrow \mathcal{J}$  and the projection  $\mathcal{J} \longrightarrow \mathcal{J} / \mathcal{J}^2$ .

Indeed, the question being local on  $S$ , one may suppose that  $S$  is affine with ring  $R$  and that  $L = \mathcal{L}(S)$  is free with basis  $(x_1, \dots, x_r)$ . Identifying  $L$  with its image in  $U = U_p(L)$ , let  $K$  be the sub- $R$ -module of  $U$  which is the direct sum of  $[L, L]$  and the submodule with basis the monomials  $x_1^{n_1} \cdots x_r^{n_r}$  such that  $n_1 + \cdots + n_r \geq 2$ ; one then verifies that  $K$  is a two-sided ideal of  $U$ . Since  $K$  is contained in  $J^2$  (where  $J$  is the augmentation ideal of  $U$ ) and contains all the products  $x_i x_j$  (which generate  $J^2$ ), one deduces that  $K = J^2$ , whence  $J^2 \cap L = [L, L]$ , and one has exact sequence  $(\dagger)$ .

On the other hand, one knows according to 6.4.2 that  $\underline{\text{Lie}} \mathbf{W}(\mathcal{F})$  is none other than  $\mathbf{W}(\mathcal{F})$ , the Lie bracket and symbolic  $p$ -th power being zero. From this and what precedes one deduces that

$$(2) \quad \text{Hom}_p(\mathcal{L}, \mathcal{F}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{L} / [\mathcal{L}, \mathcal{L}], \mathcal{F}) = \text{Hom}_{\mathcal{O}_S}(\mathcal{J} / \mathcal{J}^2, \mathcal{F})$$

and this, combined with (1), completes the proof of theorem 7.2.

**Lemma 7.3.** *If  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra, the morphism  $j_{\mathcal{L}} : \mathcal{L} \longrightarrow \text{Lie } G_p(\mathcal{L})$  of 5.5 is invertible.*

<sup>(68)</sup> For the proof, see 5.5.1.

<sup>(66)</sup>**N.D.E.** In what follows, the original has been expanded (and simplified), taking account of VI B, 11.6.1.

<sup>(67)</sup>**N.D.E.** What follows has been added. (The original indicated “the case of  $\mathbf{W}(\mathcal{F})$  is analogous”.)

<sup>(68)</sup>**N.D.E.** To avoid changing the numbering, statement 7.3 has been retained, although it was included, with its proof, in 5.5.1.

**7.4.** — To conclude this section, we shall give a characterization of the  $S$ -group schemes of the form  $G_p(\mathcal{L})$ , where  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra.

Let  $G$  be an  $S$ -group scheme,  $\varepsilon_G$  its unit section, and  $\mathcal{I}'$  the kernel of the morphism  $\varepsilon_G^{-1}(\mathcal{O}_G) \rightarrow \mathcal{O}_S$  corresponding to  $\varepsilon_G$ . The image of  $\underline{\mathrm{Lie}}(G/S)(S)$  in  $U(G)$  identifies, according to 2.5 and 1.3.1, with the morphisms of  $\mathcal{O}_S$ -modules from  $\varepsilon_G^{-1}(\mathcal{O}_G)$  to  $\mathcal{O}_S$  which vanish on the unit section of  $\varepsilon_G^{-1}(\mathcal{O}_G)$  and on  $\mathcal{I}'^2$ . One thus recovers the canonical isomorphism from  $\underline{\mathrm{Lie}}(G/S)(S)$  onto  $\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{I}'/\mathcal{I}'^2, \mathcal{O}_S)$  of II, 3.3 and 4.11.4.<sup>(69)</sup> We shall put  $\omega_{G/S} = \mathcal{I}'/\mathcal{I}'^2$  as in loc. cit., so that the sheaf  $\mathcal{L}\mathrm{ie}(G/S)$  identifies with  $\omega_{G/S}^* = \mathcal{H}\mathrm{om}_{\mathcal{O}_S}(\omega_{G/S}, \mathcal{O}_S)$ .<sup>(70)</sup> Moreover, if  $G = G_p(\mathcal{L})$ , where  $\mathcal{L}$  is a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra, we saw in 5.5.3 that  $\omega_{G/S} = \mathcal{L}^*$ .

**Theorem.** *If  $G$  is a group scheme over a scheme  $S$  of characteristic  $p > 0$ , the following assertions are equivalent:*

- (i) *There exists a finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebra  $\mathcal{L}$  such that  $G \simeq G_p(\mathcal{L})$ .*
- (i') *The  $\mathcal{O}_S$ -Lie  $p$ -algebra  $\mathcal{L}\mathrm{ie}(G)$  is finite locally free and  $G \simeq G_p(\mathcal{L}\mathrm{ie}(G))$ .*
- (ii)  *$G$  is affine over  $S$ ,  $\omega_{G/S}$  is a locally free  $\mathcal{O}_S$ -module of finite type, and the affine algebra of  $G$  is locally isomorphic to the quotient of the symmetric algebra  $S_{\mathcal{O}_S}(\omega_{G/S})$  by the ideal generated by the  $p$ -th powers of sections of  $\omega_{G/S}$ .*
- (iii)  *$G$  is locally of finite presentation over  $S$ , of height  $\leq 1$ , and  $\omega_{G/S}$  is locally free.*
- (iii')  *$G$  is locally of finite type over  $S$ , of height  $\leq 1$ , and  $\omega_{G/S}$  is locally free.*
- (iv)  *$G$  is locally of finite presentation and flat over  $S$ , of height  $\leq 1$ .<sup>(71)</sup>*

**7.4.1.** — The equivalence (i)  $\Leftrightarrow$  (i') follows from 5.5.3 (i), the implications (ii)  $\Rightarrow$  (iii)  $\Rightarrow$  (iii') are clear, and one has (i)  $\Rightarrow$  (iv) since  $G_p(\mathcal{L})$  is finite locally free and of height  $\leq 1$ , according to 5.5.2 and lemma 7.2. Let us show that (i) implies (ii). Denote by  $\mathcal{I}$  the augmentation ideal of  $\mathcal{A} = \mathcal{U}_p(\mathcal{L})^*$ . We already saw in 5.5.3 (ii) that  $\omega_{G/S} = \mathcal{I}/\mathcal{I}^2$  identifies with  $\mathcal{L}^*$ , hence is finite locally free.

Now suppose that  $S$  is affine. There is then a section  $\sigma : \omega_{G/S} \rightarrow \mathcal{I}$  of the projection  $\mathcal{I} \rightarrow \mathcal{I}/\mathcal{I}^2$ ; it induces an algebra morphism  $\sigma' : S_{\mathcal{O}_S}(\omega_{G/S}) \rightarrow \mathcal{A}$ , and, according to lemma 7.2,  $\sigma'$  factors into a morphism

$$\phi : S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} \rightarrow \mathcal{A},$$

where  $\mathcal{K}$  denotes the ideal generated by the  $p$ -th powers of sections of  $\omega_{G/S}$ . If one filters  $\mathcal{A}$  (resp.  $S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K}$ ) by the powers of  $\mathcal{I}$  (resp. of the ideal generated by  $\omega_{G/S}$ ), it is clear that  $\phi$  induces an epimorphism of the associated graded objects. Thus  $\phi$  is an epimorphism of locally free  $\mathcal{O}_S$ -modules of the same rank (cf. 5.3.3); hence  $\phi$  is an isomorphism. This proves that (i)  $\Rightarrow$  (ii).

**7.4.2.** — Now suppose that  $G$  is of height  $\leq 1$  and locally of finite presentation over  $S$ .<sup>(72)</sup> Since the Frobenius morphism  $\mathrm{Fr} : G \rightarrow G^{(p)}$  is integral and factors through the unit section of  $G^{(p)}$ ,  $G$  is integral (hence affine) over  $S$ . Put  $\mathcal{A} = \mathcal{A}(G)$ ; since  $G$  is assumed locally

<sup>(69)</sup>**N.D.E.** If  $G$  is affine over  $S$  and  $\mathcal{I}$  denotes the augmentation ideal of  $\mathcal{A}(G)$ , then  $\mathcal{I}'/\mathcal{I}'^2$  identifies with  $\varepsilon_G^*(\mathcal{I}/\mathcal{I}^2)$ , cf. loc. cit.

<sup>(70)</sup>**N.D.E.** The following sentence has been added.

<sup>(71)</sup>**N.D.E.** On the one hand, assertion (i'), implicit in the original, has been added, and on the other hand assertions (iii') and (iv), pointed out by O. Gabber; assertion (iv) takes up a footnote of the original, which indicated: "The condition on  $\omega_{G/S}$  is in fact unnecessary, as one sees easily by reducing to the case where  $S$  is local with residue field  $k$ , and applying the theorem to the group  $G_k$ ." As pointed out by Gabber, this is incorrect without a flatness hypothesis: if  $A$  is a local artinian ring of characteristic  $p > 0$  and  $J$  a proper nonzero ideal of  $A$ , let  $H$  be the subgroup  $\mathrm{Spec} A[x]/(x^p, Jx)$  of  $\alpha_{p,A}$  (i.e. for every  $A$ -algebra  $R$ ,  $H(R) = \{x \in R \mid x^p = 0 \text{ and } Jx = 0\}$ ); then  $H$  is not flat over  $A$ , hence is not of the form  $G_p(\mathcal{L})$ , where  $\mathcal{L}$  is a free Lie  $p$ -algebra of finite rank over  $A$ .

<sup>(72)</sup>**N.D.E.** The original has been expanded (and simplified) in what follows.

of finite presentation over  $S$ , it follows that  $G$  is finite and of finite presentation over  $S$ , hence that  $\mathcal{A}(G)$  is a finitely presented  $\mathcal{O}_S$ -module (cf. EGA IV<sub>1</sub>, 1.4.7). Let  $\mathcal{I}$  be the augmentation ideal of  $\mathcal{A}$ ; since  $\mathcal{A} = \eta_{\mathcal{A}}(\mathcal{O}_S) \oplus \mathcal{I}$  (where  $\eta_{\mathcal{A}}$  is the unit section of  $\mathcal{A}$ ),  $\mathcal{I}$  is a finitely presented  $\mathcal{O}_S$ -module, and therefore so is  $\omega_G = \mathcal{I}/\mathcal{I}^2$ . When one assumes that  $G$  is of height  $\leq 1$  and locally of finite type over  $S$ , one obtains similarly that  $\mathcal{A}(G)$ ,  $\mathcal{I}$ , and  $\omega_G = \mathcal{I}/\mathcal{I}^2$  are  $\mathcal{O}_S$ -modules of finite type.

Thus, under hypothesis (iii'), one obtains that  $\omega_{G/S}$  is finite locally free over  $\mathcal{O}_S$ , as is  $\mathcal{L} = \mathcal{L}e(G/S) = \omega_{G/S}^*$ . Put  $\mathcal{B} = \mathcal{U}_p(\mathcal{L})^*$  and  $H = G_p(\mathcal{L}) = \text{Spec } \mathcal{B}$ . According to theorem 7.2, the identity map of  $\mathcal{L}$  corresponds to a group morphism from  $H = G_p(\mathcal{L})$  to  $G$ , hence to a morphism of  $\mathcal{O}_S$ -algebras  $\theta : \mathcal{A} \rightarrow \mathcal{B}$ . One must show that  $\theta$ , which by definition induces an isomorphism from  $\omega_{G/S}$  to  $\omega_{H/S}$ , is an isomorphism.

For this purpose one may restrict to the case where  $S$  is affine. There is then a section  $\tau$  of the projection  $\mathcal{I} \rightarrow \omega_{G/S}$ ; it induces an algebra morphism  $\tau' : S_{\mathcal{O}_S}(\omega_{G/S}) \rightarrow \mathcal{A}$ , and since every local section of  $\mathcal{I}$  has zero  $p$ -th power (because  $\text{Fr} : G \rightarrow G^{(p)}$  factors through the unit section of  $G^{(p)}$ ),  $\tau'$  induces a morphism of  $\mathcal{O}_S$ -algebras  $\psi$  occurring in the commutative diagram below:

$$\begin{array}{ccc} S_{\mathcal{O}_S}(\omega_{G/S})/\mathcal{K} & \xrightarrow{\psi} & \mathcal{A} \\ & \searrow \phi & \downarrow \theta \\ & & \mathcal{B} \end{array}$$

where  $\mathcal{K}$  is the ideal generated by the  $p$ -th powers of sections of  $\omega_{G/S}$ . On the one hand, one shows as in 7.4.1 that  $\psi$  is an epimorphism of  $\mathcal{O}_S$ -modules. On the other hand, we saw in 7.4.1 that  $\phi = \theta \circ \psi$  is an isomorphism. The same is therefore true of  $\theta$ . This proves that (iii')  $\Rightarrow$  (i).

**7.4.3.** — <sup>(73)</sup> Finally, let us show that (iv) implies (iii). It suffices to show that  $\omega_{G/S}$  is locally free, so one may suppose that  $S$  is affine with ring  $R$ . As observed at the beginning of 7.4.2, hypothesis (iv) then implies that  $G = \text{Spec } A$ , for an  $R$ -algebra  $A$  which is a finitely presented  $R$ -module, as is  $\omega_{G/A} = I/I^2$  (where  $I$  is the augmentation ideal of  $A$ ). Since one also assumes that  $G$  is flat over  $S$ ,  $A$  is a finite locally free  $R$ -module (cf. [BAC] II, § 5.2, Th. 1 and cor. 2), and, according to loc. cit., to show that  $\omega_{G/A}$  is locally free of finite rank, it suffices to show that  $(\omega_{G/A})_{\mathfrak{m}}$  is flat for every maximal ideal  $\mathfrak{m}$  of  $R$ . Thus one may suppose that  $R$  is local and  $A$  free of rank  $n + 1$ , hence  $I$  free of rank  $n$ . Let  $\mathfrak{m}$  be the maximal ideal of  $R$  and  $k = R/\mathfrak{m}$ .

Denote by  $I_k$  the augmentation ideal of  $A_k$  and by  $r$  the dimension of  $\omega_{G_k/k} = I_k/I_k^2$ . Let  $(e_1, \dots, e_n)$  be a basis of  $I_k$  such that  $(e_{r+1}, \dots, e_n)$  is a basis of  $I_k^2$ , and let  $x_1, \dots, x_n$  be elements of  $I$  lifting the  $e_i$ . According to Nakayama's lemma,  $(x_1, \dots, x_n)$  is a basis of  $I$  over  $R$ . Let  $N$  be the sub- $R$ -module of  $I$  with basis  $(x_1, \dots, x_r)$ , and let  $B$  be the quotient of the symmetric algebra of  $N$  by the ideal generated by the elements  $x^p$ , for  $x \in N$ . Since every element of  $I$  has zero  $p$ -th power, one obtains a morphism of  $R$ -algebras

$$\psi : B \rightarrow A.$$

According to 7.4.2,  $\psi \otimes k$  is an isomorphism. It follows that  $\text{Coker } \psi = 0$  and that, denoting  $\text{Ker } \psi$  by  $K$ , the morphism  $\tau : K \otimes k \rightarrow B \otimes k$  is zero. But since  $\psi$  is surjective and  $A$  flat over  $R$ ,  $\tau$  is also injective, whence  $K \otimes k = 0$ . On the other hand, since  $A$  is a finitely presented  $R$ -module,  $K$  is an  $R$ -module of finite type (cf. [BAC] I, § 2.8, Lemma 9), whence  $K = 0$  according to Nakayama. Thus  $\psi$  is an isomorphism of  $R$ -algebras, and since  $\psi^{-1}(I)$  contains the augmentation ideal  $J$  of  $B$ , it follows that  $\psi^{-1}(I) = J$ , and hence  $\psi^{-1}$  induces an isomorphism of  $R$ -modules from  $I/I^2$  to  $J/J^2 = N$ . This proves that  $\omega_{G/S}$  is finite locally free, whence the implication (iv)  $\Rightarrow$  (iii). This completes the proof of theorem 7.4.

<sup>(73)</sup>**N.D.E.** This paragraph has been added to prove that (iv)  $\Rightarrow$  (iii), cf. N.D.E. (71).

**Remark 7.5.** <sup>(74)</sup> It clearly follows from theorems 7.2 and 7.4 that the functors  $G \mapsto \mathcal{L}ie(G)$  and  $\mathcal{L} \mapsto G_p(\mathcal{L})$  induce equivalences, quasi-inverse to one another, between the category of  $S$ -groups locally of finite presentation and flat, of height  $\leq 1$ , and the full subcategory of that of  $\mathcal{O}_S$ -Lie  $p$ -algebras consisting of finite locally free  $\mathcal{O}_S$ -Lie  $p$ -algebras.

## 8. Case of a base field

**8.1.** — Let us now summarize the results obtained in the case where  $S$  is the spectrum of a field  $k$  of characteristic  $p > 0$ . Let us say that a  $k$ -group scheme is *algebraic* if the underlying scheme is of finite type over  $k$ . In this case, according to theorem 7.2, one obtains:<sup>(75)</sup>

**Theorem 8.1.1.** *The functor  $G_p$ , which associates to every Lie  $p$ -algebra  $\mathcal{L}$  of finite dimension over  $k$  the  $k$ -group  $G_p(\mathcal{L})$ , is left adjoint to the functor which associates to every algebraic  $k$ -group  $G$  its  $\mathcal{L}ie(G)$ .*

Combining this with theorem 7.4, one obtains:

**Theorem 8.1.2.** *The functors  $G_p : \mathcal{L} \mapsto G_p(\mathcal{L})$  and  $G \mapsto \mathcal{L}ie(G)$  induce equivalences, quasi-inverse to one another, between the category of Lie  $p$ -algebras of finite dimension over  $k$  and that of algebraic  $k$ -groups of height  $\leq 1$ .*

Then, since  $G_p$  is a left adjoint functor, it commutes with inductive limits,<sup>(76)</sup> hence in particular with the formation of cokernels. On the other hand, if one has two morphisms  $\phi : G \rightarrow H$  and  $\phi' : G' \rightarrow H$  between algebraic  $k$ -groups of height  $\leq 1$ , the fibered product  $G \times_H G'$  is again an algebraic  $k$ -group of height  $\leq 1$  (for the morphism  $\text{Fr} : G \rightarrow G^{(p)}$  commutes with fibered products). Thus the inclusion of the category of algebraic  $k$ -groups of height  $\leq 1$  into that of all algebraic  $k$ -groups commutes with fibered products, hence in particular with the formation of kernels. One deduces the:

**Corollary 8.1.3.** *The functor  $G_p$  is exact in the following sense. If  $\pi : \mathcal{L}_1 \rightarrow \mathcal{L}_2$  is a surjective morphism between Lie  $p$ -algebras of finite dimension over  $k$  and  $i$  is the inclusion of  $\mathcal{L}_0 = \text{Ker } \pi$  into  $\mathcal{L}_1$ , there is an exact sequence of algebraic  $k$ -groups:*

$$1 \rightarrow G_p(\mathcal{L}_0) \xrightarrow{G_p(i)} G_p(\mathcal{L}_1) \xrightarrow{G_p(\pi)} G_p(\mathcal{L}_2) \rightarrow 1.$$

(77)

Indeed, according to what precedes,  $G_p(i)$  induces an isomorphism from  $G_p(\mathcal{L}_0)$  onto  $\text{Ker}(G_p(\pi))$  (this kernel being the same in the category of all algebraic  $k$ -groups  $H$  or in that of the  $H$  of height  $\leq 1$ ), and  $G_p(\pi) : G_p(\mathcal{L}_1) \rightarrow G_p(\mathcal{L}_2)$  identifies  $G_p(\mathcal{L}_2)$  with the quotient of  $G_p(\mathcal{L}_1)$  by  $G_p(\mathcal{L}_0)$  in the category of algebraic  $k$ -groups.

**Remark 8.1.4.** <sup>(78)</sup> Let  $\phi : G \rightarrow H$  be a morphism of  $k$ -groups and  $K = \text{Ker}(\phi)$ . One assumes that  $\phi$  is covering for the (fpqc) topology (this will be the case, in particular, if  $\phi$  is covering for a coarser topology, for example the (fppf) topology). Then, on the one hand,  $\phi$  is a  $K$ -torsor over  $H$  (cf. IV 5.1.7.1). On the other hand (cf. IV 6.3.1), there exists a covering of  $H$  by affine open subsets  $S_i$ , and for each  $i$  an affine faithfully flat morphism  $T_i \rightarrow S_i$  factoring through  $\phi$ . Then  $G \times_H T_i$  is  $T_i$ -isomorphic to  $K \times T_i$ , hence faithfully flat over  $T_i$ , and therefore, by (fpqc) descent,  $G \times_H S_i \rightarrow S_i$  is faithfully flat, so that  $\phi$  is faithfully flat.

<sup>(74)</sup>**N.D.E.** This remark has been added.

<sup>(75)</sup>**N.D.E.** The numbering 8.1.1 to 8.1.3 has been added to bring out the results stated there.

<sup>(76)</sup>**N.D.E.** What follows, as well as the proof of the corollary below, has been expanded.

<sup>(77)</sup>**N.D.E.** Moreover, according to VI A, 3.2,  $G_p(\mathcal{L}_2)$  represents the (fppf) quotient sheaf of  $G_p(\mathcal{L}_1)$  by  $G_p(\mathcal{L}_0)$ .

<sup>(78)</sup>**N.D.E.** This remark, pointed out by O. Gabber, has been added; it will be useful in 8.3.1.

Conversely, if  $\phi$  is faithfully flat and quasi-compact (resp. and locally of finite presentation), it is covering for the (fpqc) (resp. (fppf)) topology, cf. IV 6.3.1. Finally, recall that a sheaf morphism is covering if and only if it is an epimorphism, cf. IV 4.4.3. One therefore obtains, in particular, that a quasi-compact morphism of  $k$ -groups is faithfully flat if and only if it is an epimorphism of (fpqc) sheaves.

**Proposition 8.2.** *Consider an exact sequence<sup>(79)</sup> of algebraic groups over a field  $k$  of characteristic  $p > 0$*

$$1 \longrightarrow G' \xrightarrow{\tau} G \xrightarrow{\pi} G'' \longrightarrow 1$$

and the following assertions:

- (i) *The morphism  $\pi$  is smooth.*
- (ii)  *$G'$  is smooth.*
- (iii) *For every integer  $n > 0$ , the sequence below, induced by  $\tau$  and  $\pi$ , is exact:*

$$1 \longrightarrow {}_{\mathrm{Fr}^n} G' \longrightarrow {}_{\mathrm{Fr}^n} G \longrightarrow {}_{\mathrm{Fr}^n} G'' \longrightarrow 1.$$

- (iv) *The morphism  ${}_{\mathrm{Fr}} \pi : {}_{\mathrm{Fr}} G \longrightarrow {}_{\mathrm{Fr}} G''$  is an epimorphism of (fppf) sheaves.*
- (v) *The morphism  $\mathrm{Lie}(\pi) : \mathrm{Lie}(G) \longrightarrow \mathrm{Lie}(G'')$  is surjective.*

Then one has the implications (i)  $\Leftrightarrow$  (ii)  $\Rightarrow$  (iii)  $\Rightarrow$  (iv)  $\Leftrightarrow$  (v), and all the assertions are equivalent when  $G$  is smooth over  $k$ .

Indeed, (i) is equivalent to (ii) according to VI B 9.2 (vii), and it is clear that (iii) implies (iv). On the other hand, the equivalence of (iv) and (v) follows from 8.1.3.

The implication (ii)  $\Rightarrow$  (iii) follows from the diagram:

$$\begin{array}{ccccccc} 1 & \longrightarrow & G' & \xrightarrow{\tau} & G & \xrightarrow{\pi} & G'' \longrightarrow 1 \\ & & \downarrow \mathrm{Fr}^n(G'/k) & & \downarrow \mathrm{Fr}^n(G/k) & & \downarrow \mathrm{Fr}^n(G''/k) \\ 1 & \longrightarrow & G'^{(p^n)} & \xrightarrow{\tau^{(p^n)}} & G^{(p^n)} & \xrightarrow{\pi^{(p^n)}} & G''^{(p^n)} \longrightarrow 1 \end{array}$$

whose two rows are exact: since  $\mathrm{Fr}^n(G'/k)$  is an epimorphism of (fppf) sheaves according to corollary 8.3.1 below,  $\pi$  induces an epimorphism from  ${}_{\mathrm{Fr}^n} G$  onto  ${}_{\mathrm{Fr}^n} G''$  (generalize the snake lemma to sheaves in groups which are not necessarily commutative).

Similarly, when  $G$  is smooth over  $k$ ,  $\mathrm{Fr}(G/k)$  is an epimorphism, hence if in addition  ${}_{\mathrm{Fr}} \pi$  is an epimorphism, the same snake lemma applied to the diagram above for  $n = 1$  shows that  $\mathrm{Fr}(G'/k)$  is an epimorphism, hence that  $G'$  is smooth over  $k$ , according to 8.3.1 below.

**Proposition 8.3.** *If  $G$  is a group locally of finite type<sup>(80)</sup> over a field  $k$  of characteristic  $p > 0$ , there exists an integer  $n_0$  such that  $G/({}_{\mathrm{Fr}^n} G)$  is smooth over  $k$  for  $n \geq n_0$ .*

Since the construction of  $G/({}_{\mathrm{Fr}^n} G)$  commutes with extension of the base field (4.1.1 and VI A, 3.3.2), we may suppose that  $k$  is perfect. In this case  $G_{\mathrm{red}}$  is a  $k$ -group locally of finite type (cf. VI A 0.2), and one has the following commutative and exact diagram, where  $H$  denotes the  $k$ -scheme  $G_{\mathrm{red}} \backslash G$ :

$$\begin{array}{ccccccc} 1 & \longrightarrow & G_{\mathrm{red}} & \longrightarrow & G & \longrightarrow & H \\ & & \downarrow \mathrm{Fr}^n(G_{\mathrm{red}}/k) & & \downarrow \mathrm{Fr}^n(G/k) & & \downarrow \mathrm{Fr}^n(H/k) \\ 1 & \longrightarrow & G_{\mathrm{red}}^{(p^n)} & \longrightarrow & G^{(p^n)} & \longrightarrow & H^{(p^n)}. \end{array}$$

<sup>(79)</sup>**N.D.E.** i.e.  $\pi$  is faithfully flat and  $i$  is an isomorphism from  $G'$  onto  $\mathrm{Ker} \pi$ , so that  $G''$  represents the (fppf) quotient sheaf of  $G$  by  $G'$ , cf. VI A, 3.2 and 5.2.

<sup>(80)</sup>**N.D.E.** “Algebraic” has been replaced by “locally of finite type”.

Now  $H$  is the spectrum of a finite local  $k$ -algebra with residue field  $k$  (cf. VI A, 5.6.1). Consequently, there exists an integer  $n_0$  such that, for every  $n \geq n_0$ ,  $\mathrm{Fr}^n(H/k)$  factors through the “unit” section of  $H^{(p^n)}$ . It follows that, for  $n \geq n_0$ ,  $\mathrm{Fr}^n(G/k)$  factors through  $G_{\mathrm{red}}^{(p^n)}$ , and hence, according to VI A, 5.4.1, there is a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{\mathrm{Fr}^n(G/k)} & G_{\mathrm{red}}^{(p^n)} \\ \pi \downarrow & \nearrow i & \\ G/(\mathrm{Fr}^n G) & & \end{array}$$

where  $i$  is a closed immersion (and  $\pi$  the canonical projection). Since, in addition,  $i$  induces a homeomorphism of the underlying topological spaces, it is therefore an isomorphism. Since  $k$  is perfect,  $G_{\mathrm{red}}^{(p^n)}$  is smooth over  $k$  (VI A, 1.3.1), and hence  $G/(\mathrm{Fr}^n G)$  is smooth over  $k$ , for every  $n \geq n_0$ .

**Corollary 8.3.1.** *Let  $G$  be a group locally of finite type<sup>(80)</sup> over a field  $k$  of characteristic  $p > 0$ , and let  $n$  be an integer  $\geq 1$ .<sup>(81)</sup> The following conditions are equivalent:*

- (i)  $G$  is smooth over  $k$ .
- (ii)  $\mathrm{Fr}^n(G/k) : G \rightarrow G^{(p^n)}$  is an epimorphism of (fppf) sheaves.
- (iii)  $\mathrm{Fr}^n(G/k) : G \rightarrow G^{(p^n)}$  is faithfully flat.

First, since  $G$  is locally of finite type over  $k$ ,  $\mathrm{Fr}^n(G/k)$  is of finite presentation, so the equivalence of (ii) and (iii) follows from 8.1.4. Suppose  $G$  smooth over  $k$ , hence  $G$  reduced. Then, since  $\mathrm{Fr}^n(G/k)$  is surjective, it is faithfully flat (cf. VI A, 6.2 or VI B, 1.3).

Conversely, suppose  $\mathrm{Fr}^n(G/k)$  faithfully flat. Since  $\mathrm{Fr}^n(G^{(p^n)}/k)$  is deduced from  $\mathrm{Fr}^n(G/k)$  by base change (cf. 4.1.3), it too is faithfully flat, as is the composite:

$$\mathrm{Fr}^{2n}(G/k) : G \rightarrow G^{(p^n)} \rightarrow G^{(p^{2n})}.$$

One thus obtains that, for every  $m \in \mathbf{N}$ ,  $\mathrm{Fr}^{mn}(G/k) : G \rightarrow G^{(p^{mn})}$  is faithfully flat, hence induces an isomorphism  $G/(\mathrm{Fr}^{mn} G) \simeq G^{(p^{mn})}$  (cf. VI A, 5.4.1). Now, according to proposition 8.3,  $G^{(p^{mn})}$  is smooth over  $k$  for large  $m$ , hence so is  $G$ , by (fpqc) descent (cf. EGA IV<sub>4</sub>, 17.7.1).

**8.4.** — In the two statements which conclude this exposé, we return to the case of a field  $k$  of arbitrary characteristic.

When  $k$  is of characteristic 0 (resp.  $p > 0$ ), let  $n$  be an integer  $\geq 1$  (resp. an integer  $\geq 1$  prime to  $p$ ); in both cases, we say simply that  $n$  is prime to the characteristic of  $k$ . Moreover, if  $G$  is a group scheme over  $k$ , we denote by  $n_G : G \rightarrow G$  the morphism of  $k$ -schemes which sends an element  $x$  of  $G(T)$  to  $x^n \in G(T)$ , as  $T$  ranges over  $k$ -schemes.

**Proposition.** *Let  $G$  be an algebraic group over a field  $k$  and  $n$  an integer prime to the characteristic of  $k$ . Then  $n_G : G \rightarrow G$  is an étale morphism.*

<sup>(82)</sup> According to VI B 1.3, it suffices to show that  $n_G$  is étale at the origin. Let  $A$  be the local ring of  $G$  at the origin and  $I$  the maximal ideal of  $A$ . According to II 3.9.4, the map  $\mathrm{Lie}(n_G) : \mathrm{Lie}(G) \rightarrow \mathrm{Lie}(G)$  induced by  $n_G$  is the homothety of ratio  $n$ . It is therefore an isomorphism, as is the endomorphism induced by  $n_G$  on  $I/I^2 = \mathrm{Lie}(G)^*$ . If  $k$  is of characteristic 0,  $G$  is smooth over  $k$  (VI B 1.6.1, see also VII B 3.3.1), so the canonical morphism  $S(I/I^2) \rightarrow \mathrm{gr}_I(A)$  is an isomorphism, where  $\mathrm{gr}_I(A)$  denotes the graded object associated to the  $I$ -adic filtration. It follows that  $n_G$  induces an automorphism of  $\mathrm{gr}_I(A)$ ,

<sup>(81)</sup>**N.D.E.** The equivalence between (ii) and (iii) has been made explicit, and the proof expanded.

<sup>(82)</sup>**N.D.E.** In the statement, “étale at the origin” has been changed to “étale”, and the following sentence added.

hence also of the completion  $\hat{A}$  of  $A$ , and therefore  $n_G$  is étale at the origin (cf. EGA IV<sub>4</sub>, 17.6.3).

If the characteristic is  $p > 0$  and  $G$  is of height  $\leq 1$ , then  $A$  is isomorphic to the quotient of the symmetric algebra of  $\omega_{G/k} = I/I^2$  by the ideal generated by the  $p$ -th powers of elements of  $\omega_{G/k}$  (cf. 7.4); one can then apply the “same” argument as in characteristic 0, and one obtains that  $n_G$  induces an automorphism of  $A$ .

If  $G$  is of height  $\leq r$  and we suppose our assertion proved for groups of height  $\leq r - 1$ , denote by  $B, A, A'$  the affine algebras of  ${}_{\text{Fr}}G$ ,  $G$ , and  $G' = {}_{\text{Fr}}G \setminus G$ , and by  $n_B, n_A, n_{A'}$  the endomorphisms of  $B, A, A'$  induced by  $n_{{}_{\text{Fr}}G}$ ,  $n_G$ , and  $n_{G'}$ .<sup>(83)</sup> Let  $I' = I \cap A$  be the maximal ideal of  $A'$ ; since there is a cartesian square

$$\begin{array}{ccc} {}_{\text{Fr}}G & \longrightarrow & G \\ \downarrow & & \downarrow \\ e & \longrightarrow & G' \end{array}$$

one has  $B = A/I'A$ . Observe that  $n_{A'}$  (resp.  $n_B$ ) is none other than the endomorphism induced by  $n_A$  on  $A'$  (resp. on  $B$ ). According to VI A 3.2,  $A$  is a faithfully flat  $A'$ -module, and since  $A'$  is a local artinian ring ( $G'$  being an algebraic  $k$ -group of height  $\leq r - 1$ ), it follows that  $A$  is a free  $A'$ -module. Since the restriction of  $n_A$  to  $A'$  is  $n_{A'}$ , which is an isomorphism by the induction hypothesis, it follows from Nakayama’s lemma that  $n_A$  will be an automorphism if the endomorphism it induces on  $A/I'A$  is one. Now this endomorphism is none other than  $n_B$ , which is an automorphism since  $B$  is of height  $\leq 1$ . Thus  $n_A$  is an automorphism.

Finally, when  $G$  is an arbitrary algebraic group over a field of characteristic  $p > 0$ , what precedes shows that  $n_G$  induces automorphisms of the  $k$ -schemes  ${}_{\text{Fr}}^r G$ ; these schemes are affine over  $k$  and have as algebras the quotients of the local algebra  $A$  by the ideal  $I^{\{p^r\}}$  generated by the  $p^r$ -th powers of elements of  $I$ . Since  $n_G$  defines automorphisms of the algebras  $A/I^{\{p^r\}}$ , one sees, by passage to the projective limit, that  $n_G$  induces an automorphism of  $\hat{A}$ , hence  $n_G$  is étale at the origin (EGA IV<sub>4</sub>, 17.6.3).

**Proposition 8.5.** *Let  $G$  be a finite algebraic group of rank  $n$  over the field  $k$ . Then  $n_G : G \rightarrow G$  is the zero morphism of  $G$ .*

Let us point out at once the following corollary, obtained by combining 8.4 and 8.5:<sup>(84)</sup>

**Corollary 8.5.1.** *Let  $G$  be a finite algebraic group of rank  $n$  over the field  $k$ . If  $n$  is prime to the characteristic of  $k$ , then  $G$  is étale over  $k$ .*

Let us now prove 8.5. Let  $H$  be a normal subgroup of  $G$  of rank  $m$  over  $k$ . Denote by  $\lambda : H \times G \rightarrow G$  the morphism induced by the multiplication of  $G$ . Then, with the notation of VI A 3.2, there is a cartesian square:

$$\begin{array}{ccc} H \times G & \xrightarrow{\lambda} & G \\ \text{pr}_2 \downarrow & & \downarrow \pi \\ G & \xrightarrow{\pi} & H \setminus G \end{array}$$

Since  $\pi : G \rightarrow H \setminus G$  is faithfully flat and quasi-compact (VI A 3.2), and  $\text{pr}_2$  is locally free of rank  $m$ , it follows from EGA IV<sub>2</sub>, 2.5.2 that  $G \rightarrow H \setminus G$  is locally free of rank  $m$ . Denoting  $\text{rg}_k(H \setminus G)$  by  $r$ , one therefore has  $n = \text{rg}_k(G) = rm$ .

<sup>(83)</sup>**N.D.E.** The original has been expanded in what follows.

<sup>(84)</sup>**N.D.E.** This corollary, indicated implicitly in the original by “(confer 8.4)”, has been added. For another proof of the corollary, not using 8.5, see for example [T070], Lemma 5.

On the other hand, one has an exact sequence of “abstract” groups

$$1 \longrightarrow H(T) \longrightarrow G(T) \longrightarrow (H \backslash G)(T)$$

for every  $k$ -scheme  $T$ ; it is therefore clear that  $n_G$  is zero if  $m_H$  and  $r_{H \backslash G}$  are. Taking for  $H$  the neutral component  $G^0$  of  $G$ ,  $G^0 \backslash G$  is étale (cf. VI A 5.5.1), so that one may suppose  $G$  étale over  $k$  or infinitesimal (cf. 7.0).

If  $G$  is étale, one reduces, by extension of the base field, to the case where  $k$  is algebraically closed. In this case,  $G$  is a constant group (cf. I 4.1), and the statement is classical.

If  $G$  is infinitesimal and nonzero,  $k$  is necessarily of characteristic  $p > 0$  (cf. VI B 1.6.1 or VII B 3.3.1); the subgroups  ${}_{\mathbb{F}_p^n} G$  then form a composition series of  $G$  whose quotients are of height  $\leq 1$ .

This reduces us to the case where  $G$  is of height  $\leq 1$ . Let  $A$  (resp.  $L$ ) be the affine algebra (resp. Lie algebra) of  $G$ , and  $U = U_p(L)$ . According to 7.4, one has  $G = G_p(L)$ , whence  $A = U^*$ ; thus, if  $\dim_k L = r$ , the rank of  $G$  over  $k$  is  $p^r$  (cf. 5.3.3). We shall therefore study the morphism  $p_G : G \longrightarrow G$  defined by raising to the power  $p$ ; it induces an endomorphism  $p_A$  of  $A$  and, by duality, an endomorphism  $p_U$  of  $U$ .

Let  $I$  be the augmentation ideal of  $A$ ; we shall show that  $p_A(I) \subset I^p$ . Assuming this established, one will therefore have  $p_A^r(I) \subset I^{p^r}$ . On the other hand, one knows that  $I^{r(p-1)+1} = 0$  (since  $I$  is generated by  $r$  elements with zero  $p$ -th power). Since  $p^r > r(p-1)$ , it follows that  $p_A^r(I) = 0$ , hence  $p_G^r$  is the zero morphism. It therefore remains to show the assertion:

$$(*) \quad p_A(I) \subset I^p.$$

For every integer  $s \geq 1$ , we shall denote by  $m_A^{s-1} : A^{\otimes s} \longrightarrow A$  (resp.  $\Delta_U^s : U \longrightarrow U^{\otimes s}$ ) the map induced by the multiplication  $m_A$  of  $A$  (resp. the comultiplication  $\Delta_U$  of  $U$ ). Then  $p_A$  is equal to the following composite:

$$A \xrightarrow{\Delta_A^{p-1}} A^{\otimes p} \xrightarrow{m_A^{p-1}} A,$$

and since the transpose of  $m_A$  (resp.  $\Delta_A$ ) is  $\Delta_U$  (resp.  $m_U$ ), the endomorphism  $p_U = {}^t p_A$  of  $U$  is the composite below:

$$U \xrightarrow{\Delta_U^{p-1}} U^{\otimes p} \xrightarrow{m_U^{p-1}} U.$$

(85) Let  $J$  be the augmentation ideal of  $U$ ; one has  $U = k1_U \oplus J$ , and we shall denote by  $\pi$  the projection  $U \longrightarrow J$  with kernel  $k1_U$ . For every integer  $s \geq 1$ , denote by  $(I^s)^\perp$  the orthogonal of  $I^s$  in  $A^* = U$ , i.e.  $(I^s)^\perp$  is the set of  $u \in U$  such that the composite below is zero:

$$I^{\otimes s} \xrightarrow{m_A^{s-1}} I \xrightarrow{u} k.$$

Since the transpose of  $m_A$  is  $\Delta_U$ , one sees that  $(I^s)^\perp$  is the vector subspace  $P_{s-1}$  consisting of the  $u \in U$  such that  $\Delta_U^{s-1}(u)$  vanishes on  $I^{\otimes s}$ , i.e. denoting by  $\overline{\Delta_U^{s-1}}$  the composite of  $\Delta_U^{s-1}$  and the projection  $\pi^{\otimes s} : U^{\otimes s} \longrightarrow J^{\otimes s}$ , one obtains that

$$(I^s)^\perp = P_{s-1} = \text{Ker } \overline{\Delta_U^{s-1}}$$

(see also VII B, 1.3.6). Thus, to show assertion (\*), one must show that the transpose map  $p_U = {}^t p_A$  maps  $P_{p-1}$  into  $I^\perp = k1_U$ . Since  $p_U(1_U) = 1_U$ , it suffices to show the assertion below, where  $P_{p-1}^+$  denotes  $J \cap P_{p-1}$ :

$$(**) \quad p_U(P_{p-1}^+) = 0.$$

On the other hand, one shows easily, by induction on  $s$ , that  $P_{s-1}^+$  is the vector subspace of  $U$  generated by the products  $x_1 \cdots x_t$ , with  $1 \leq t \leq s-1$  and  $x_i \in L$  (cf. VII B 4.3). Now, if

(85) **N.D.E.** The original has been expanded in the following paragraph.

$x_1, x_2, \dots, x_t$  are elements of  $L$ , one has:

$$p_U(x_1 x_2 \cdots x_t) = m_U^{p-1} \left( \prod_{j=1}^t \sum_{i=1}^p 1 \otimes \cdots \otimes \overset{i}{\underset{\downarrow}{x_j}} \otimes \cdots \otimes 1 \right).$$

It is clear that the expression  $\prod_j \sum_i 1 \otimes \cdots \otimes x_j \otimes \cdots \otimes 1$  is a sum of  $p^t$  terms  $x_h$  indexed by the maps  $h$  from  $\{1, \dots, t\}$  to  $\{1, \dots, p\}$ .<sup>(86)</sup> Such a map  $h$  defines an *ordered partition*  $\mathbf{p}_h$  of  $\{1, \dots, t\}$  into at most  $p$  parts. Indeed, denote by  $i_1 < \cdots < i_r$  the elements of the image of  $h$ , and, for  $s = 1, \dots, r$ , put  $I_s = h^{-1}(i_s)$  and  $x_{I_s} = \prod_{j \in I_s} x_j$ , the product being taken in increasing order. Then  $h$  corresponds to the  $p$ -tensor

$$1 \otimes \cdots \otimes x_{I_1} \otimes \cdots \otimes x_{I_r} \otimes \cdots \otimes 1$$

(where each  $x_{I_s}$  is in position  $i_s$ ), and its image under  $m_U^p$  is the product:

$$x_{I_1} \otimes \cdots \otimes x_{I_r}$$

which depends only on the ordered partition  $\mathbf{p} = (I_1, \dots, I_r)$ , and which will be denoted by  $x_{\mathbf{p}}$ . For fixed  $\mathbf{p}$ ,  $x_{\mathbf{p}}$  is obtained for all choices of  $i_1 < \cdots < i_r$  in  $\{1, \dots, p\}$ , and one therefore obtains the equality

$$p_U(x_1 x_2 \cdots x_t) = \sum_{\mathbf{p}} \binom{p}{n(\mathbf{p})} x_{\mathbf{p}},$$

where  $\mathbf{p}$  ranges over the set of ordered partitions of  $\{1, \dots, t\}$  into at most  $p$  parts, and  $n(\mathbf{p})$  denotes the number of parts of  $\mathbf{p}$ . (One has  $1 \leq n(\mathbf{p}) \leq \min(t, p)$ .) When  $t < p$ , all the terms  $\binom{p}{n(\mathbf{p})}$  are therefore zero, so that  $p_U(x_1 \cdots x_t) = 0$ . Thus  $p_U$  vanishes on  $P_{p-1}^+$ , which proves assertion (\*\*), hence (\*), and completes the proof of 8.5.

**Corollary 8.5.2.** <sup>(87)</sup> *Let  $S$  be a reduced scheme and  $G$  a finite locally free  $S$ -group of rank  $n$ . Then  $n_G : G \rightarrow G$  is the zero morphism of  $G$ .*

Indeed, let  $S'$  be the sum of the  $\text{Spec } \mathcal{O}_{S, \eta}$ , as  $\eta$  ranges over the maximal points of  $S$ . Since  $S$  is reduced, the morphism  $S' \rightarrow S$  is schematically dominant, and the same is true of the morphism  $f : G_{S'} \rightarrow G$ , since  $G$  is finite locally free over  $S$  (cf. EGA IV<sub>3</sub>, 11.10.5). Since  $G \rightarrow S$  is affine, hence separated, the coincidence locus of  $n_G$  and the zero morphism is a closed subscheme of  $G$ ; now it contains  $f$  according to 8.5, hence equals  $G$ , i.e.  $n_G$  is the zero morphism.

**Remark 8.5.3.** Let us also point out that, according to a theorem of P. Deligne (see [TO70], p. 4), if  $G$  is a finite locally free *commutative*  $S$ -group of rank  $n$  over an arbitrary base  $S$ , then  $n_G = 0$ .

<sup>(86)</sup>**N.D.E.** The original has been expanded in what follows, replacing the notion of preorder by the equivalent notion of ordered partition.

<sup>(87)</sup>**N.D.E.** This corollary, mentioned in Exp. VIII, Remark 7.3.1, has been added.

## Bibliography

- [BAI<sub>g</sub>] N. Bourbaki, *Algèbre*, Chap. I–III, Hermann, 1974, Chap. X, Masson, 1980.
- [BAC] N. Bourbaki, *Algèbre commutative*, Chap. I–IV, Masson, 1985.
- [BLie] N. Bourbaki, *Groupes et algèbres de Lie*, Chap. I, Hermann, 1971.
- [DG70] M. Demazure, P. Gabriel, *Groupes algébriques*, Masson & North-Holland, 1970.
- [Ja03] J. C. Jantzen, *Representations of algebraic groups*, Academic Press, 1987; 2nd edition, Amer. Math. Soc., 2003.
- [TO70] J. Tate, F. Oort, *Group schemes of prime order*, *Ann. scient. Éc. Norm. Sup.* **3** (1970), 1–21.